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Product-Service System Designing for Environmental Sustainability:

Insights into Design Cognition of Experienced Product Developers

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The cover was designed by the author and developed by Shwetha VC. The central figure of the cover symbolically represents planet Earth, the constituent artificially developed human societies, and the effects of such human constructs on the environment of the planet. The right half of the figure represents a scenario in which our societies are polluted due to unsustainable manufacturing practices. The left half of the figure represents a scenario in which, our societies are environmentally sustainable and prosperous. The wind-mill and the service technicians represent product-service systems and their potential to support the transition towards an environmentally sustainable society. The human face fused with the planet symbolically suggests that the key to unlock the doors to either scenario, lies with our ability to think and design our societies.

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"*The world as we have created it is a process of our thinking. It cannot be changed without changing our thinking*."

– Albert Einstein

ABSTRACT

The overall objective of this research is to support practitioners within the manufacturing industry to effectively design product-service system (PSS)-based solutions with relatively lower environmental impacts than typical product-centric solutions. Design research methodology (DRM) is applied to systematically address this objective with both descriptive and prescriptive approaches. The research draws empirical insights from a pool of 68 experienced product developers from the manufacturing industry. Initially, a generic model to predict the variant nature of domain-specific design cognition is conceptualized.

Using this conceptual framework, differences in design cognition between conceptual product and PSS designing are theoretically predicted and empirically tested using protocol analysis. Based on a statistical comparison of the findings, the cognitive nature of PSS designing is descriptively characterized in relation to that of typical product designing. The results suggest that PSS designing is more problem-focused and applies a similar level of focus on the highest level of systems abstraction as product designing. A deeper investigation into PSS designing revealed that compared to product aspects, relatively less focus was afforded to the service aspects, interactions between product and service aspects, and the aspects regarding the overall PSS. Based on these findings, it was inferred that the cohort might need PSS domain-specific support to effectively design such systems. Three design support prescriptions were developed based on the improved understanding of the cognitive characteristics of PSS designing. One of these prescriptions, termed the PSS design schema, was developed using the method of literature analysis and synthesis. The other two prescriptions, termed lifecycle function deployment (LFD) and systemic lifecycle engineering, were developed using the method of action design research (ADR). Subsequently, the effects of the PSS design schema were descriptively tested using protocol analysis, and those of the other two prescriptions were assessed with a lifecycle assessment (LCA) of the design outcomes.

The effects of the PSS design schema point towards significant changes in the cognitive characteristics of PSS designing. The effects of the other two prescriptions point towards significant reductions in the potential environmental impacts of the concepts when compared with the respective existing product-centric offerings that were redesigned using the support. Based on these empirical findings, the conceptual framework is refined and extended to conceptualize and hypothesize the interplays of pre-existing domain-specific knowledge of the designers and the effects of design support on design cognition.

Keywords: Design cognition, Product-service system designing, Product designing, Design support, Design for environmental sustainability.

Sammanfattning

Det övergripande målet med denna forskning är att stödja utvecklare inom tillverkningsindustrin att effektivt skapa lösningar baserat på produkttjänstesystem (eng. product-service system (PSS)) som har en lägre miljöpåverkan än typiska produktcentrerade lösningar. För att uppfylla detta syfte tillämpas en systematisk forskningsmetodik kallad DRM. Detta tillvägagångssätt möjliggör både beskrivande och föreskrivande forskning. I forskningen nyttjas en resurs bestående av 68 erfarna produktutvecklare från tillverkningsindustrin för att uppnå empiriska resultat. I ett första stadie utvecklas en generisk modell för att förutsäga karaktären av domänspecifik designkognition.

Genom att tillämpa detta konceptuella ramverk förutsägs och undersöks skillnader i designkognition vid utveckling av både produkter och PSS. I ramverket används protokollanalys för att sätta empiri mot teori. Sedan, baserat på statistisk bearbetning och jämförelse av forskningsdata, beskrivs kognitiva kännetecken för PSS-utformning i förhållande till rutinmässig produktutveckling. Det indikeras att utformning av PSS-lösningar är mer problemfokuserat än för produkter. Däremot kräver båda lika stort fokus vid hög abstraktionsnivå. Vid djupdykning inom PSS-utformning visas att begränsat och inkonsekvent fokus tilldelas aspekter inom tjänster, interaktioner mellan produkter och tjänster samt övergripande för PSS. Baserat på dessa resultat drogs slutsatsen att utvecklare kan behöva domänspecifikt stöd för att effektivt kunna utforma PSS.

Med grund i den förbättrade förståelsen av kognitiva kännetecken för PSS-utformning utvecklades tre metoder som stödjer under designstadiet. En av dessa, benämnd designschemat för PSS, utvecklades genom litteraturanalys och -syntes. De andra två metoderna – driftsättning av livscykelfunktionalitet och systematisk livscykelteknik – utvecklades genom en aktionsforskningsmetod. Effekterna av designschemat för PSS undersöktes och beskrevs genom protokollanalys medan de andra två evaluerades genom att tillämpa livscykelanalys (LCA) på det som utformats.

De identifierade effekterna av designschemat för PSS pekar på betydande förändringar i de kognitiva kännetecknen för PSS-utformning. Genom att använda det utvecklade stödet pekar de andra två metoderna mot betydande minskning av konceptens potentiella miljöpåverkan jämfört med respektive omarbetat produkterbjudande. Baserat på de empiriska designkognitionsresultaten förfinades och utökades det konceptuella ramverket till att konceptualisera och hypotetisera samspelet mellan redan existerande domänspecifik kunskap hos utvecklare och effekten av stöd vid utformning PSS-lösningar.

ಸಾರಾಂಶ

ವಿಶಿಷ್ಟ ಉತ್ಪನ್ನ-ಕೇಂದ್ರಿತ ಪರಿಹಾರಗಳಿಗಿಂತ ತುಲನಾತ್ಮಕವಾಗಿ ಕಡಿಮೆ ಪರಿಸರ ಪರಿಣಾಮಗಳೊಂದಿಗೆ ಉತ್ಪನ್ನ-ಸೇವಾ ವ್ಯವಸ್ಥೆ (PSS) ಆಧಾರಿತ ಪರಿಹಾರಗಳನ್ನು ಪರಿಣಾಮಕಾರಿಯಾಗಿ ವಿನ್ಯಾಸಗೊಳಿಸಲು ಉತ್ಪಾದನಾ ಉದ್ಯಮದೊಳಗಿನ ಅಭ್ಯಾಸಕಾರರನ್ನು ಬೆಂಬಲಿಸುವುದು ಈ ಸಂಶೋಧನೆಯ ಒಟ್ಟಾರೆ ಉದ್ದೇಶವಾಗಿದೆ. ಡಿಸೈನ್ ರಿಸರ್ಚ್ ಮೆಥಡಾಲಜಿ (DRM) ಅನ್ನು ಕ್ರಮಬದ್ಧವಾಗಿ ಈ ಗುರಿಯನ್ನು ವಿವರಣಾತ್ಮಕ ಮತ್ತು ಪ್ರಿಸ್ಕ್ರಿಪ್ಟಿವ್ ಆಪ್-ಪ್ರೋಚ್‌ಗಳೊಂದಿಗೆ ಪರಿಹರಿಸಲು ಅನ್ವಯಿಸಲಾಗುತ್ತದೆ. ಸಂಶೋಧನೆಯು ಉತ್ಪಾದನಾ ಉದ್ಯಮದಿಂದ 68 ಅನುಭವಿ ಉತ್ಪನ್ನ ಡೆವಲಪರ್‌ಗಳ ಪೂಲ್‌ನಿಂದ ಪ್ರಾಯೋಗಿಕ ಒಳನೋಟಗಳನ್ನು ಸೆಳೆಯುತ್ತದೆ. ಆರಂಭದಲ್ಲಿ, ಡೊಮೇನ್-ನಿರ್ದಿಷ್ಟ ವಿನ್ಯಾಸದ ಅರಿವಿನ ಭಿನ್ನ ಸ್ವರೂಪವನ್ನು ಊಹಿಸಲು ಜೆನರ್-ಐಸಿ ಮಾದರಿಯನ್ನು ಪರಿಕಲ್ಪನೆ ಮಾಡಲಾಗಿದೆ.

ಈ ಪರಿಕಲ್ಪನಾ ಚೌಕಟ್ಟನ್ನು ಬಳಸಿಕೊಂಡು, ಪರಿಕಲ್ಪನಾ ಉತ್ಪನ್ನ ಮತ್ತು ಪಿಎಸ್ಎಸ್ ವಿನ್ಯಾಸದ ನಡುವಿನ ವಿನ್ಯಾಸದ ಅರಿವಿನ ವ್ಯತ್ಯಾಸಗಳನ್ನು ಸೈದ್ಧಾಂತಿಕವಾಗಿ ಊಹಿಸಲಾಗಿದೆ ಮತ್ತು ಪ್ರೋಟೋಕಾಲ್ ವಿಶ್ಲೇಷಣೆಯನ್ನು ಬಳಸಿಕೊಂಡು ಪ್ರಾಯೋಗಿಕವಾಗಿ ಪರೀಕ್ಷಿಸಲಾಗುತ್ತದೆ. ಸಂಶೋಧನೆಗಳ ಅಂಕಿಅಂಶಗಳ ಹೋಲಿಕೆಯ ಆಧಾರದ ಮೇಲೆ, ಪಿಎಸ್ಎಸ್ ವಿನ್ಯಾಸದ ಅರಿವಿನ ಸ್ವರೂಪವು ವಿಶಿಷ್ಟ ಉತ್ಪನ್ನ ವಿನ್ಯಾಸಕ್ಕೆ ಸಂಬಂಧಿಸಿದಂತೆ ವಿವರಣಾತ್ಮಕವಾಗಿ ನಿರೂಪಿಸಲ್ಪಟ್ಟಿದೆ. ಫಲಿತಾಂಶಗಳು ಪಿಎಸ್ಎಸ್ ವಿನ್ಯಾಸವು ಹೆಚ್ಚು ಸಮಸ್ಯೆ-ಕೇಂದ್ರಿತವಾಗಿದೆ ಮತ್ತು ಉತ್ಪನ್ನ ವಿನ್ಯಾಸದಂತೆ ಉನ್ನತ ಮಟ್ಟದ ಸಿಸ್ಟಮ್ ಅಮೂರ್ತತೆಯ ಮೇಲೆ ಒಂದೇ ರೀತಿಯ ಗಮನವನ್ನು ಅನ್ವಯಿಸುತ್ತದೆ ಎಂದು ಸೂಚಿಸುತ್ತದೆ. PSS ವಿನ್ಯಾಸದ ಆಳವಾದ ತನಿಖೆಯು ಉತ್ಪನ್ನದ ಅಂಶಗಳಿಗೆ ಹೋಲಿಸಿದರೆ, ಸೇವಾ ಅಂಶಗಳು, ಉತ್ಪನ್ನ ಮತ್ತು ಸೇವಾ ಅಂಶಗಳ ನಡುವಿನ ಪರಸ್ಪರ ಕ್ರಿಯೆಗಳು ಮತ್ತು ಒಟ್ಟಾರೆ PSS ಗೆ ಸಂಬಂಧಿಸಿದ ಅಂಶಗಳಿಗೆ ತುಲನಾತ್ಮಕವಾಗಿ ಕಡಿಮೆ ಗಮನವನ್ನು ನೀಡಲಾಗಿದೆ ಎಂದು ಬಹಿರಂಗಪಡಿಸಿತು. ಈ ಸಂಶೋಧನೆಗಳ ಆಧಾರದ ಮೇಲೆ, ಅಂತಹ ವ್ಯವಸ್ಥೆಗಳನ್ನು ಪರಿಣಾಮಕಾರಿಯಾಗಿ ವಿನ್ಯಾಸಗೊಳಿಸಲು ಸಮೂಹಕ್ಕೆ PSS ಡೊಮೇನ್-ನಿರ್ದಿಷ್ಟ ಬೆಂಬಲ ಬೇಕಾಗಬಹುದು ಎಂದು ಊಹಿಸಲಾಗಿದೆ. PSS ವಿನ್ಯಾಸದ ಅರಿವಿನ ಗುಣಲಕ್ಷಣಗಳ ಸುಧಾರಿತ ತಿಳುವಳಿಕೆಯ ಆಧಾರದ ಮೇಲೆ ಮೂರು ವಿನ್ಯಾಸ ಬೆಂಬಲದ ಪ್ರಿಸ್ಕ್ರಿಪ್ಷನ್ಸ್‌ಗಳನ್ನು ಅಭಿವೃದ್ಧಿಪಡಿಸಲಾಗಿದೆ. PSS ವಿನ್ಯಾಸ ಸ್ಕೀಮಾ ಎಂದು ಕರೆಯಲ್ಪಡುವ ಈ ಪ್ರಿಸ್ಕ್ರಿಪ್ಷನ್‌ಗಳಲ್ಲಿ ಒಂದನ್ನು ಸಾಹಿತ್ಯ ವಿಶ್ಲೇಷಣೆ ಮತ್ತು ಸಂಶ್ಲೇಷಣೆಯ ವಿಧಾನವನ್ನು ಬಳಸಿಕೊಂಡು ಅಭಿವೃದ್ಧಿಪಡಿಸಲಾಗಿದೆ. ಲೈಫ್‌ಸೈಕಲ್ ಫಂಕ್ಷನ್ ಡಿಪ್ಲಾಯ್‌ಮೆಂಟ್ (ಎಲ್ಎಫ್‌ಡಿ) ಮತ್ತು ಸಿಸ್ಟಮಿಕ್ ಲೈಫ್‌ಸೈಕಲ್ ಇಂಜಿನಿಯರಿಂಗ್ ಎಂದು ಕರೆಯಲ್ಪಡುವ ಇತರ ಎರಡು ಪ್ರಿಸ್ಕ್ರಿಪ್ಷನ್‌ಗಳನ್ನು ಆಕ್ಷನ್ ಡಿಸೈನ್ ರಿಸರ್ಚ್ (ADR) ವಿಧಾನವನ್ನು ಬಳಸಿಕೊಂಡು ಅಭಿವೃದ್ಧಿಪಡಿಸಲಾಗಿದೆ. ತರುವಾಯ, ಪಿಎಸ್ಎಸ್ ವಿನ್ಯಾಸದ ಸ್ಕೀಮಾದ ಪರಿಣಾಮಗಳನ್ನು ಪ್ರೋಟೋಕಾಲ್ ವಿಶ್ಲೇಷಣೆಯನ್ನು ಬಳಸಿಕೊಂಡು ವಿವರಣಾತ್ಮಕವಾಗಿ ಪರೀಕ್ಷಿಸಲಾಯಿತು, ಮತ್ತು ಇತರ ಎರಡು ಪ್ರಿಸ್ಕ್ರಿಪ್ಷನ್‌ಗಳನ್ನು ವಿನ್ಯಾಸದ ಫಲಿತಾಂಶಗಳ ಜೀವನಚಕ್ರ ಮೌಲ್ಯಮಾಪನ (LCA) ಯೊಂದಿಗೆ ಮೌಲ್ಯಮಾಪನ ಮಾಡಲಾಯಿತು.

ಪಿಎಸ್‌ಎಸ್ ವಿನ್ಯಾಸದ ಸ್ಕೀಮಾದ ಪರಿಣಾಮಗಳು ಪಿಎಸ್‌ಎಸ್ ವಿನ್ಯಾಸದ ಅರಿವಿನ ಗುಣಲಕ್ಷಣಗಳಲ್ಲಿ ಗಮನಾರ್ಹ ಬದಲಾವಣೆಗಳತ್ತ ಗಮನ ಹರಿಸುತ್ತವೆ. ಬೆಂಬಲವನ್ನು ಬಳಸಿಕೊಂಡು ಮರುವಿನ್ಯಾಸಗೊಳಿಸಲಾದ ಸಂಬಂಧಿತ ಅಸ್ತಿತ್ವದಲ್ಲಿರುವ ಉತ್ಪನ್ನ-ಕೇಂದ್ರಿತ ಕೊಡುಗೆಗಳೊಂದಿಗೆ ಹೋಲಿಸಿದರೆ ಪರಿಕಲ್ಪನೆಗಳ ಸಂಭಾವ್ಯ ಪರಿಸರ-ಮಾನಸಿಕ ಪರಿಣಾಮಗಳಲ್ಲಿ ಗಮನಾರ್ಹವಾದ ಕಡಿತದ ಕಡೆಗೆ ಇತರ ಎರಡು ಪ್ರಿಸ್ಕ್ರಿಪ್ಷನ್‌ಗಳ ಪರಿಣಾಮಗಳು ಸೂಚಿಸುತ್ತವೆ. ಈ ಪ್ರಾಯೋಗಿಕ ಆವಿಷ್ಕಾರಗಳ ಆಧಾರದ ಮೇಲೆ, ವಿನ್ಯಾಸಕಾರರ ಪೂರ್ವ-ಅಸ್ತಿತ್ವದಲ್ಲಿರುವ ಡೊಮೇನ್-ನಿರ್ದಿಷ್ಟ ಜ್ಞಾನದ ಪರಸ್ಪರ ಸಂಬಂಧಗಳು ಮತ್ತು ಡಿ-ಸೈನ್ ಅರಿವಿನ ಮೇಲೆ ವಿನ್ಯಾಸ ಬೆಂಬಲದ ಪರಿಣಾಮಗಳನ್ನು ಪರಿಕಲ್ಪನೆ ಮಾಡಲು ಮತ್ತು ಊಹಿಸಲು ಪರಿಕಲ್ಪನಾ ಚೌಕಟ್ಟನ್ನು ಪರಿಷ್ಕರಿಸಲಾಗಿದೆ ಮತ್ತು ವಿಸ್ತರಿಸಲಾಗಿದೆ.

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Lastly, I would like to dedicate this thesis to my parents, wife, brother and the rest of my family and friends. Pappa and Amma, you are my biggest role models, and I cannot thank you enough for all that you have done for me. Your constant source of support, love and belief has driven me throughout my life. Shwetha, I cannot express how thankful I am to have you as my best friend and my life partner. Thank you for patiently supporting me throughout my PhD journey and for constantly making me a better person. I look forward to a life full of quirky adventures with you!

List of appended papers

I. Neramballi, A., Sakao, T., & Gero, J.S. (2022). ***Is product-service system designing different from product designing? A cognitive study of experienced practitioners.***
(Working journal manuscript).

Author's contribution: *As the first author, I led the conceptualisation and writing of the majority of the article with the support of the two co-authors. Empirical data collection was carried out by all the three authors.*

II. Neramballi, A., Sakao, T., Gero, J.S. (2022). ***How Do Designers Think in Systems? – Empirical Insights from Protocol Studies of Experienced Practitioners Designing Product-Service Systems.*** In: Gero, J.S. (eds) Design Computing and Cognition'20. Springer, Cham. https://doi.org/10.1007/978-3-030-90625-2_27
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Author's contribution: *As the first author, I led the conceptualisation and writing of the majority of the article with the support of the two co-authors. I along with the second co-author carried out the empirical data collection.*

III. Sakao, T., & Neramballi, A. (2020). ***A Product/Service System Design Schema: Application to Big Data Analytics.*** Sustainability, 12(8), 3484. https://doi.org/10.3390/su12083484
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Author's contribution: *The first author led the conceptualization, data-collection and writing of several parts of the paper, with a focus on big data analytics. I supported the conceptualization of the paper, was responsible for the literature analysis and synthesis of prescriptive PSS design literature and led the development of the product-service system (PSS) design schema.*

IV. Neramballi, A., Sakao, T., Willskytt, S., & Tillman, A. M. (2020). ***A design navigator to guide the transition towards environmentally benign product/service systems based on LCA results.*** Journal of Cleaner Production, 277, 124074.
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Author's contribution: *I and the second co-author led the conceptualisation of the paper. I led the writing of the several parts of the paper (with a focus on PSS design) with support from the all the co-authors. The second co-author drafted the introduction chapter, the third and fourth co-authors took the lead on the lifecycle assessment (LCA) aspects of the paper. The first three authors were responsible for empirical data collection.*

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Related work

Neramballi, A., Sakao, T., & Gero, J. S. (2018). ***What do experienced practitioners discuss when designing Product/Service Systems?*** *In International Conference on Design Computing and Cognition, Lecco, Italy, July 2018*. (pp. 361-380). Springer, Cham. https://doi.org/10.1007/978-3-030-05363-5_20

Neramballi, A., Sakao, T., & Gero, J. S. (2019). ***Effects of a design support on practitioners designing a Product/Service System–a case study.*** *In Human Behavior in Design Conference, Tutzing, Germany*, April, 2019.

Sakao, T., Liu, Y., & Neramballi, A. (2019). ***Enhancing PSS design through big data, IoT and bigdata analytics.*** In *Spring Servitization Conference, Linköping, Sweden, May 2019* (pp. 197-205).

Neramballi, A. (2020). ***Understanding and Supporting Product-Service System Designing: Preliminary Insights and Support for Designing Resource-Efficient and Effective Solutions*** (Vol. 1877). *Linköping University Electronic Press.*

Neramballi, A., & Sakao, T. (2021). ***Zooming in on product-service system ecodesigning: a novel analysis approach and application to a case involving experienced practitioners.*** *In International Conference on Engineering Design, Gothenburg, Sweden, August 2021*. *Proceedings of the Design Society*. Cambridge University Press, 1, pp. 3461–3470. doi: 10.1017/pds.2021.607.

Sakao, T., Neramballi, A., Matschewsky, J. (2022) ***Avoid Service Design Trap by Guiding Product/Service System Design with Product-Service Dependency Knowledge Base.*** *In Design 2022, Dubrovnik, Croatia, May 2022. Proceedings of the Design Society*. Cambridge University Press, 2, pp. 1955–1964. doi: 10.1017/pds.2022.198.

Glossary

Cognitive effort ..... Cognitive effort is referred to as the available amount of processing capacity of the working memory space that is utilized to perform an information processing task (based on Sweller (1988) and Tyler et al. (1979)).

Design cognition ..... The mental processes underlying the activity of designing (based on (Hay et al., 2017, 2020)).

Domain ..... Anything a specific constraint can be generalized to and from, such as a design object, design process or the experiences of the designer (based on Visser (2009)).

Design problem ..... A design challenge that is defined by certain requirements, conditions, and constraints. Domain specific design problem is a design challenge that is constrained by requirements, design objects and conditions specific to the domain.

Design support ..... Any kind of design method, procedure, tool, means or aid that can improve the design process (based on (Blessing & Chakrabarti, 2009)). A design support can be considered to be domain-specific, if its internal logic is catered to the goals of the domain of the object being designed. Design process support prescribes a formally specified sequence of activities to aid designers to develop a design or a class of designs specific to a domain, which will often be a customization of a methodology to a domain specific problem (based on Gerrike et al. (2015)). Design support methods are specifications of how a specified result is to be achieved, which can include specifications of how information is to be shown, how a task should be decomposed, what information is to be used as an input to the

method, what tools are to be used, what actions are to be performed and how actions should be sequenced (based on Gerrike et al. (2015)).

Experienced practitioners..... Design professionals with an average of 10 years of experience.

Environmental sustainability ... "Maintenance of natural capital" (Goodland, 1995). Sustainability refers to "meeting the needs of the present without comprising the ability of future generations to meet their own needs" (WCED, 1987).

Product-service system..... "A system consisting of integrated combinations of tangible products and intangible services, that is designed from a lifecycle perspective to create value for customers and to reduce its environmental impact" (Neramballi et al., 2020).

Product ..... A tangible and physical artefact manufactured to address specific exogenous requirements upon use.

Lifecycle..... Consecutive and interlinked activities a product goes through, starting from material acquisition, design and development, manufacturing, delivery and installation, use (including activities such as reuse, maintenance, repair, remanufacturing, refurbishing and upgrading), end-of-life treatment and finally, disposal (ISO:14006, 2020).

Schema..... A higher-level element of knowledge consisting of lower-level elements of information stored in the long term memory space (based on (Sweller et al., 1998)).

Schema construction ..... Linking of different lower-level specific elements of information processed in the working memory space, as a single higher-level element of knowledge schema in the long term memory space (based on (Sweller et al., 1998)).

Schema automation ..... Process in which lower-level elements of processed information are repeatedly compiled under a domain-compatible higher-level knowledge schema. This can lead to automatic processing of domain compatible information (processing not requiring significant conscious effort) (based on (Sweller et al., 1998)).

Action design research ..... "A research method for generating prescriptive design knowledge through building and evaluating IT artefacts in an organizational setting" (Sein et al., 2011).

Protocol analysis ..... A research method for collecting and analysing verbal think-aloud data (Ericsson & Simon, 1993; van Someren et al., 1994).

Micro frame ..... A temporary construct that is developed within the working memory space to interpret, manipulate, and organize exogenous design data to simultaneously solve and learn from design problems.

Abbreviations

STS.....Socio-technical systems

GHG.....Green-house gas

PSS.....Product-Service System

LCA.....Lifecycle assessment

QFD.....Quality function deployment

FBS.....Function-Behavior-Structure

HSA.....Hierarchical levels of systems abstraction

TIPS or TRIZ.....Theory of inventive problem-solving

DbA.....Design by Analogy methods

PBSA.....Pahl and Beitz systematic approach

CAD.....Computer aided design

CAM-D.....Cognitive architecture and mechanism of designing

CE.....Circular economy

DRM.....Design Research Methodology

RC.....Research clarification

DS-1.....Descriptive studies – 1

DS-2.....Descriptive studies – 2

PS.....Prescriptive studies

AR.....Action research

ADR.....Action design research

LFD.....Lifecycle function deployment

DP.....Design parameter

P-S space.....Problem-Solution space

P-S DP.....Product-Service DP

RET.....Solution retirement

MUL.....Multiple stages

DEV ..... Solution development

DEP..... Solution deployment

VAE ..... Vertical access equipment

Symbols

x & y.....Representations of two different domains
K.....Total knowledge
Kx.....Knowledge schema specific to domain x
n.....Degree of automation specific to domain x
F.....Micro-frame
j.....Number of micro-frames
Ky.....Knowledge schema specific to domain y
m.....Degree of automation specific to domain y
o.....Information output
i.....Knowledge input
Ey.....Exogenous data specific to domain y
Ry.....Design representations specific to domain y

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Chapter 1 – Introduction

1.1 Systemic approaches to solve systemic challenges

Human societies can be considered as socio-technical systems (STSs) with a variety of interacting social and technical subsystems and components (Geels, 2004). The intensifying demands of these artificial systems have contributed to the steady acceleration of the rate of global resource extraction and the destabilization of planetary systems that are conducive to human development (Steffen & Rockström, 2015). A significant portion of these resources are procured and is utilized by the manufacturing industry (International resource panel, 2017). The manufacturing industry is one of the major drivers of the growth of STSs as they contribute to the development of product artefacts and services which serve as the building blocks of such systems (Hughes, 1987). Although the increasing rate of global resource extraction has fuelled the growth of the manufacturing industry, it has also had devastating impacts on the environment.

Such an unsustainable practice of resource extraction, manufacturing, consumption, and subsequent disposal is expected to drive up the rate of resource extraction from 92 billion tons per year in 2017 (UNEP, 2019) to around 183 billion tons per year by 2050 (UNEP, 2016). This accelerating rate of resource extraction contributes to over half of the total green-house gas (GHG) emissions (UNEP, 2019). These trends increase the risk of causing irreversible damage to the environment, and consequently, threaten the wellbeing of humanity (Steffen et al., 2015). Manufacturing companies are further challenged by inter-linked issues such as increasing market competitiveness, resource supply risks, and resource costs (Jackson, 2009; UNEP, 2019).

To effectively address such complex and inter-linked challenges, significant changes in design approaches are necessary in terms of re-consideration of what is being designed, how it is being designed, and the values to be delivered (Dorst, 2011). Environmental sustainability potential is expected to increase with the consideration and involvement of a wider range of societal stakeholders and the application of a higher level of a systems perspective in the design processes (Ceschin & Gaziulusoy, 2016). In recognition of this potential many manufacturing companies are increasingly adapting their design and business approaches. For example, several manufacturers are adopting more systemic design approaches (Charnley et al., 2011) and are increasingly integrating human-based service activities into their core product offerings (EU Business Innovation Observatory, 2016). These types of systemic solutions are commonly referred to as product-service systems (PSSs) (Mont, 2002; Tischner et al., 2002).

In this thesis, a PSS is defined as *"*a system consisting of integrated combinations of tangible products and intangible services, that is designed from a lifecycle perspective to create value for customers and to reduce its environmental impact*"* (Neramballi et al., 2020). These systems are demanded (Brissaud et al., 2022; Goedkoop et al., 1999; Mont, 2002) and are also shown, in certain cases, to have relatively benign environmental impacts and a higher potential for resource and cost efficiency in comparison to the traditional product-based offerings (Lindahl et al., 2014; Matschewsky, 2019; Vasantha et al., 2015). The PSS is also expected to be a source of market differentiation and a steady source of revenue for the manufacturers (Baines et al., 2017; Neely, 2008; Oliva & Kallenberg, 2003).

1.2 From product to product-service system designing for environmental sustainability and competitiveness

PSS-based solutions are widely considered to have high potential to simultaneously address societal needs and contribute to the demanded societal transition towards a resource-efficient and environmentally sustainable economy (Tukker, 2015). In traditional business settings, manufacturers primarily generate revenue by selling the ownership of products to customers (Aurich et al., 2006; Mont, 2002). In stark contrast, PSS-based businesses provide the possibility to sell functionality or performance of the system that can effectively address customer needs without necessarily transferring the ownership of product elements (Alonso-Rasgado et al., 2004). Thus, PSS-based solution providers will be incentivised to minimize resource use of the system, to reduce potential economic costs throughout the lifecycle (Vezzoli et al., 2015). Despite its widely acknowledged potential benefits, the process of designing these types of systemic solutions has often not been straightforward for manufacturers (Brissaud et al., 2022; Cavalieri & Pezzotta, 2012; Isaksson et al., 2009).

Compared to a traditional focus of design disciplines on the development of pure material artefacts, PSS designing requires a shift in perspective and the framing of the design practice (Brissaud et al., 2022; Meier et al., 2010; Morelli, 2003). Here, framing refers to the application of certain working principles to design specific artefacts or activities that can lead to the attainment of specific values (Dorst, 2011). The designing of PSS-based solutions requires a more systemic approach with the simultaneous consideration and integration ("the how") of interdependent tangible product and intangible service design objects ("the what") that can together deliver functional performance (Alonso-Rasgado et al., 2004; Dewberry et al., 2013; Meier et al., 2010) with minimal environmental impacts ("the aspired values"). However, a mere transition from the practice of selling products to offering PSSs may not necessarily lead to an absolute decoupling of economic growth from resource use or environmental costs (Kjaer et al., 2019; Tukker, 2015).

Unless PSS solutions are designed to have minimal environmental impacts, the environmental sustainability performance of such systems may be limited (Vasantha et al., 2015). Literature suggests that manufacturers have struggled to effectively reframe their design practices for PSS development (Beuren, Ferreira and Miguel, 2013; Annarelli, Battistella, and Nonino, 2016), due to a variety of issues, such as, a lack of suitable know-how and unsuitable organizational structures (Isaksson et al., 2009; Matschewsky et al., 2018).

In terms of know-how, designers are required to possess a broader range of knowledge and capabilities during PSS designing than during either product or service designing and the former is expected to be more complex (Maussang et al., 2009; Vasantha et al., 2015). For example, product and service design objects have fundamentally different characteristics: there are differences in the tangibility of the design objects, differences in terms of the separability of production and consumption of products and services, and the temporal freedom to design the respective objects along the lifecycle (Maussang et al., 2009; Meier et al., 2010). Due to the higher degree of intangibility (Maussang et al., 2009) and the higher degree of the social construction of PSS than its product part (Morelli, 2002), PSSs are considered to have higher degrees of uncertainties, which designers need to address (Sakao et al., 2020).

In addition, the potentially extended control over the lifecycle (Schweitzer & Aurich, 2010) and increasing trends of digitalization have contributed to the possibility for the providers to collect and leverage vast amounts of data concerning product use or behaviour, services, and product-service relationships throughout the PSS lifecycle (Geum & Park, 2011; Lerch & Gotsch, 2015). Thus, PSS designers need to be able to distil and extract useful knowledge from such vast amounts of data to improve the design, delivery and environmental performance of the PSS-based solutions (Geum & Park, 2011; Meier et al., 2010). For these and other reasons, designers are expected to need a wide range of new competencies, capabilities and support to design environmentally benign PSS solutions (Isaksson et al., 2009; Vasantha et al., 2015).

1.3 Missing pieces of the puzzle of product-service system design research for environmental sustainability

The main objectives of design research are (Blessing & Chakrabarti, 2009; Gero, 1990) 1) to improve the empirically based understanding of the design activity, 2) to effectively support the activity based on the developed empirical understanding and, 3) to empirically test and improve the efficacy of the developed support. The activity of designing is multi-dimensional and can be investigated in various scales such as the macro, meso- and micro levels of the activity (Cash et al., 2015; Wynn & Clarkson, 2018).

Micro-scaled insights into the design activity can refer to the design-related actions carried out by the individual designers (ibid) within specific phases (e.g., conceptual design). Several prescriptive approaches, design methods and tools have also been developed to support PSS designing with a focus on environmental sustainability (referred to as PSS designing from here on) (Brambila-Macias et al., 2018; Vasantha et al., 2012, 2015). These support prescriptions are mainly adaptations of traditional engineering design support (Baines et al., 2007; Cavalieri & Pezzotta, 2012; Cook et al., 2006) or are based on PSS design models defined a priori or even on macro- or meso-scaled empirical observations of the activity of designing such systemic solutions (for examples, see Manzini & Vezzoli (2003) and Matschewsky et al. (2018)). Such available support is not necessarily grounded upon an empirical understanding of the micro-scaled characteristics of the PSS design process. In order to effectively support any design activity, it is crucial to cater any prescriptive support being developed to the characteristics of the design activity (Daalhuizen & Cash, 2021; Visser, 2009).

Product and PSS designing are often assumed to be similar problem-solving activities on the micro-scale; see such an example in Vasantha et al. (2015). However, the situated notion of design (Gero & Kannengiesser, 2004) and more specifically, the augmented generic design hypothesis (Visser, 2009) assert that multiple factors such as the characteristics of the design object or design problem that characterize the design domain, education and the level of design expertise of the designers are expected to significantly influence the design activity on the micro-scale. Although extant literature reports significant differences in the characteristics of the product and PSS design objects and the design problems of the respective domains (see Section 1.2), there is limited empirical evidence to suggest if these differences will have any influence on the respective design activities in the micro-scale (Sakao et al., 2020). PSS designing needs to be compared with product designing in order to ascertain if it is different from the latter and to determine whether there is a need to develop dedicated support for the former (based on Kannengiesser & Gero (2022)).

In particular, there is a lack of empirical insights into how practitioners experienced in product development-related domains and with limited PSS design knowledge would address PSS design problems. More specifically, there is limited insight into how their lack of PSS design knowledge and their prior experiences grounded in typical product development would influence their design ability on the micro-scale, while addressing the characteristically distinct PSS design problems. As a consequence of this knowledge gap, it is currently unclear if such prior experiences and knowledge of the designers gathered over the years would help them or restrict their ability to effectively explore the design space while addressing the relatively unfamiliar PSS design problems (see Figure 1).

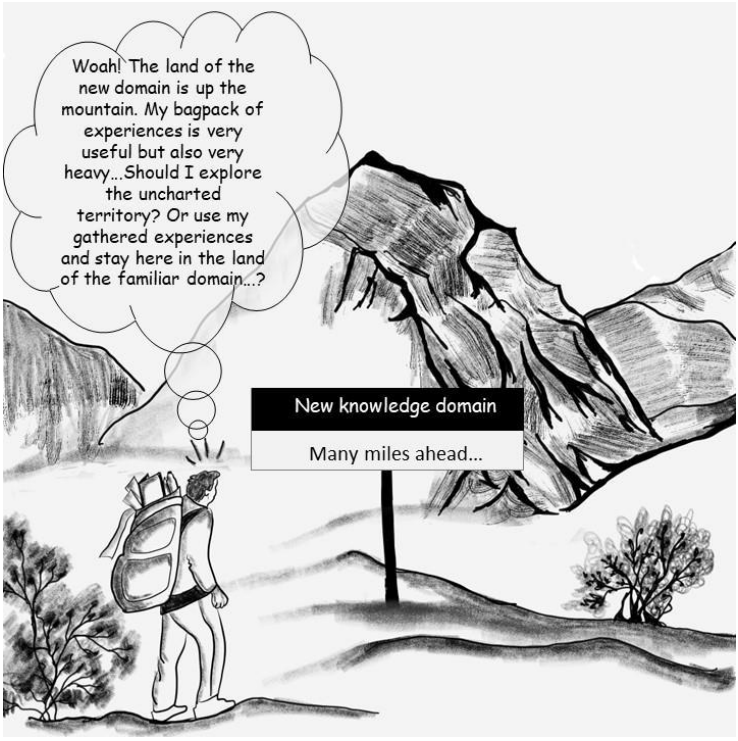


Figure 1. Pre-existing design knowledge: A value or a burden? This figure metaphorically represents the situation of a practitioner who is experienced in designing specifically to one domain and attempting to explore a relatively unfamiliar and characteristically distinct design domain. The bag of experiences represents the design knowledge the practitioner gathered during the previous experiences and how it may potentially influence the practitioner's journey.

The design activity on the micro-scale can be represented by the design-specific cognitive processes of human designers (Wynn & Clarkson, 2018). Empirical insights into design cognition are crucial to improve the overall design process, develop effective design support and pedagogy, and thus, enhance the outcomes of designing (Dinar et al., 2015; Gero & Milovanovic, 2020). There is a substantial set of research that has described the cognitive characteristics of product designing or designing of similar material-based artefacts (Dinar et al., 2015; Hay et al., 2017). In contrast, there are only a few studies that have carried out a micro-scaled investigation of PSS designing (for a few examples see Bertoni (2013), Sakao et al. (2020) and Shimomura et al. (2015)) and, virtually none of the PSS design process with a focus on environmental sustainability.

Moreover, the few works that have studied PSS designing on the micro-scale have mainly used novice designers (Vasantha et al., 2015). Due to a lack of an empirically based understanding of the cognitive characteristics of PSS designing, it is currently not possible to ascertain the efficacy of the available support and further develop effective support catered to its cognitive characteristics. Additionally, the efficacy of the majority of the available PSS design support prescriptions in the state of the art has been insufficiently evaluated in relevant environments within industrial contexts. The validation of such prescriptions has been typically performed through single case studies of the application and not through controlled trials (comparative studies of the design process and design outcomes, with and without support) (Boehm & Thomas, 2013; Tukker, 2015; Vasantha et al., 2012). Empirically informed knowledge of these issues is crucial to effectively support the practitioners within manufacturing companies to transition from designing products to designing PSSs with reduced environmental impacts.

1.4 Aim, Scope, and Research Questions

The overarching objective of this research is to effectively support practitioners within the product manufacturing industry to conceptually design PSSs with comparatively lower environmental impacts than typical product centric offerings.

To address this objective, this research first aims to descriptively develop an improved understanding of how experienced product developers carry out conceptual PSS designing on the micro-scaled cognitive level. By building upon an improved understanding of the cognitive characteristics of the design process, the research also aims to prescriptively develop support, and descriptively investigate the effects of prescriptive support for conceptual PSS designing in the context of environmental sustainability.

Cognitive characteristics refer to the attributes of the mental processes underlying the activity of designing on the micro-scale (based on Hay et al. (2017)). Prescriptive support in this context refers to any kind of design method, procedure, tool, means or aid that can improve the design process (Blessing & Chakrabarti, 2009). The conceptual stage of PSS designing is under the lens of this research, as the conceptual design is claimed to play a critical role in shaping the final outcomes of design (based on French (1985)). Furthermore, several authors have also suggested that the cognitive processes associated with conceptual designing are relatively less understood compared to the other stages of design (Hay et al., 2017), especially in the domain of PSS design which requires systemic thinking and simultaneous product and service design (Sakao et al., 2020).

Experienced practitioners from the product manufacturing industry are the chosen cohort of this research. Experienced practitioners were chosen due to their high level of design experience, and studying their design activities can potentially provide direct insights into design expertise (Cross, 2004) in industrial contexts. Such insights into design expertise may not be obtained through the study of other types of cohorts, such as students. Furthermore, the potential of design practice to address societal challenges is widely acknowledged (Zielhuis et al., 2022). Design practitioners are expected to drive the industrial transition from designing standalone products to PSSs, and to have the highest potential to influence the environmental impacts of the solutions to be developed. Thus, practitioners in the product manufacturing industry are the primarily targeted end users of the prescriptive support to be developed in this research.

The research aim is operationalized with the following research questions (RQs):

RQ1: *How do practitioners experienced in domains associated with product development conceptually design product-service systems on the cognitive level?*

The answer to this RQ is expected to provide insights into if and how product designing and PSS designing are different from each other, and subsequently to provide guidance to develop effective prescriptive support catered to the cognitive characteristics of PSS designing.

RQ2: *How could conceptual designing of product-service systems be supported based on an understanding of their cognitive characteristics?*

Grounded on the empirical understanding of the cognitive processes of PSS designing in relation to typical product designing, to be established by answering RQ1, RQ2 aims to inform how to effectively support the different dimensions of PSS designing with prescriptive approaches, methods, and tools.

RQ3: *How could domain-specific prescriptive support influence cognitive characteristics of conceptual product-service system designing and the environmental impacts of the resulting design outcomes?*

The answer to this RQ is expected to reveal the effects of the developed prescriptive support in terms of the changes in the PSS design-specific cognitive processes and potential environmental impacts of the design outcomes.

1.5 Thesis disposition

The remainder of the thesis is laid out as follows: Chapter 2 presents the theoretical background, which describes the various concepts, models, and theories used in this research. Chapter 3 then presents the author's own generic conceptual model of design cognition, termed cognitive architecture and mechanisms of designing (CAM-D). Proposition 1 and associated Hypotheses 1-4 of CAM-D are inductively formulated in Chapter 3. Next, Chapter 4 presents the research design. Chapters 5, 6, and 7 follow with a presentation of the results and related discussions of the research, mainly derived from the appended papers, and answer RQs 1, 2, and 3, respectively. Further, Proposition 1 and associated Hypotheses 1-4 developed in Chapter 3 are deductively tested in Chapter 5. The findings of Chapters 5 and 6 serve as inputs to Chapter 7.

In Chapter 8, based on the results of Chapters 5 and 7, Proposition 2 (Hypotheses 5-7) and Proposition 3 (Hypotheses 8-12) are abductively and inductively formulated, respectively. Thus, the conceptual model of CAM-D is refined and extended in Chapter 8. Chapter 9 presents the assessment of research quality and limitations and discusses the scientific contributions and practical implications of the results. Finally, Chapter 10 summarizes the answers to the RQs and presents the concluding remarks and future research directions.

The thesis disposition is visualized in Figure 2.

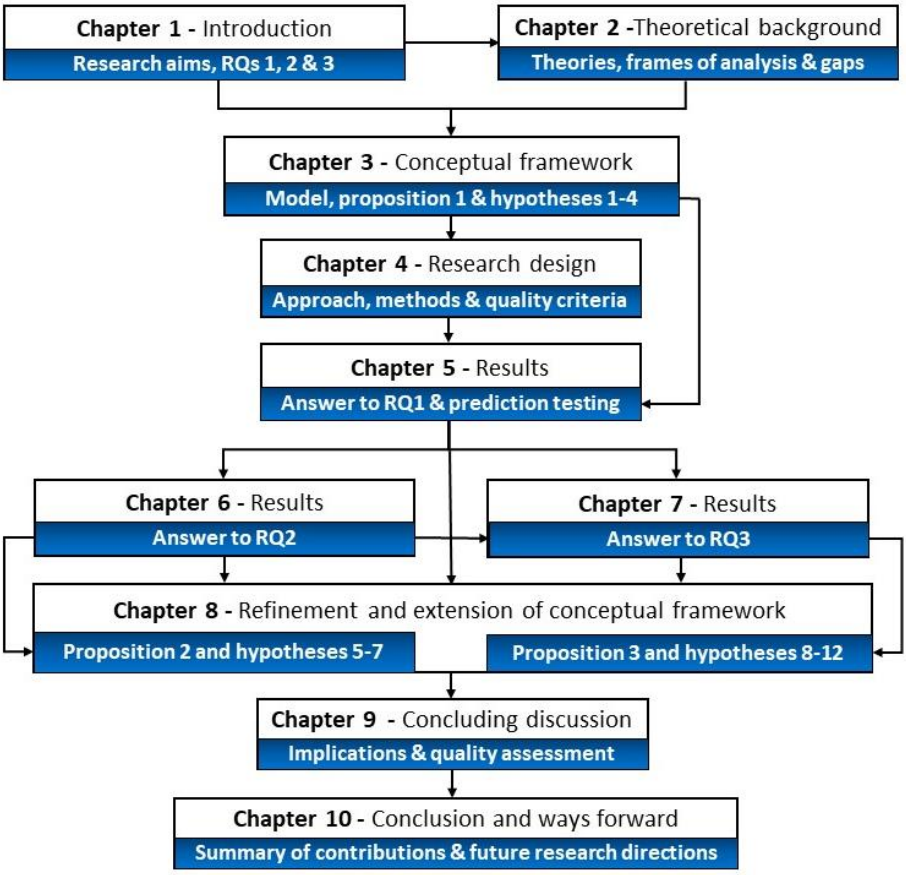


Figure 2. Visualization of thesis disposition. Note: white boxes represent chapters of the thesis, blue boxes represent outcomes of respective chapters and black arrows represent the flow of information.

Chapter 2 – Theoretical background

2.1 Design: Concept, History, and Impact

Designing can be defined as an activity of devising courses of action that aims to change existing situations into preferred ones (Simon, 1969). The meta-goal of this activity is to transform exogenous requirements in terms of functions that embody the expectations of the teleology of the artefact or activity to be developed, into a set of design descriptions (Gero, 1990). Gero (1990) further characterized this activity as constrained goal-oriented decision-making and explorative learning activity that happens within a context and is simultaneously influenced by the designer's perception of the context. Based on these definitions, a designer can be considered a change agent who devises courses of action to change existing situations into preferred ones by transforming the external environment (based on Gero (1990) and Simon (1969)).

Design is considered all-encompassing, as almost everything around us in our artificial environments such as buildings, roads, and manufactured products such as vehicles, clothes, furniture, machines, software, and communication systems have been designed (Cross, 2006). Some of the earliest accounts of this activity can be traced back thousands of years in ancient India (Lipman et al., 2022; Nathan, 2015), the Roman empire, Babylonia (Gero, 1990) and in other ancient civilizations. During the early 20th century and before, design was mainly considered to be an intuitive and mysterious activity performed mainly by artisans and there were limited insights into the principles that guided it (Buchanan, 2001).

Formal efforts to scientise design as a process can be traced back to the mid-20th century, during which the need to develop rational and systematic approaches to create artefacts started emerging (Cross, 2006, p. 95). This emergence is primarily attributed to the onset of global events such as World War II and the Cold War (Bayazit, 2004). The formal launch of design methodology as a field of scientific enquiry can be linked to a conference on design methods held by Jones and Thornley (1963) at the Imperial College, London in 1962 (Cross, 2006; Lloyd, 2017). Soon after, several seminal contributions to design research emerged such as the book chapter titled *Design Science* (Gregory, 1966) and the *Sciences of the Artificial* (Simon, 1969) (based on Cross (2006).

Recognizing the harmful effects of unsustainable manufacturing practices, Victor Papanek argued that design approaches must be compatible with the general human socio-economic order in which it is to operate (Papanek, 1985). However, the current state of affairs does not follow this logic. Scientific evidence suggests that the environmental impacts of socio-economic activities such as manufacturing are increasingly threatening the resilience of the Earth system that is conducive to human development (based on Steffen & Rockström

(2015)). More than 80% of such environmental impacts originating from manufactured products may be linked to the decisions taken during the early stages of design (British Standards Institution, 2017; Ramani et al., 2010), as illustrated in Figure 3. This points towards the need for designing for environmental sustainability within the manufacturing industry.

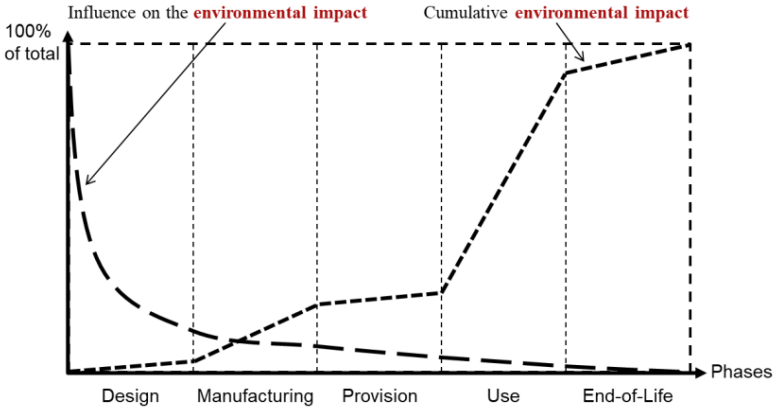


Figure 3. Illustration of influence on the environmental impacts of products by the different stages of the lifecycle. Adapted from ((British Standards Institution, 2017; Lindahl & Sundin, 2013)).

2.2 Evolution of designing for environmental sustainability

Fuller (1969) is considered to be one of the first to have highlighted the importance of and also the role of design in mitigating the harmful environmental impacts of unsustainable manufacturing practices (Ceschin & Gaziulusoy, 2016). One of the earliest examples of systematic approaches to address the environmental impacts of products is the practice of green design (Burall, 1991). The aim of green design is to reduce environmental impacts through targeted individual changes to the design of individual products (Ceschin & Gaziulusoy, 2016). Green design eventually gave way to the practice of eco-design.

The international standards organization (ISO) defines *eco-design* as a *"*systematic approach, which considers environmental aspects in design and development with the aim to reduce adverse environmental impacts throughout the lifecycle of a product[1]*"* (ISO:14006, 2020). The key difference between green design and eco-design is in the application of a lifecycle perspective.

Until the late 1980s, lifecycle aspects were a peripheral issue of research and design practice, as aspects such as ease of manufacturing, disposability and

[1] A product can refer to any good or service (ISO:14006, 2020).

serviceability were considered after important design decisions were made, which had led to sub-optimized designs from the lifecycle perspective (Finger & Dixon, 1989b). Application of a lifecycle perspective during eco-design entails the consideration of environmental impacts throughout the lifecycle of products. *Lifecycle stages* refer to the consecutive and interlinked activities a product goes through, starting from material acquisition through design and development, manufacturing, delivery and installation, use (including activities such as reuse, maintenance, repair, remanufacturing, refurbishing, and upgrading), end-of-life treatment and finally, disposal (ISO:14006, 2020).

There are other similar terms for the same approach, such as sustainable life cycle design (Alting & Legarth, 1995) or design for the environment (Fiksel, 1996). Furthermore, several methods and tools have been developed to support these approaches (for examples, see reviews by Ramani et al. (2010) and Rossi et al. (2016)). Most of these support prescriptions tend to be checklist, lifecycle assessment (LCA), or quality function deployment (QFD)-based tools (Ramani et al., 2010).

LCA is a systems analysis method that can be used to perform quantitative assessments of the environmental impacts of products or other systems, throughout the lifecycle of the system in question (Baumann & Tillman, 2004; Finnveden et al., 2009; ISO, 2006). Although LCA has been widely used to support the eco-design of products (for an example, see Muñoz et al. (2009)), it is not design-oriented and there are several issues that impede its effective use in eco-design (Ramani et al., 2010). For example, the process of translating LCA results into actionable information for design tends to be case specific, non-systematic, and non-generic.

QFD is widely used in industry (Booker, 2012) to translate design requirements into prioritized characteristics and components of either new or existing design artefacts for design or redesign, respectively (Fargnoli & Sakao, 2017). It was originally developed by Akao (1990), but since then, there have been multiple adaptations of the method (Fargnoli & Sakao, 2017; Vinayak & Kodali, 2013). It has also been extensively used to support eco-design (Masui et al., 2003; Ramani et al., 2010; Zhang et al., 1998). One major drawback of QFD-based eco-design tools is that the process of co-relation of environmental requirements and product characteristics is generally reliant on the expert judgements of the designers and not on quantitative information on the environmental impacts of the product throughout its lifecycle (Ramani et al., 2010). In order to address this issue, LCA-integrated QFD-based eco-design tools have also been developed (see Sakao (2007), for an example). However, the process of translating the environmental impact assessment derived from the LCA into design requirements remains a grey-box.

More recently, the approach of circular design has been gaining prominence in both industry and academia (Bocken et al., 2016; EEA, 2017; M.

Moreno et al., 2016), which is a key pillar of the circular economy (CE) concept (EMF, 2013; Geissdoerfer et al., 2017). Compared to eco-design and other similar design initiatives, circular design is mainly resource-centric. The focus of circular design is on maintaining the useful value and utility of products or materials at the highest possible levels at all times (Webster, 2015) using different design strategies such as enhanced product durability, material efficiency and the different "R" strategies such as re-using, repairing, remanufacturing and recycling (Bakker et al., 2014; EEA, 2017; Lindahl, 2018).

Although such product-level design strategies are vital to reduce potential environmental impacts, these tend to be symptomatic solutions as long as the priority of the manufacturing industry is fixated on the generation of revenue through product sales (Roy, 2000). In order to enhance the environmental sustainability potential of design approaches, it is necessary to extend the level of systems perspective to be applied while also considering and involving a wider range of societal actors in the design process (Ceschin & Gaziulusoy, 2016; Wasserbaur & Sakao, 2020). Moreover, it is necessary to decouple the economic performance of the manufacturing industry from resource use through changes in the design of business models (Roy, 2000; Tukker, 2004).

In order to effectively address environmental sustainability and resource related challenges faced by our societies, it is vital to dematerialise the economy by significantly restructuring manufacturing and consumption patterns (Mont, 2002). One way for manufacturers to achieve this could be through a transition from a typical product-based business model to a PSS-based one, as it is considered to have a high potential to open up a wide array of opportunities to reduce resource consumption and enhance the environmental sustainability potential of manufacturing activities (Annarelli et al., 2016; Tukker, 2004, 2015).

2.3 Product-service system

2.3.1 Business models

Value proposition is a key construct of the PSS concept (Morelli, 2003; Tukker & Tischner, 2006) and depending on the product/service content ratio from which the value is to be generated, PSSs can be classified under three main types of business models (Tukker, 2004): 1) product-oriented, where value is mainly sourced from the tangible product content with some services added on; 2) use-oriented, where the value is mainly derived from the product content, however, the ownership of the products stays with the providers and customers pay for the use of the products; and 3) result-oriented, where the emphasis is on selling results to the customers that address their needs and provide value with no fixed product in focus. The PSSs with relatively higher service content are hypothesized and also shown in some cases to have higher environmental sustainability

potential (Tukker, 2004, 2015). These three main types of PSSs and their respective sub-categories (Tukker, 2004) are visualized in Figure 4.

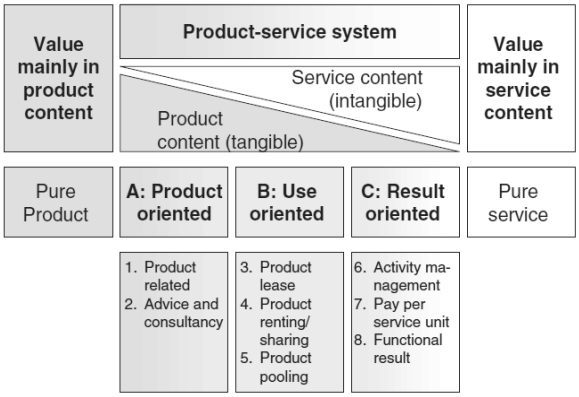


Figure 4. Main categories and sub-categories of the PSS (adopted from Tukker (2004)). © Reused with permission from the publisher "John Wiley and Sons".

2.3.2 Properties, design knowledge, and the design object

PSSs have similar properties as that to an open STS, including complex interactions of technological artefacts, human actors and the environment (Annarelli et al., 2016; Morelli, 2002; Roy, 2000). More specifically, a PSS is expected to have the following open systems properties: an internal environment with separate but interdependent technical and social sub-systems or elements, the ability of the overall system to adapt collectively and to address goals in external environments, the property of equifinality that allows it to address system goals in more than one way, and reliance of the system performance on the joint optimization of the social and technical sub-systems or elements (Badham et al., 2000; Baxter & Sommerville, 2011).

The core PSS design object includes tangible technical product and intangible social service elements that are to be designed, integrated and optimized as a system to jointly address customer needs and provide value (Maussang et al., 2009; Meier et al., 2010; Sakao et al., 2020). The overall system goal of the PSS is to design solutions that can deliver the demanded functional performance (Alonso-Rasgado et al., 2004; Meier et al., 2010). In addition, the PSS has the open system property of *equifinality* as the overall system goal can be achieved through multiple means that may include any suitable combinations of (as discussed by Maussang et al. (2009)) and through the exchangeability of the product and service elements (Sakao & Lindahl, 2015). *Exchangeability* refers to the ability to transfer the design efforts between the product and service elements in

order to address the overall system goals (ibid). Product and service elements are characteristically different (Maussang et al., 2009; Morelli, 2002; Sakao et al., 2020), and these differences are listed in Table 1.

Table 1. Comparison of the characteristics of product and service elements.

Characteristics of product elements	**Characteristics of service elements**
Tangible and physical elements	Intangible human activities
Separated production and use	Simultaneous production and use
Design frozen in the use phase	Can be designed in the use phase
Temporal behaviour governed by physical laws (i.e., deterioration)	Temporal behaviour is governed by the heterogeneity of the human behaviour
Transferable ownership	Non-transferable ownership
Higher degree of technological orientation	Higher degree of social orientation

While designing a PSS, a lifecycle perspective is prescribed to be applied, as the responsibility of the provider is extended throughout the lifecycle of the constituent product elements (Aurich et al., 2006; Brissaud et al., 2022; Cavalieri & Pezzotta, 2012). With the lifecycle perspective, potential interdependencies of the constituent product and service elements need to be anticipated, and subsequently leveraged by integrating them from a systems perspective during the early stages of design (Meier et al., 2010; Sakao, Neramballi, & Matschewsky, 2022). Furthermore, the potential behaviour of the system elements throughout the lifecycle and use scenarios need to be simulated to obtain actionable design insights (Alonso-Rasgado et al., 2004; Maussang et al., 2009).

In addition, the *delivery channels* that can be used to control, transfer and upscale the system elements should be simultaneously taken into account (Sakao & Shimomura, 2007). The delivery channel may include the needed infrastructure and network of companies (Mont, 2002). Along with the consideration of such core constructs, the designers also have to simultaneously consider *the business model* for the solutions (Meier et al., 2010; Meier & Massberg, 2004; Tukker, 2004). PSSs have a relatively higher degree of *social construction* than typical products as a wide range of value chain actors, such as networks of customers, companies, and other collaborating actors, need to be actively involved throughout the lifecycle to ensure the desired quality of PSS solutions (Annarelli et al., 2016; Morelli, 2003).

2.3.3 Origin, drivers, barriers, and potential for environmental sustainability

Although the PSS concept has one of its roots in marketing research (Vandermerwe & Rada, 1988), one major motivation for researching this concept in academia is its environmental sustainability potential (Goedkoop *et al.*,

1999; Annarelli, Battistella and Nonino, 2016). Several scholars have also argued that a transition to PSS-based business models can potentially offer several business incentives such as higher and more durable profit margins, market differentiation, production and consumption efficiency, reduction in costs and increase in cost efficiency, and the ability to lock in customers (see the reviews by Annarelli et al. (2016), Baines et al. (2009) and Beuren et al. (2013)).

In recognition of the potential of the PSS, many manufacturers have adopted PSS-based business models (Isaksson et al., 2009; Meier et al., 2010). Rolls-Royce, a major player in the aerospace sector, is a well-known example of a manufacturer who has transitioned to a PSS-based business model (Rolls-Royce, 2012). In the past, technological advancements at Rolls-Royce had contributed to significant enhancement of their product durability and reliability, which in turn, had led to a substantially reduced demand for their spare parts (Smith, 2013). Since spare part supply was one of their most profitable revenue streams, they adopted PSS-based business models and offered systemic solutions in order to overcome the resulting significant losses in revenue (ibid).

Several challenges and barriers associated with PSS development in industry have also been documented in case studies (Durugbo, 2013; Martinez et al., 2010; Matschewsky et al., 2018) and in reviews of such studies (Annarelli et al., 2016; Beuren et al., 2013; Brissaud et al., 2022). Examples of documented challenges include organizational, economic, capability and market-related issues that hinder PSS development. Although the environmental sustainability potential of the PSS has been widely heralded by academia, until 2007, almost no empirical studies performed PSS lifecycle sustainability assessment (Vasantha et al., 2015).

Since then, there have been a few empirical investigations and confirmations of the positive lifecycle environmental sustainability potential of the PSS documented in case studies (Amaya et al., 2014; Lelah et al., 2011; Lindahl et al., 2014) and a review (Blüher et al., 2020). The need for a more rigorous environmental sustainability assessment of PSSs in industrial contexts, has been deemed necessary (Tukker, 2015; Vasantha et al., 2015). Furthermore, the process of carrying out the LCA of PSSs has been found to be riddled with methodological challenges and complications such as the identification of reference systems and definition of the functional unit of the PSS (Blüher et al., 2020; Kjaer et al., 2016). However, efforts to address such issues are underway (Kjaer et al., 2018).

Furthermore, several authors have argued, and previous empirical research has shown that a mere transition to PSS-based business models may not necessarily lead to achieving environmental sustainability and reducing resource use (Kjaer et al., 2019; Tukker, 2015). The design, development, and delivery of the PSS may have to be tailored specifically to realize its high potential for environmental sustainability and business competitiveness (Vezzoli

et al., 2014, 2015). This points towards the need for more research into PSS design with a focus on environmental sustainability. Design has been researched across a range of different disciplines, topics, and levels (Christensen & Ball, 2019; Cross, 1982). The focus of such past research has ranged from the characteristics of design goals or requirements, design objects or artefacts, design support methodologies, design support tools, designer characteristics, design processes, and design thinking, among many more (Horvath, 2004).

2.4 Sciences of designing the artificial

An important area of design research is the investigation of the human ability to design, both through the empirical investigation of designer behaviour and theoretical reflections on the nature of the ability of designers (Bayazit, 2004; Cross, 1999). This body of research can be termed *science of designing*, which refers to the research that aims to enhance the understanding of the activity of designing through systematic and reliable approaches to investigation (Cross, 2006). Blessing and Chakrabarti (2009) suggest that to develop a scientific understanding of designing, it is first necessary to develop and validate theories and models of the phenomenon. There are several theories of design with both mathematical and symbolic formalisms (Hubka & Eder, 1988; Suh, 1995; Yoshikawa, 1981); for more examples, see reviews by Chakrabarti & Blessing (2015), Finger & Dixon (1989a) and Tomiyama et al. (2009). The design activity has also been modelled on its various levels of scope and types of dimensions (Wynn & Clarkson, 2018). Such models of design can be descriptive, prescriptive, or a combination of both (Chakrabarti & Blessing, 2015; Finger & Dixon, 1989a).

The framework used by Wynn and Clarkson (2018) in their review of existing design process models is partly adopted here to explore and structure the highly varied perspectives available in the design literature. The first level of categorization of design process models reviewed in this framework is in terms of the scope, that is, the depth of coverage of the models. The scope is further broken down into three levels: micro-level, meso-level, and macro-level models. The micro-level models focus on individual process steps and their immediate contexts, the meso-level models focus on end-to-end flows of progression of design tasks, and the macro-level models focus on project structures and the overall design activities in context. The second level of categorization of the design process models is in terms of the type or purpose of the models.

This categorization is further broken down into procedural, analytical, and abstract models. Procedural models offer guidance to real-world practitioners with best practices, while analytical models provide context-specific insights and improvements by representing particular instances of design process models. Abstract models communicate theories and provide insights, which form the

basis for developing more pragmatic support for practitioners (Wynn & Clarkson, 2018).

2.5 Cognitive architecture and cognitive load theory

In order to understand the underlying cognitive mechanisms of designing, it is first necessary to understand the human cognitive architecture (Sweller et al., 1998). Sweller et al. propose that human cognition functions on two basic types of memory: 1) working memory and 2) long-term memory. They equate working memory to human consciousness and suggest that it is a form of limited memory space used to process a limited amount of external information by organizing, contrasting, comparing or through other forms of processing. They further suggest that humans can only be conscious of the information pulled into the working memory space and that the other cognitive functions tend to be out of view unless brought into that space. Compared to short-term memory, long-term memory is considered to have relatively unlimited capacity and stores information in the form of schematized knowledge or schemas (Sweller et al., 1998).

Schema theory suggests that *schema* is a cognitive construct that stores multiple abstracted discrete elements of information processed in the working memory space as a higher-level element of knowledge in the long-term memory (Chi et al., 1981; Schnotz & Kürschner, 2007; Sweller et al., 1998). The working memory can only process a limited number of discrete elements of information, but the size and complexity of the elements stored in a schema need not be limited. Even though there are limitations on the number of independent elements of information that can be processed separately in the working memory, vast amounts of such lower-level elements of information organized as a higher-level schema will be processed as a single entity of knowledge by the working memory. This type of processing requires relatively lower cognitive load than the processing of individual elements of information. *Cognitive load* is a construct that represents the working memory resources required to perform or learn certain tasks (Schnotz & Kürschner, 2007; Sweller et al., 1998; Sweller & Chandler, 1994).

Thus, two of the key functions of schemas are reducing the cognitive load in the working memory and storing knowledge in the long-term memory (Sweller, 1988; Sweller et al., 1998; Van Merriënboer & Paas, 1990). *Schema construction* occurs as the working memory actively processes and combines lower-level elements of the information as higher-level knowledge schemas. During schema construction, the lower-level elements of information are categorized in the long-term memory space in the specific configuration or patterns that they are processed with in the working memory space. *Schema automation* progresses when increasing number of lower level elements information are compiled in the constructed knowledge schemas, during repeated engagement

in a particular problem-solving activity. This leads to the automation of knowledge, and thus the development of expertise (Sweller et al., 1998; Van Merriënboer & Paas, 1990). Automated schemas help reduce cognitive load on the limited working memory capacity and thus, ensure that enough cognitive capacity is available to address complex problem-solving challenges (Sweller et al., 1998).

It is further suggested that the cognitive load of novices can be greatly reduced by introducing procedural frameworks (Paas et al., 2004). According to cognitive load theory (Paas et al., 2004; Sweller, 1988; Sweller et al., 1998), *intrinsic cognitive load* is expended on the processing of novel elements of information in the working memory space, while the *germane cognitive load* is expended on schema construction and automation. *Extrinsic cognitive load* is expended on processing out irrelevant or unneeded exogenous information (ibid). A closely linked cognitive science theory, that has seen recent emergence in design research is dual process theory (Kahneman, 2011; Kannengiesser & Gero, 2019), which suggests that there are two forms of thinking: 1) fast and intuitive thinking (which can be linked to automated information processing), and 2) slow and analytical thinking (which can be linked to conscious information processing).

2.6 Domain-independent micro-scaled abstract theories and models of designing

2.6.1 Overview

Wynn and Clarkson (2018) suggest that abstract theories and models present descriptive insights into the design process. These theories and models tend to be domain independent and thus are considered to be useful for generating generalisable insights into designing (Wynn & Clarkson, 2018). The empirical investigation of the nature of designing on the micro-scaled abstract level, as an internal cognitive process of the designer has been at the core of design research for the past several decades and significant advances have been made in this area (Dinar et al., 2015; Gero & Milovanovic, 2020; Hay et al., 2017). Some of the key theoretical insights derived from this vast body of literature are briefly discussed below (partly based on the reviews by (Ball & Christensen, 2019; Hay et al., 2017; Wynn & Clarkson, 2018)).

Designing on the abstract micro-scaled level is often classified as a form of problem-solving activity (Simon, 1969). However, most design processes are considered to be distinct from other typical problem-solving activities (Goel and Pirolli, 1992; Visser, 2009), as design problems tend to include unknown variables (Dorst, 2011) which are to be discovered by the designer over time (Gero and Milovanovic, 2020). Gero (2000) suggested that routine problem solving,

or routine designing (e.g., design optimization), involves the search of parameters for known variables, in line with arguments made by Dorst (2011). In contrast, most design problems are considered to be ill-structured or non-routine, which tend to have ambiguous goals and no pre-set solution principles (Gero, 2000; Simon, 1973). Designing involves the manipulation of the constraints or even the discovery of new variables (Gero, 2000). Based on a systematic review of design literature, Hay *et al.* (2017) categorized designing into four main design activities: 1) problem structuring/analysis, 2) generation and synthesis, 3) evaluation, and 4) decision making.

Problem structuring/analysis may entail problem formulation, which refers to the process of reflecting on what the problem is through the introduction, definition, and redefinition of the goals and constraints of the design task, based both on the exogenous information made available to the designers and the domain knowledge of the designers (Ball & Christensen, 2019). After the problem formulation, designers typically move on to framing. Akin and Akin (1996) suggest that a typical frame consists of aspects of design (or the design task/object), goals to achieve within each aspect, methods, procedures for achieving each goal, and a robust representation mechanism. They further suggest that such frames can be externally imposed, for example, via the externally given design task. The frames can also be constructed by the designers based on the given information or their own knowledge (Akin and Akin, 1996).

Looking more closely into the cognitive processes underlying design activities, two main viewpoints that reflect two paradigmatically different views that have evolved over the past two decades emerge (Dorst & Dijkhuis, 1995; Hay et al., 2017): as a process of search (based on the problem-solving paradigm (Newell & Simon, 1972)), and as a process of exploration (based on the reflective paradigm (Schön, 1983)). It is suggested that studies with the view of "design as search" tend to describe designing as a linear and sequential set of elementary processes (for example, processes involving retrieval and manipulation of operators, inferring, analogical reasoning, problem structuring, and decomposition) that transform information from input to output states, thus changing the states of design knowledge (Dorst & Dijkhuis, 1995; Hay et al., 2017); for examples see Liikkanen & Perttula (2009), Lloyd & Scott (1994) and Stauffer & Ullman (1991).

Whereas from the viewpoint of "design as exploration", designing is considered an iterative and situated process of the co-evolution of problem and solution spaces (Dorst & Dijkhuis, 1995; Hay et al., 2017); for examples, see Dorst & Cross (2001) and Maher & Tang (2003). Furthermore, designing is considered a situated activity, as the interactions between the designer and the environment are said to recursively influence the design process (Gero & Kannengiesser, 2004). Gero and Kannengiesser (2004) introduced the notion of situatedness in the context of design by suggesting that while designing,

designers interpret and observe the outcomes of their actions, which in turn influence the direction of their new actions on the environment.

Gero and Kannengiesser (2004) further elaborate on cognitive mechanisms underlying this phenomenon by suggesting that designing relies on constructive memory. More specifically, they suggest that the memory being recalled is newly constructed at the time of the original experience, upon memories of previous experiences, while the entire process is influenced by the situation (including the actions of the designer) surrounding the time of the construction of the memory (Gero & Kannengiesser, 2004). Thus, the memory upon construction is added to the designer's experience, becomes part of the situation, and influences the construction of further memories. In line with the situated notion of designing, the augmented generic design hypothesis suggests that the characteristics of the design object, design process and the designer can significantly influence design cognition (Visser, 2009). Furthermore, it is also widely suggested that designing is an iterative and progressive process that enables the development of knowledge, concurrency and the introduction of relevant changes (Wynn & Eckert, 2017).

2.6.2 A generic model of knowledge transformation during designing

There are several theories of design (Hatchuel & Weil, 2009; Smithers, 1998; Yoshikawa, 1981). Building on the works of Newell (1980) and Simon (1982), Sim and Duffy (2003) present a knowledge-level ontological representation of designing. A simplified version of this model is described as follows: In this model, the input and output to the design activity (A_d) at the knowledge level can be represented by the symbolic structures: input knowledge (I_k) and output knowledge (O_k). They suggest that input knowledge may be influenced by the situated designer's perception of the external design context and the tacit and explicit domain knowledge of the designer. They broadly categorize the tacit domain knowledge (also termed experiential knowledge) of the designer into declarative (knowledge of the design object), procedural (knowledge of the design process) and causal knowledge (the design rationale).

In contrast, explicit domain knowledge of the designers refers to codified or formalized transmittable systematic knowledge. Based on the principle of bounded rationality (Simon, 1982), Sim and Duffy (2003) suggest that the design goals (G_d) trigger the retrieval of relevant tacit and explicit domain knowledge. The design goals can either be specified or derived. Specified design goals are inferred from externally defined design requirements or tasks, while derived design goals are invoked by the designers during the design process. The output of the design activity may include symbolic representations of the design solutions that fulfil the design goals and the knowledge gained by the

designer through the entire process of designing. Thus, the characteristics of the gained design knowledge are shaped by both the design activity and the input knowledge (Sim & Duffy, 2003). The element of design activity can be further examined in terms of the multi-dimensional transformation of the design states and problem-solving strategies undertaken by the designers. The relationships between the different elements of this model are elucidated in Figure 5 (Sim & Duffy, 2003).

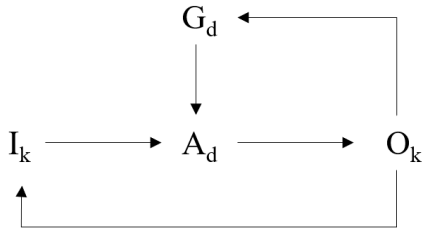


Figure 5. Knowledge-level formalism of the design activity (Sim & Duffy, 2003). © Reused (re-drawn) with permission from the publisher "Springer Nature".

The phenomenon of knowledge reuse and effects of previous design knowledge on design cognition, such as design fixation, has been widely investigated in previous works (Ball & Christensen, 2019; Visser, 2009). While domain-specific design knowledge is a crucial construct of design expertise, it can also have undesirable effects in terms of design fixation (Ball & Christensen, 2019; Crilly & Cardoso, 2017; Visser, 2009). Design fixation is defined to as 'a state in which someone engaged in a design task undertakes a restricted exploration of the design space due to an unconscious bias resulting from prior experiences, knowledge or assumptions' (Crilly & Cardoso, 2017). Since one of the early works by Jansson & Smith (1991), this phenomenon has been extensively investigated; as reviewed by (Crilly & Cardoso, 2017; Purcell & Gero, 1998).

2.6.3 A generic model of abstract elementary processes of designing

Certain models represent design activity as the abstract elementary transformation of states or issues of design that link the design goals and aspects of the design artefact (Sim & Duffy, 2003; Wynn & Clarkson, 2018). It is suggested that there is a high degree of ontological fragmentation across such models of design (Hay et al., 2017). For example, several such micro-scaled descriptive models of the design activity are based on different perspectives (Chakrabarti & Blessing, 2015; Wynn & Clarkson, 2018). Since the use of such different representation models has resulted in a high degree of ontological variability across the different design cognition studies, there is a widespread agreement concerning the need for the use of a common ontological framework to enable

comparison and meta-analysis and thus, to foster the generalizability of the findings (Cash, 2018; Dinar et al., 2015; Hay et al., 2020).

One such widely used domain-independent framework is the Function-Behaviour-Structure (FBS) ontological model of the design activity (Gero, 1990). This model has been extensively used to conceptually, computationally and empirically study designing across a wide range of different domains (Gero & Milovanovic, 2020; Wynn & Clarkson, 2018). It has also been shown to be applicable for the analysis of PSS designing, and its potential to compare PSS designing with the product and other domains of designing has been empirically validated (Sakao et al., 2020). The FBS ontology has its roots in cognitive science, as it is partly based on prototype theory (Osherson & Smith, 1981).

Based on prototype theory, this model proposes a knowledge representation schema termed *design prototypes* that bring together individual design experiences or design knowledge of designers under generalized heterogeneous classes of variables, which represent the characteristics of design artefacts or objects (Gero, 1990). It includes three main classes of variables, that describe the different issues of the design object and which are described as follows (Gero, 1990): 1) Function (**F**): Describes the teleology of an aspect of the design object. i.e., what it is for; 2) Structure (**S**): Describes the components of the design object, i.e., what it is; and 3) Behaviour (**B**): Describes the attributes that are derived (**Bs**) from the structure or are expected to be derived (**Be**) from the function of the design object. Design Description (**D**) represents documentation or described outcomes of the design activity. It also includes the variable Requirements (**R**) that is directly linked to exogenous design goals. This framework is visualized in Figure 6 below.

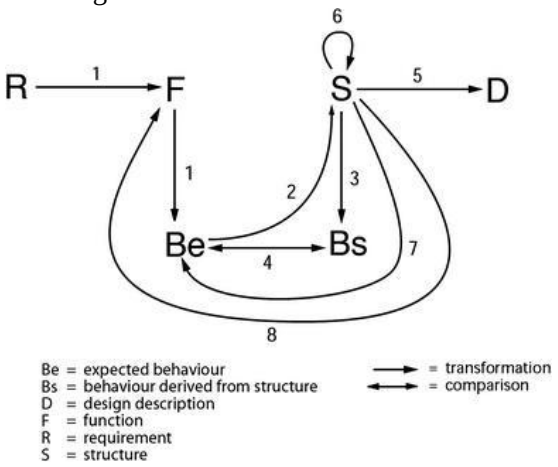


Figure 6. Design issues and processes in the FBS framework (Gero, 1990), figure adopted from Kannengiesser & Gero (2015). © Reused with permission from the publisher "Springer Nature".

This framework represents the design activity as a series of eight transformation processes linking the start, continuation and end of designing through the different design issues (Gero, 1990). The eight processes are claimed to be fundamental in designing and are described as follows (ibid):

1) **Formulation:** Transforms the design requirements in terms of function (**F**) into behaviour that is expected (**Be**) to facilitate this function.
2) **Synthesis:** Transforms the expected behaviour (**Be**) into a solution structure (**S**) which displays the expected behaviour (**Be**).
3) **Analysis:** Generates the behaviour (**Bs**) of the solution structure (**S**).
4) **Evaluation:** Assessment of generated solution by comparing behaviour of structure (**Bs**) and expected behaviour (**Be**).
5) **Documentation:** Documentation of design description (**D**) for further stages of development.
6) **Reformulation 1:** If the assessed behaviour of generated structure solutions (**Bs**) is unsatisfactory, generated structure variables are subjected to changes in the design state space.
7) **Reformulation 2:** If the assessed behaviour (**Bs**) of the generated structure solutions is unsatisfactory, expected behaviour (**Be**) variables are subjected to changes in the design state space.
8) **Reformulation 3:** If the assessed behaviour of generated structure solutions (**Bs**) is unsatisfactory, derived function variables are subjected to changes in the design state space.

Gero and Kannengiesser (2004) extended this model to represent the situated nature of the design activity by mapping out twenty additional transformation design processes spread across three world-views: the external world, the external world as interpreted by the designer, and the expected world-view of the designer. In order to investigate the designing styles of designers, FBS design issues and processes have also been mapped onto problem and solution spaces, which facilitates the meta-level semantic analysis of the design activity (Gero et al., 2013). It is suggested that during problem formulation, designers mainly reason with Requirements, Function, and Expected Behaviour, while they reason with Structure and Behaviour of Structure as they develop solutions in terms of the design artefact (Gero, 1990; Gero & Kannengiesser, 2004; Jiang et al., 2014).

In terms of FBS design processes, Formulation, Reformulation 2 and 3 are associated with the problem space, while Synthesis, Analysis, Evaluation, and Reformulation 1 are linked to the solution space (Gero et al., 2013). Such a meta-level semantic analysis of the design activity on the cognitive level can be operationalized by determining the relative distribution of cognitive effort on the problem (P) and solution (S) spaces using a measurement technique referred to as the P-S index (Gero et al., 2013).

2.6.4 A generic model of problem-solving strategies specific to designing

From a problem-solving point of view, the activity of designing is widely considered to entail navigation through different hierarchical structures of the design problem and solution spaces at different levels of abstraction (Chandrasekaran, 1990; Gero & Mc Neill, 1998; Wynn & Clarkson, 2018). This process of navigation through different hierarchical levels of systems abstraction is widely referred to as problem decomposition and composition or re-composition (Ho, 2001; Liikkanen & Perttula, 2009; Sim & Duffy, 2003). It is suggested that (Liikkanen & Perttula, 2009; Wynn & Clarkson, 2018) these problem-solving techniques may have their roots in cognitive psychology (Newell & Simon, 1972), with applications in domains of decision-making (Gettys et al., 1987) and instructional science (Sweller, 1988) and the development of normative theories of design processes across different domains of design (Pahl & Beitz, 1984; VDI Society for Product Development, 1987). These models have been previously used to study how designers with different levels of expertise address design problems, and the findings of such studies have indicated that there are significant differences in the problem-solving strategies of experienced and novice designers (Ho, 2001; Song et al., 2016)

Problem decomposition (also referred to as working forward strategy) is often referred to as the process of breaking down design problems (or tasks/solutions/objects) into a set of smaller independent entities or sub-problems (Ho, 2001; Liikkanen & Perttula, 2009; Sim & Duffy, 2003). *Problem re-composition* (also referred to as working backward strategy) refers to the bottom-up process of combining sub-solutions that can achieve the design requirements and or goals (Chandrasekaran, 1990; Sim & Duffy, 2003; Song et al., 2016). Although there are some conflicting reports, several studies of both problem-solving and design-specific problem-solving suggest that while experts primarily apply working forward strategies, novices mainly tend to apply working backward strategies (Ho, 2001; Song et al., 2016).

Based on previous works (Gero & Mc Neill, 1998; Song et al., 2016), Gero and Milovanovic (2020) categorize the levels of systems abstraction of the design problem into the 1) systems level: designers focus on the design problem as a whole, 2) interactions level: designers focus on the interactions between design sub-systems, and 3) sub-systems level: designers focus on the design sub-systems. Problem decomposition is referred to as a change of level from a higher to a lower level (e.g. systems to sub-systems), and vice-versa for problem re-composition (Gero & Milovanovic, 2020).

Previous studies that used similar frameworks to study the problem-solving strategies of experts and novices suggest that, the former tend to apply more of a breadth-first approach during problem decomposition (primarily

decomposing to sub-systems level), while the latter mainly tend to apply a depth-first problem decomposition strategy (primarily decomposing to element level) (Ball et al., 1997; Ho, 2001; Song et al., 2016).

2.7 Product-service system eco-design domain-specific abstract micro-scaled models

2.7.1 Descriptive models to represent product-service system designing

There are a few descriptive abstract micro-scaled models of PSS design available in the literature. These have been developed mainly based on either a priori models of PSSs or on the analysis of design sessions. For example, Sakao, Paulsson and Mizuyama (2011) use a prescriptive model proposed by Müller *et al.* (2009) that describes the key facets of PSSs and use it to analyse PSS designing on the micro-scale, along with an adaptation of a model that has been used previously to study the problem-solving micro-strategies of product designing (Gero & Mc Neill, 1998; Mc Neill et al., 1998).

This model included various dimensions relevant to the development and delivery of the PSS such as customer value, deliverables, lifecycle activities, actors, core products, peripherals, contracts, finance, and business model aspects. The results of this study suggested that PSS designing follows a typical problem-solving pattern with a strong focus on solution analysis, and lifecycle activities was the most discussed PSS dimension, while the designers began the design session focusing on needs and ended with a focus on value (Sakao et al., 2011).

Another previous work by Shimomura, Nemoto and Kimita (2015), used a similar frame of analysis along with a more detailed view of the PSS dimensions, while trying to highlight the relationships between the aspects of PSS solutions being generated and the design process on the micro-scale. Based on the results, they hypothesized that aspects related to customer, value-proposition and product-service architecture need to be spirally designed (Shimomura et al., 2015).

2.7.2 Descriptive models to represent eco-designing

Vallet *et al.* (2013) presented an approach to model the extent of environmental consciousness of the designers during eco-designing on the micro-scale of the design process, to study the effects of eco-design tools. This model entailed measuring cognitive effort expended on different eco-designing strategies (through an adaptation of traditional design strategies), which included goal clarification, initial environmental assessments, applied environmental

strategies, solution generation and assessment, decision, and control actions. The initial environmental assessment category included sub-categories related to the product lifecycle.

A micro-scaled abstract model to represent the distribution of cognitive effort on different lifecycle activities of solutions during the eco-designing of any solution-providing system (products or PSS) was proposed and validated (Neramballi & Sakao, 2021). It includes three main lifecycle stages: 1) solution development, 2) solution deployment and 3) solution retirement. A fourth category referred to as multiple stages, was also included in this model, to represent eco-design-specific issues or processes that addressed more than one lifecycle stage. It also includes a framework to represent the interactions of the solution under focus with the environment in terms of energy input, material input, and emission output (ibid). This model is based on the product lifecycle model developed for LCA (ISO, 2006) and PSS lifecycle activities. This model is represented in Figure 7.

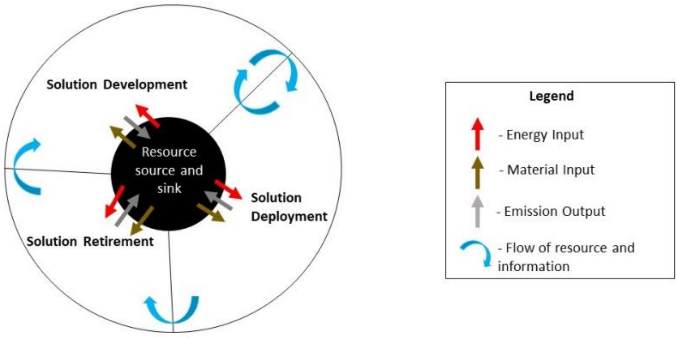


Figure 7. Representation of solution lifecycle stages and environmental interactions (Neramballi & Sakao, 2021). © Reused under CC-BY License.

Based on this framework, a coding scheme to carry out protocol analysis was developed and validated by Neramballi & Sakao (2021). The different lifecycle stages, the respective qualifying lifecycle activity and the respective code are presented in Table 2 below.

Table 2. Framework and coding scheme for solution lifecycle stages (Neramballi & Sakao, 2021).

Lifecycle stage	Qualifiers - Lifecycle activities	Code
Multiple stages	- Activities that are not limited to one lifecycle stage	1
Solution development	- Business model design - Product and service design - Material procurement and processing - Product fabrication/adaptation - Filling/packaging/distribution	2
Solution deployment	- Logistics - Instalment - Product use - Service execution (spare or consumable supply/repair/maintenance) - Product reuse	3
Solution retirement	- Remanufacture - Recycle - Logistics (takeback) - Waste management	4

2.8 Prescriptive micro-level procedural models and methods

Micro-level procedural models provide prescriptive guidelines and systematic support for specific design steps (Wynn & Clarkson, 2018); similar to design support methods (Gericke et al., 2016). Examples of such models include a wide range of methods, such as the morphological matrix that can be used to develop suitable combinations of solution/working principles (Pahl & Beitz, 2007), axiomatic design (Suh, 1995), QFD (Akao, 1990), theory of inventive problem-solving (TIPS or TRIZ) (Al′tshuller, 1999), and other Design by Analogy (DbA) methods (McAdams & Wood, 2002), among many more. Since its conception in Japan in the late 1960s, QFD has been widely used by several leading companies from different sectors around the world to ensure quality control and customer satisfaction of products and services (Akao & Mazur, 2003).

QFD is primarily used to translate prioritized requirements into functions and quality or performance characteristics of systems being developed (Akao, 1990). Akao and Mazur (2003) suggest that QFD offers a wide range of benefits such as effective and efficient support for the decision-making process during the early stages of product design, reduction in product development time and costs, and enhanced communication and collaboration within product development teams. There have been several adaptations of QFD (Vinayak & Kodali, 2013) and it has been employed to support a variety of Design for X strategies (Fargnoli & Sakao, 2017). One key application of QFD-based tools is eco-design (Section 2.2). It has also been previously adapted to support PSS designing with (Fargnoli et al., 2018; Sousa-Zomer & Miguel, 2017) and without

(Cavalieri & Pezzotta, 2012; Sakao, Birkhofer, et al., 2009) a focus on environmental sustainability

Analogy is a key cognitive science concept that refers to the cognitive process of developing associations across different situations by identifying potential relationships (Gentner, 1983). Design by Analogy (DbA) methods are widely used to support designers to help them identify, develop, associate and transfer variables from other appropriate cases, domains, and scenarios to derive inspiration to solve their own design problems (McAdams & Wood, 2002; Moreno et al., 2014; Qian & Gero, 1996). Designs can be considered analogous if they share at least one or more transferable functions and/ or behaviour (Qian & Gero, 1996). DbA has been used to support designers by supporting them to transfer analogous variables from a variety of other domains such as the natural world, biology and also across different engineering domains (McAdams & Wood, 2002; Sartori et al., 2010). TRIZ is a widely used analogy-based problem solving method that has been applied to support eco-design, mechanical and product design (Chang & Chen, 2004; McAdams & Wood, 2002; Sakao, 2007). This method relies on the discovery, abstraction, categorization and transference of patterns, conflicts, and solutions (best practices) from the working principles of a large data base of patents, into new design problems.

2.9 Prescriptive meso-level procedural support

Tomiyama *et al.* (2009) suggest that in practice, engineers, designers, and managers are mainly concerned with producing desirable design outcomes by following rigorous processes that ensure the final design results meet the objectives of product development. Wynn and Clarkson (2018) suggest that meso-level procedural models prescribe such systematic design processes to ensure the generation of good design outcomes. They further suggest that several prescriptive procedural models represent the design process as a series of stages. These models begin with stages that detail how to develop specific forms of design information beginning on the abstract level and ending with detailed information, while each stage sets the goals and constraints for the following stages (Wynn & Clarkson, 2018). Each or several of the stages within such models tend to have iterative feedback loops (ibid). This type of support is similar to design process support (Gericke et al., 2016).

One of the earliest prescriptive meso-level models of design was proposed by Asimow (1962), which has three main stages: feasibility study, preliminary design and detailed design, which include several design steps. More examples of such models include models for product engineering and development (Pugh, 1991; Ulrich & Eppinger, 2012) and design of technical systems (VDI Society for Product Development, 1987) for engineering design (French, 1999; Hubka & Eder, 1987; Pahl & Beitz, 2007) and mechanical design (Ullman, 1992), among many more (for more examples, see reviews by

Chakrabarti & Blessing (2015); Tomiyama et al. (2009); Wynn & Clarkson (2018)). Pahl and Beitz's (1984, 2007) systematic approach (PBSA) to engineering design, is one of the most widely used models of design in both industry and academia (Kannengiesser & Gero, 2017; Tomiyama et al., 2009). It is a stepwise prescriptive procedure for design modelled partly based on the experience of practitioners and empirical observations of the design process in practice (Kannengiesser & Gero, 2017).

Tomiyama *et al.* (2009) suggest that this model is grounded in the fundamentals of technical systems, systematic approaches, and general problem-solving approaches. The PBSA is said to have influenced several other models of design as well (Wallace & Blessing, 2000) and is widely prevalent in the design curriculums of numerous universities around the world (Kannengiesser & Gero, 2017). The different steps of this model are briefly elaborated on as follows. The overall model has four main stages (Pahl & Beitz, 2007), 1) task clarification, 2) conceptual design, 3) embodiment design and 4) detail design. This thesis focuses on the early stages of design and, more specifically, on the conceptual design phase. The overall aim of the conceptual design phase is to develop and detail basic working principles of design concepts or solutions, based on the preceding task clarification. PBSA prescribes designers carry out conceptual designing by following certain steps:

Step 1): Abstract and identify the essential problems

Step 2): Establish relevant function structures, by decomposing overall functions into subfunctions

Step 3): Develop working principles that can fulfil the subfunctions

Step 4): Combine suitable working principles into working structures

Step 5): Select suitable combinations of working structures

Step 6): Narrow down the solution variants

Step 7): Evaluate variants against technical and economic criteria

2.10 Product-service system design research

There is a significant amount of extant PSS design research, as reviewed by Brissaud et al. (2022), Meier et al. (2010) and Vasantha et al. (2015), ranging from the elaboration of the PSS concept (Mont, 2002; Tukker, 2004) to representation models for the PSS design object (McKay & Kundu, 2014; Sakao, Shimomura, et al., 2009) and the PSS design process (Alonso-Rasgado et al., 2004; Aurich et al., 2006; Morelli, 2003). Further, there are several prescriptive design methods and tools that aim to support PSS designing. For example, there are adaptations of widely used product design methods (Kimita et al., 2017; Sakao, Birkhofer, et al., 2009) and service design methods (Costa et al., 2018; Geum & Park, 2011) to support PSS designing. There are several other PSS

design methods and tools, as reviewed by Brambila-Macias et al. (2018), Qu et al. (2016) and Vasantha et al. (2012).

PSS design prescriptions have also been developed and applied in case studies, based on insights derived from meso- and macro-scaled (Wynn & Clarkson, 2018) descriptive empirical accounts of PSS designing (Doultsinou et al., 2009; Manzini & Vezzoli, 2003; Morelli, 2003). Only a few works have provided empirical insights into the micro-scaled cognitive nature of PSS designing. For example, Bertoni (2013) carried out an analysis of multiple conceptual PSS design sessions using a cohort of student designers (n=8) to provide some quantitative insights into the effects of computer-aided design (CAD) model-based informational support on conceptual PSS design thinking, in terms of time spent on different problem-solving strategies and different dimensions of PSSs. Shimomura et al. (2015) also carried out multiple protocol studies (n=3) of conceptual PSS designing, also by student designers, to provide insights into trends in the process of PSS design thinking measured in terms of time spent on different design elements of PSSs. Other exploratory studies of PSS designing on the cognitive level have also used PSS domain-specific frames of analysis and novice student designers as cohorts (Sakao, Paulsson and Mizuyama, 2011; Won Lee *et al.*, 2013).

2.11 Gaps and opportunities relevant to research objectives

Although there are few previous works (as reviewed above in Section 2.10) that provide some insight into the micro-scaled cognitive characteristics of conceptual PSS designing, such existing studies lack experimental rigour and/ or are based on small cohort sizes. Such drawbacks in the research design limit the reliability of the findings (see discussions concerning design of experiments by Dinar et al. (2015)). Furthermore, the cohorts of these past studies are primarily composed of student designers which might restrict the applicability of the knowledge in the context of industrial practice, as insights into the design cognition of expert design practitioners are crucial to improve the understanding of design expertise (Cross, 2004, 2006). Moreover, there are virtually no such works on PSS designing with a focus on environmental sustainability.

A key construct of designing for environmental sustainability is the lifecycle perspective. Research in lifecycle design has two main foci: 1) acquiring, organizing and utilizing knowledge regarding lifecycle-based issues that can be linked to the early stages of design; and 2) effectively organizing and controlling design processes to enable early and simultaneous consideration of lifecycle issues (Finger & Dixon, 1989b). An area of research that is underemphasized is how the two dimensions of eco-design mentioned above unfold on the cognitive level, which is crucial to effectively supporting design for environmental sustainability (Ramani et al., 2010; Vallet et al., 2013). Furthermore, it is suggested that PSS designers need additional support to apply PSS domain-

specific strategies, to choose and define sustainability criteria, application of a higher systems perspective and a wider lifecycle perspective while designing, and to develop environmentally sustainable design alternatives (Brissaud et al., 2022; Cavalieri & Pezzotta, 2012; Vasantha et al., 2015). Furthermore, the process of translating the results of LCA into actionable insights for PSS designing remains a grey-box (Section 2.2). Also, there is a lack of insights into how prescriptive support for PSS designing may interact with the cognitive processes of the designers.

To characterise the cognitive processes specific to conceptual PSS designing for environmental sustainability, it is effective to compare it with a different domain. Although there are previous empirical studies (Akin, 2001; Purcell & Gero, 1996) that provide support for the augmented generic design hypothesis (see Section 2.6.1), there is a lack of empirically informed theory concerning the variables and mechanisms underlying the variant nature of design cognition (Visser, 2009). Moreover, there is a lack of clear descriptions of the constructs, variables and relationships underlying conceptual design cognition in general (Cash, 2018; Hay et al., 2017). Thus, there is a crucial need and also the opportunity to describe the major constructs and variables, to postulate, hypothesize and test the relationships underlying design cognition (Hay et al., 2020). Such an opportunity can be operationalized by building on the rich exploratory design research available in the state of the art and on the established cognitive science theories (ibid).

2.12 Methods and theories relevant to research objectives

Different research methods have been used to descriptively study design cognition. For example, research methods such as interviews, surveys and black box experiments have been used to indirectly study cognitive behaviour of designers (Gero & Milovanovic, 2020). Protocol analysis (Ericsson & Simon, 1993; van Someren et al., 1994) has been widely used to directly measure design cognition (Dinar et al., 2015; Gero & Milovanovic, 2020). Various studies have utilized this method to generate and test hypotheses, and to aid the development of a better understanding of design activities (Dinar et al., 2015; Gero & Milovanovic, 2020; Hay et al., 2017). Using this method, previous studies have investigated the design cognition of a wide range of cohorts such as students, professionals and experts performing design tasks characteristic to a variety of domains such as engineering, electronic, industrial, software, systems and architectural design (Dinar et al., 2015; Gero & Milovanovic, 2020; Hay et al., 2017).

A key advantage of protocol analysis over other social science research methods is that the former allows reproducibility and statistical analysis. However, the high resource-intensiveness of protocol analysis has often restricted the ability of researchers to carry out controlled studies with large enough

samples to enable statistical analysis (Dinar et al., 2015). Although the efficacy of protocol analysis in measuring design cognition has been widely debated, it has also been argued that it could be one of the few available research methods that can provide direct access to an improved understanding of design cognition (Cross, 2006; Lloyd et al., 1995).

Commensurable dimensions in the frames of analysis are crucial to facilitate meta-analytics across different domains of design and thus, to facilitate theory development (Cash, 2018; Gero & Milovanovic, 2020; Hay et al., 2020). Generic frames of analysis based on the FBS-ontology and problem-solving based frameworks have been used to model design cognition across a wide range of domains (Gero & Milovanovic, 2020). The frames of analyses used in the past protocol studies of PSS designing are mainly specific to the PSS design domain (Section 2.10), and therefore, the insights derived may not be commensurable with studies of other domains of design. Such commensurable insights are crucial to characterise the nature of PSS design cognition.

In order to facilitate cross-domain analysis, Sakao et al. (2020) proposed an FBS ontology-based commensurable approach of protocol analysis to compare the cognitive characteristics of product and PSS designing. Furthermore, the potential of building on established theories of cognitive science has been highlighted (Cash, 2018; Hay et al., 2020). Theories of cognitive architecture and cognitive load (Section 2.5) can potentially be built upon to improve the understanding of cognitive structures and mechanisms underlying designing. Although these theories have mainly been utilized in the domain of instructional science, their utility in studying and also supporting the cognitive mechanisms of design has been briefly discussed (Cash, 2020; Curry, 2014).

It has been widely reported that although academia has been developing a tremendous amount of design methods, their adoption in industry has been fairly limited (Gericke et al., 2020; Schønheyder & Nordby, 2018; Tomiyama et al., 2009). One potential reason for the low deployment of the design methods could potentially be attributed to the top-down development of such methods, without considering the real-world adaptability, needs and requirements of the method users (Lindahl, 2006; Schønheyder & Nordby, 2018). The method action design research (ADR) (Sein et al., 2011) is considered to be useful to address case-specific challenges that entails the development of a design object within an organization. It is also expected to contribute to the development of a generic prescriptive method that focuses on the building, intervention and evaluation of the design object in other similar contexts (Sein et al., 2011). As opposed to using typical top-down approaches entailing pure desktop research based development and case study applications of the design methods, ADR offers a bottom-up approach to develop prescriptive design support.

Chapter 3 – Conceptual framework: Cognitive architecture and mechanisms of designing

3.1 Overview

In this chapter, a conceptual model of cognitive architecture and mechanism of designing (CAM-D) is developed by integrating a widely used generic model of cognitive architecture and mechanism of problem-solving activities (Section 2.5) with generic theories and models of the design process (Section 2.6). This conceptual model is partly grounded in the situated notion of designing (Gero & Kannengiesser, 2004) and more specifically, in the claims of the augmented generic design hypothesis (Visser, 2009). These theoretical viewpoints of design cognition argue that there are different forms of design cognition specific to different conditions of design situations or domains.

Quoting other previous works, Visser (2009) suggests that a domain is anything a specific constraint can be generalized to and from, such as a design object, design process or the experiences of the designer. Since this research primarily aims to characterize the cognitive processes underlying conceptual PSS designing, the major constructs, variables, and relationships under lying conceptual design cognition are initially conceptualized (see Section 2.11). Based on this conceptualization, a predictive proposition and associated hypotheses concerning the structures and mechanisms underlying domain-specific design cognition are formulated.

3.2 Cognitive architecture and mechanisms of designing CAM-D

3.2.1 Theoretical conceptualization and model

Here, it is considered that the activity of designing unfolds in two realms: the external realm, and the internal realm. The external realm refers to the system external to the human designer, which can include the source of the design problem and the external environment that is to be manipulated by the design outcomes. The internal realm includes the working and long-term memory spaces of the designer (based on the theory of cognitive architecture by Sweller et al. (1998)).

Prior to the design activity in focus, the designer may have pre-existing knowledge stored in the long-term memory space of the internal realm. The design process unfolds within the working memory space as it processes the incoming exogenous data pertaining to the design problem in terms of lower-level elements of information (based on Sweller et al. (1998)), by building on the pre-existing knowledge of the designer (partly based on (Gero & Kannengiesser, 2004; Gero, 2000; Sim & Duffy, 2003)).

The processed lower-level elements of information are then categorized and organized as higher-level knowledge structures within the long-term memory space, termed design knowledge schemas (see Section 2.5). Such domain-specific design knowledge schemas are constructed through domain-based experience (based on Sweller et al. (1998) and Ball et al. (2004)). Thus, the configurations of such knowledge schemas being constructed, are considered to be influenced by the specific configurations of the information processing unfolding within the working memory space during the design process. As the design process progresses, newly processed domain-specific lower-level elements of information are categorized and abstracted under previously constructed domain-compatible design knowledge schemas leading to schema automation (Section 2.5).

With highly automated knowledge schemas, the intrinsic cognitive load required to process the incoming exogenous design data during the respective design process is expected to be significantly lower (see Section 2.5). Partly based on the arguments of previous literature (Ball et al., 2004; Sweller, 1988; Sweller et al., 1998), it is suggested that the process of schema automation in designing may also be domain-specific. That is, if the present design problem being addressed is characteristically similar to the previously addressed design problems grounded on which the pre-existing knowledge schemas were constructed upon, then increase in schema automation will occur through repeated practice within the domain. Therefore, it is suggested that the degree of automation of a domain specific design knowledge schema is proportional to the level of domain-based experience of the designer. For example, if an architect has spent several years practicing architectural design and relatively fewer years practicing industrial design, then the degree of automation of knowledge schemas specific to architectural design is expected to be higher than that of the latter. Such highly automated schemas can be utilized to address different types of architectural design challenges with relatively lesser cognitive load, compared to designing specific to industrial design challenges.

In the absence of pre-existing schemas or sufficiently automated schemas, the design process may proceed through the application of generic problem-solving strategies (e.g., means-end analysis) within the working memory space (based on Sweller (1988)). More specifically, during the design process, effective and efficient sequences, or configurations of operations to manipulate the elements of information (the number of configurations increases in terms of permutations of the number of discrete information elements) need to be found in a non-automated manner (based on Schnotz & Kürschner (2007) and Sweller et al. (1998)). This type of cognitive behaviour is typically exhibited by domain-novices, as opposed to domain-experts who tend to have knowledge schemas with higher degrees of automation (Sweller, 1988).

The domain-specific design knowledge schemas are also expected to be transferable to designing specific to other domain-specific design problems. Such transferability depends upon the proximity of the characteristics of the current design problem being addressed to that of the previously addressed design problems upon which the constructed and automated schemas are grounded (based on (Van Merriënboer & Paas, 1990)). For example, insights gained from architectural design experiences may be transferable to mechatronics product design but may not necessarily be optimal and comprehensive. In contrast, insights gained from mechatronics product design experiences may be relatively more readily transferable to electronic design problems.

Here, it is proposed that as the design process unfolds, the exogenous design data and, subsequently, information are processed within the working memory space or more specifically within the construct termed *design micro-frame*. In this case, a design micro-frame is considered a fundamental unit of the design process. It is defined as a temporary construct developed within the working memory space to interpret, manipulate, and organize exogenous design data to simultaneously solve and learn from design problems.

The design micro-frame serves the following purposes (based on Sections 2.5 and 2.6.1): 1) based on the incoming exogenous data pertaining to the design problem, it filters and retrieves the relevant design knowledge schemas stored in the long-term memory; 2) based on the retrieved schemas, it interprets the incoming exogenous data pertaining to the design problem in terms of design information, thus carrying out problem formulation; 3) based on the retrieved schemas, the interpreted design information is manipulated through design-specific processes and 4) it supports the development of design outcomes and the construction or automation of knowledge schemas.

The design process is assumed to progress through the construction of a series of micro-frames. Since the development of domain expertise is linked to the extent of practice within the domain on the temporal scale (based on Cross (2006, p.75)), it is proposed that the increase in the degree of automation of a domain-specific design knowledge schema is a function of the number of micro-frames generated in the working memory space during the respective domain-specific design process. However, the increase in degree of automation may also be influenced by other factors such as the extent of germane cognitive load available to support automation and the relative extent of intrinsic and extraneous cognitive load imposed by the design problem (Section 2.5).

Based on the above-presented conceptualizations, it is assumed that prior to the design process, along with other knowledge, the designer has a pre-existing design knowledge schema. This knowledge schema ($Kx_{n=p}$) is built upon previous design experiences specific to a domain *x*, after the creation of any a^{th} micro-frame ($F_{j=a}$). Here, *j* represents the number of micro-frames, *a* is a value of the number of micro-frames, *n* represents the degree of automation of the

knowledge schema specific to domain *x* and *p* is a value of the degree of automation at the ath micro-frame. As the designer is introduced to a design problem specific to a novel domain *y* originating from the external realm, the design process unfolds as the respective exogenous data (*Ey*) is pulled into the working memory space. The total knowledge ($K_{j=0}$) at the 0th micro-frame (j=0) during the initial state of the design process specific to domain *y*, stored in the long-term memory space within the internal realm can be represented as:

$$K_0 = \{Kx_{n=p}, Ky_{m=0} \ldots.\} \quad (1)$$

Here, ($Ky_{m=0}$) represents the initial null-state of the knowledge schema specific to the novel domain *y*, where *m* represents the degree of automation of the knowledge schema specific to domain *y*. It has 0 degrees of automation since the designer has no pre-existing experience nor knowledge of the design domain *y*. The retrieved knowledge schema coming into any b^{th} micro-frame ($F_{j=b}$) (where j = b > 0) from the long-term memory is referred to as input knowledge (i_{b-1}) and each iteration of micro-framing recursively adds new output knowledge (o_b) to the long-term memory space or the total available knowledge. The total knowledge can also include other schemas. The designer may also engage in the creation of external design representations (*Ry*) throughout the design process, which form the design outcomes. The above-described conceptual model of cognitive architecture and mechanisms of the design process is visualized in Figure 8.

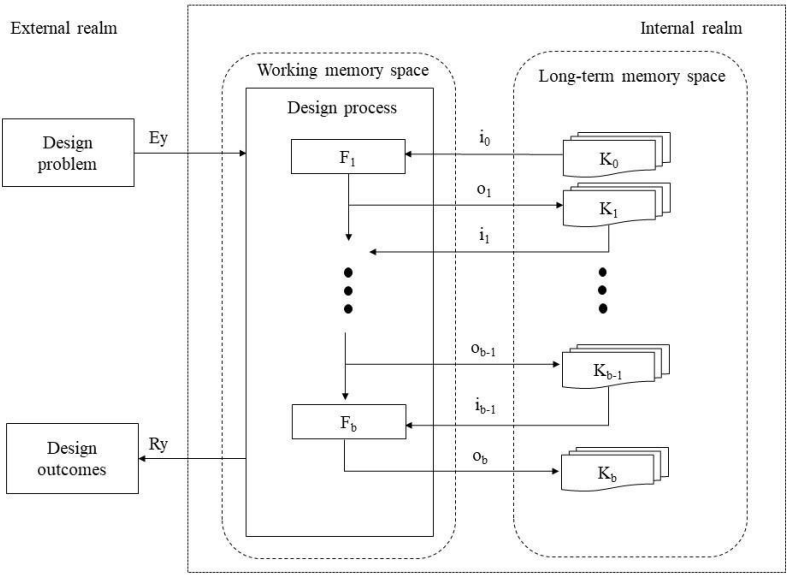


Figure 8. Formalism of the cognitive architecture and mechanism of designing (author's own).

3.3 Predictive proposition

3.3.1 An approach to model design cognition

Here it is assumed that the cognitive characteristics of a domain-specific design process can be modelled by measuring how designers distribute their cognitive effort during the processing of information within the design micro-frames, in relation to that of a different domain-specific design process. *Cognitive effort* is referred to as the amount of available processing capacity of the working memory space that is utilized to perform an information processing task (based on Sweller (1988) and Tyler et al. (1979)).

Since the design activity is a multi-faceted phenomenon, information processing within the design micro-frame is expected to unfold in terms of the changes in the variables of the design process such as co-evolution of problem-solution spaces, the transformation of the design state issues and processes linked to the problem-solution spaces, differing levels of systems abstraction, problem decomposition and re-composition (see Sections 2.4 and 2.6 for more details). Here, changes in total cognitive effort required to carry out information processing are not considered, but rather patterns of distribution or organization of cognitive effort across the different variables of the micro-frames are under consideration.

3.3.2 Influence of characteristics of design problems

Differences in design cognition can be observed across previous empirical studies of designing specific to the domains of industrial design (Kruger & Cross, 2006), mechanical design (Lloyd & Scott, 1994) and in a study entailing a comparison of the two (Jiang et al., 2014). The differences that can be observed across those studies are in terms of differing focus over time on the following variables (see Section 2.6.3):

- Problem-solution spaces
- Design state issues
- Design state transformation processes

One potential reason for the differences in the design cognition of the respective design processes can be attributed to the differences in the tangibility of the objects being designed. Designing specific to the domains of industrial design and engineering design requires the consideration of aspects with characteristically distinct degrees of tangibility (Kim & Lee, 2016). More specifically, while mechanical designing requires the consideration of tangible aspects such as material quality and geometric dimensions, industrial designing requires the consideration of more intangible aspects such as ergonomics and aesthetics.

Furthermore, it has been suggested that the structuredness of the design problem in terms of its social embeddedness can influence the focus on the extent of application of ‘‘weaker’’ criteria or the highest level of systems abstraction (see Section 2.6.4) (for an example, see the discussion concerning the differences in the applied levels of systems perspective between social and engineering design problems (Simon, 1969)). These previous empirical findings and the lines of reasoning can be inductively linked to the situated notion of designing (Gero & Kannengiesser, 2004) and more specifically, in terms of the augmented generic design hypothesis (Visser, 2009). It can be inferred from these theoretical view-points that context specific variables such as the structuredness of the design problem being addressed, in terms of tangibility and social embeddedness of the design object, are expected to significantly influence design cognition.

3.3.3 Proposition 1 and hypotheses

By applying the conceptual model of CAM-D (Section 3.2.1) and the above presented arguments (Section 3.3.2), the following proposition is inductively postulated:

Proposition 1:

It is postulated that the characteristics of the domain-specific design problem being addressed, influence the configuration of information processing within the design micro-frames. Thus, the respective knowledge schemas developed within the design micro-frame and subsequently stored in the long-term memory space, are also influenced by the characteristics of the domain-specific design problem being addressed. Two conditions that postulate the state of the resulting total knowledge (K_b , where b > 0) stored in the long-term memory space after receiving o_b from F_b during the design process specific to domain *y*, are presented as follows:

Condition 1: The characteristics of the present design problem specific to domain *y* being addressed are highly similar to the characteristics of the design problems specific to domain *x* that have been addressed during the previous design experiences of the designer. Then, the pre-existing design knowledge schema $Kx_{n=p}$ will be retrieved to effectively support the information processing within the design micro-frames. Furthermore, at the bth micro-frame during the design process specific to domain *y*, the new output knowledge o_b being added to the long-term memory space will build upon the pre-existing design knowledge schema $Kx_{n=p,}$ thus contributing to its automation by transforming it to $Kx_{n=p+z}$. Here, *z* is a value of the degree of automation of the knowledge schema specific to domain *y* at the bth micro-frame.

Condition 2*:* The characteristics of the present design problem specific to domain *y* are significantly different from those of the design problem specific to domain *x* addressed during the previous design experiences of the designer. Then, after the creation of the bth micro-frame during the design process specific to domain *y*, the new output knowledge being added to the long-term memory space contributes to the construction of the knowledge schema ($Ky_{m=z}$) specific to the domain *y*. Here, *z* is a value of the degree of automation at the bth micro-frame. After the creation of the bth micro-frame F_b, two different knowledge schemas $Kx_{n=p}$ and $Ky_{m=z}$, will be part of the total knowledge K_b within the long-term memory space, along with other pre-existing knowledge schemas. In this condition, the configuration of knowledge schema specific to domain *y* is expected to be influenced by the configuration of information processing within the bth micro-frame while addressing the design problem specific to domain *y*.

For each of the above two conditions, the values of the total state of the knowledge K_b, after receiving o_b from F_b during the design process specific to domain *y* can be represented as follows:

$$K_b = \begin{cases} \{ Kx_{n=p+z}, \dots \} & Condition\ 1: if\ x \sim y \\ \{ Kx_{n=p}\ ,\ Ky_{m=z}, \dots \} & Condition\ 2: if\ x\ !\sim y \end{cases} \quad (2)$$

Based on Condition 2, the following hypotheses concerning the influence of characteristics of the design problem on different variables of information processing within the design micro-frame (see Section 3.3.2) during the design process are formulated. One type of the characteristic of the design problem can be represented in terms of the degree of structuredness. It can be measured in terms of the relative characteristics such as tangibility and social embeddedness of the design objects to be developed (based on Simon (1969, 1973) and Visser (2009)).

H1 : There are significant differences in the distribution of cognitive effort in the problem-solution design space between designing specific to two design problems with characteristically distinct degrees of structuredness.

H2 : There are significant differences in the distribution of cognitive effort on the design state issues between designing specific to two design problems with characteristically distinct degrees of structuredness.

H3 : There are significant differences in the distribution of cognitive effort on the design state transformation processes between designing specific to two design problems with characteristically distinct degrees of structuredness.

H4 : There are significant differences in the distribution of cognitive effort on the highest hierarchical level of systems abstraction between designing specific to two design problems with characteristically distinct degrees of structuredness.

Chapter 4 – Research Design

4.1 Research methodology

The overall approach of this research is dualistic in nature, as it includes both descriptive and prescriptive dimensions. Since the overarching objective of this thesis is to effectively support practitioners of the product manufacturing industry to conceptually design PSSs with relatively lower environmental impacts, it is first necessary to descriptively investigate how such a process unfolds before prescribing any support. Subsequently, the developed prescriptions need to be assessed to ensure the efficacy of the support provided.

To effectively address the descriptive and prescriptive aims of this research, the research approach was systematically structured by implicitly following the design research methodology (DRM) (Blessing & Chakrabarti, 2009). This methodology was chosen as it prescribes a systematic way to descriptively study and prescriptively support the phenomenon of designing of interest, in an iterative manner. The links between the three RQs (Section 1.4), the different dimensions of this research framed under the three stages of DRM, the research methods used within those three stages of research, and the appended publications are visualized in Figure 9.

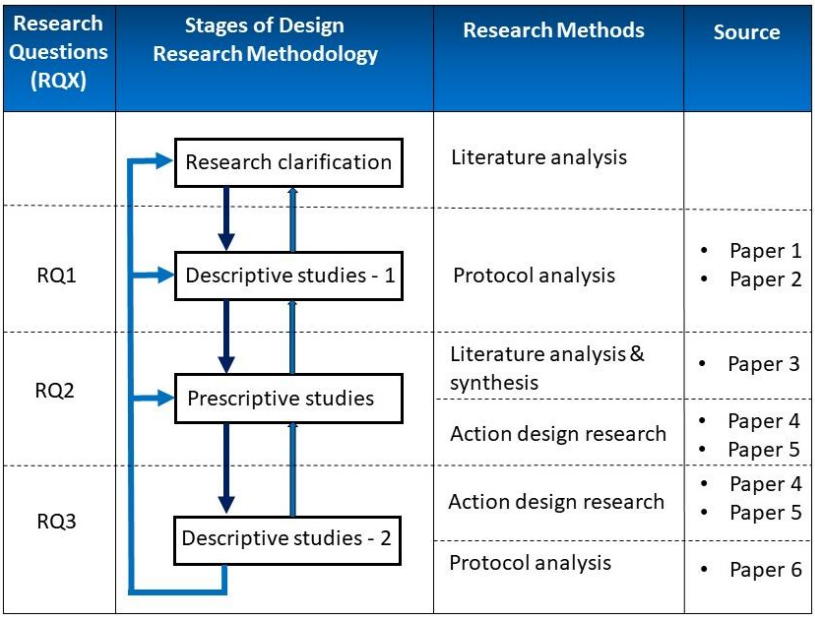


Figure 9. Research outline based on design research methodology (DRM) (adapted from Blessing & Chakrabarti (2009)).

4.1.1 Research clarification

The input to the research clarification (RC) stage is the research problem and aim established in Chapter 1, along with the theoretical background presented in Chapter 2. The main outcome of the RC stage is the generic conceptual model, CAM-D. Such a conceptual framework is deemed necessary as it can 1) provide an outline of how the defined research problem can be best explored, 2) lay out the key constructs and variables of the phenomenon being studied, and 3) provide postulations of the relationships between the variables and the respective constructs (Grant & Osanloo, 2014). The subsequent stages within which the empirical research of this thesis is carried out are briefly described below.

4.1.2 Descriptive studies - 1

This stage of the thesis includes descriptive studies – 1 (DS-1), documented in Papers 1 and 2. The outcomes of the DS-1 contribute to the development of an improved understanding of the cognitive characteristics of PSS designing and provide implications for support development, thus contributing to the answering of RQ1. Protocol analysis is used for the DS-1.

4.1.3 Prescriptive studies

This stage includes three prescriptive studies (PS) covered in Papers 3, 4 and 5. These studies present three different types of prescriptive support catered to the implications for support development, derived from the improved understanding of the cognitive characteristics of PSS designing developed in DS-1. Thus, the three PSs together contribute to the answering of RQ2. While the prescriptive design support presented in Paper 3 was developed using a literature analysis and synthesis method, principles of the ADR method were followed to develop the prescriptive design support presented in Papers 4 and 5.

4.1.4 Descriptive studies - 2

This stage includes the descriptive studies (DS-2) covered in Papers 4, 5, and 6. Papers 4 and 5 report the measured effects of the prescriptive support developed within the same respective papers by implicitly following the principles of the ADR method and, more specifically, with LCA. The support presented in Paper 3 is descriptively assessed in Paper 6 using protocol analysis. Thus, these DS-2s covered in Papers 4, 5 and 6 together answer RQ3. The different research methods and respective data collection techniques utilized within all three stages are presented in the following sections.

4.2 Research methods

4.2.1 Literature review – analysis and synthesis

The literature review is an important part of this research as it facilitates the mapping of relevant established intellectual information on the phenomenon of conceptual PSS designing being studied and can enhance the established knowledge base (based on Tranfield et al. (2003)). In this thesis, the literature review is concept-centric (Webster & Watson, 2002). Using this concept-centric approach, the literature analysis of this thesis is structured around the core topics such as design, design for environmental sustainability, cognitive science, the science of designing and PSS design. Also, meta-synthesis of captured insights from the literature review is carried out to capture theories, ideas, narratives, generalizations, or interpretations elicited from integration or comparisons of qualitative studies centred around the above-mentioned core topic of interest (Sandelowski et al., 1997). This research method is implicitly used in part to answer all the research questions of the thesis. More specifically, to answer the second RQ in the PS stage of this thesis, insights gathered from the concept-centric literature review of prescriptive PSS design literature are synthesized by carrying out meta-synthesis to contribute to the creation of three different types of prescriptive support for PSS designing.

4.2.2 Protocol Analysis

In this thesis, protocol analysis is used to collect and analyse verbal think-aloud (Ericsson & Simon, 1993; van Someren et al., 1994) data gathered from conceptual product and PSS designing in empirical settings. This research method was initially prescribed by Ericsson and Simon (1993) and further developed by van Someren et al. (1994). In protocol analysis, verbal thought sequences generated from the think-aloud protocol are collected as valid sources of data on cognition. The think-aloud protocol is the process of speaking out one's thought sequences while carrying out a defined activity, such as designing (van Someren et al., 1994).

This method is primarily used in Papers 1, 2 and 6 to answer the first and third RQs within the DS-1 and DS-2 stages, respectively. To collect the protocol data, design sessions are conducted in a laboratory setting with a cohort of experienced product developers from the manufacturing industry. To address the first RQ, protocol studies of product designing, and PSS designing are carried out and compared (in Paper 1), and an in-depth protocol study of PSS designing is also carried out (in Paper 2). To partly address the third RQ, protocol analysis of designing PSSs with and without a prescriptive PSS design support is carried out (in Paper 6). The argumentation for choosing protocol analysis as the primary research method for DS-1 and 2 is highlighted in Section 2.12.

4.2.3 Action design research

Action research (AR) is a hermeneutical approach to studying human actions and social practices where a researcher can search for knowledge as an observer and simultaneously create practical benefits for the collaborating practitioner (Williamson, 2002). ADR (Sein et al., 2011), a derivative of AR, is primarily applied in the discipline of information science. ADR is described as "a research method for generating prescriptive design knowledge through building and evaluating IT artefacts in an organizational setting" (Sein et al., 2011). This research method is adapted in Papers 4 and 5 to partly address the second and third RQs.

In this thesis, ADR, together with literature analysis and synthesis (Section 3.4.1), is used to generate and evaluate prescriptive design knowledge in the form of two generic design support methods that can support the conceptual designing and evaluation of concepts of PSS-based offerings in two separate industrial cases, covered in Papers 4 and 5 of the PS and DS-2 stages of this thesis. The developed prescriptive support not only reflects relevant theoretical precursors specific to design cognition and prescriptive PSS design literature but also the context of the practitioners who use the support.

ADR also considers the situation of the end-users who use the artefact and its resulting environmental impacts (in line with the prescribed principles of ADR (Sein et al., 2011)). AR has been previously applied in multiple domains of design, such as in product design, to develop environments that support the collaboration of multi-disciplinary teams in product lifecycle engineering (Mejía et al., 2007) and in service design to enhance value co-creation in social innovation initiatives (Yang & Sung, 2016). The argumentation for choosing ADR as the primary research method for PS is highlighted in Section 2.12.

4.3 Research sample, data, and analytical procedure

4.3.1 Research sample

A criterion-based sampling approach (Cash et al., 2022) was used to define the sample of the empirical work presented in this thesis: *the sample should belong to a population of practitioners experienced (average experience of 10 years) in domains associated with product development within the product manufacturing industry*. To elaborate, the practitioners should be experienced in performing organizational functions such as product design, mechanical design, product eco-design, product-based service design, product management and other product lifecycle-related functions linked to the development and offering of physical products and related services, within a typical product manufacturing company operating on a goods-dominant logic. From here on, this cohort is referred to as experienced product developers.

This criterion for the sampling was chosen for the following reasons: 1) the overall objective of this thesis is to support practitioners within typical product manufacturing companies (operating on a goods-dominant logic), without pre-existing PSS design expertise to effectively design and offer environmentally benign PSS-based solutions; 2) experienced practitioners are chosen over other cohorts, as the study of experts can significantly improve our understanding of expertise in design rather than the alternative of studying non-experts (Cross, 2006).

4.3.2 Description of cohorts

The sampled practitioners involved in the empirical work covered in Papers 1, 2, 4, 5, and 6 work in companies that operate in a variety of industrial sectors that manufacture construction equipment, vertical-access equipment, and telecommunication, automotive, and consumable incontinence products. More specifically, the DS-1 covered in Paper 1 included a sample of 56 experienced practitioners from multiple manufacturing companies based in the US and Sweden, who were divided into two cohorts for the protocol analysis. Each cohort included multiple pairs of sampled practitioners.

One cohort who participated in the DS-1 covered in Paper 1 was given a typical product design task (Condition A, N=18 pairs), and the other, a typical PSS design task (Condition B, N=10 pairs) (see Paper 1 for more details). The DS-1 covered in Paper 2 utilized the same data collected from the cohort that was given the PSS design task in Paper 1. The data collected from the cohort that was given the PSS design task in Paper 1 was also partly used as the data for the control group (N=3 pairs) in the DS-2 covered in Paper 6. The experiment group of Paper 6 included a cohort (N=3 pairs) that was given the same PSS design task but was additionally introduced to the PSS design support presented in Paper 3.

In the PS and DS-2 covered in Paper 4, ADR was carried out with a multinational incontinence product manufacturing company based in Sweden. This study included practitioners (N=3) experienced in domains associated with product development. More specifically, these practitioners served the functions of product design, service design and environmental specialist, respectively. In the PS and DS-2 covered in Paper 5, ADR was carried out with a multi-national vertical-access equipment manufacturer based in Spain. This study also included practitioners (N=3) experienced in domains associated with product development, with a similar mix (in terms of the type of expertise and years of experience of the practitioners) as that of the study presented in Paper 4.

4.3.3 Description of data collection and analysis techniques

For the DS-1 and DS-2 that have used protocol analysis as a research method (Papers 1, 2 and 6), design sessions in a controlled laboratory setting were preferred over case studies in industrial settings, primarily because it was deemed to be impractical to shadow the participants in their real-life work settings and to maintain control in the highly varied industrial contexts. Furthermore, PSS designing requires simultaneous and unrestricted collaboration between practitioners with product and service design roles (Meier et al., 2010). However, in practice, they usually operate in silos and carry out their respective tasks in sequence, which is not ideal for PSS designing (Matschewsky et al., 2018). These design sessions were primarily carried out within the company premises of the participants. The protocol data from the design sessions was collected in both audio and video formats. The unit of analysis of these studies is the think-aloud protocols performed by each pair during their respective design tasks.

The collected protocol data was subsequently transcribed, segmented and coded using an ontologically based procedure (Kan & Gero, 2017). This procedure advocates the breakdown of chunks of transcribed think-aloud utterances into segments or the smallest possible units of data, until each segment can be assigned only one code from an ontologically based coding scheme. To ensure coding reliability, the Delphi protocol (Gero & Mc Neill, 1998) was followed in which two independent analysts segmented and coded the protocols individually using a certain *a priori* coding scheme based on relevant theoretical frameworks.

The disagreements in the segmentation and coding were resolved by a third expert arbitrator in consultation with both analysts. The coded protocols were analysed using both descriptive and inferential statistics (Holcomb, 2016) and other available quantitative techniques for studying design protocols (see Kan and Gero (2017)) in Papers 1, 2 and 6. This analysis of coded think-aloud protocol represents the focus expended or time spent by the designer on the different variables of the design process represented by each code. The distribution of time spent is equated with the distribution of cognitive effort on the different variables of the design process (see Section 3.1).

For the PS and DS-2 covered in Papers 4 and 5, under the ADR method, the retrospective data analysis technique (Blessing & Chakrabarti, 2009) of document analysis was used in part to analyse and interpret pre-existing data in order to interpret meanings and develop empirical knowledge (Bowen, 2009). The focus group technique (Krueger, 1994) was used to develop and apply the prescriptive design support methods that can support conceptual PSS designing, in collaboration with the study cohorts. In Paper 4, one focus group workshop was held on the premises of the respective case company based in Sweden, while multiple virtual workshops were also held during the ADR process. In Paper 5, multiple focus group workshops were held only in a virtual setting. The

workshops held in both studies documented in Paper 4 and Paper 5 were either audio recorded with the permission of all the participants or captured through extensive notes taken by the participating researchers. These notes taken from the multiple researchers were initially compared and then consolidated, before being consistently sent back to the participants from the respective cases for their review.

Lastly, LCA (ISO, 2006) was used to assess the reduction in environmental impacts of concepts for PSS-based solutions designed using the two design support methods, compared against that of existing product-based offerings within the case companies. Therefore, the relative environmental impacts of the concepts designed with and without the given design support, represent the other unit of analysis of these two studies.

4.4 Research quality measures

Since the empirical research documented in this thesis has both quantitative and qualitative dimensions, the following quality measures used mainly in social science research are used in this thesis to assess its quality: 1) credibility (similar to internal validity – accuracy of data that reflects the reality of the observed social phenomenon), 2) transferability (similar to external validity – level of applicability to other settings or situations), 3) dependability (corresponds to reliability – level of replicability), and 4) conformability (corresponds to objectivity – extent to which others can confirm the findings and is not influenced by researcher's own bias), based on Golafshani (2003) and Wahyuni (2012).

The quality of a theory of a complex social phenomenon such as designing can be assessed based on the following attributes (based on Bhattacherjee (2012), Chakrabarti & Blessing (2015) and Eckert & Stacey (2014)): 1) logical consistency between the constructs, variables, propositions, and assumptions; 2) predictive and explanatory ability of the theoretical model; 3) testability and falsifiability of the theory that ensures that the theory is empirically testable and disprovable if the empirical data does not match the theoretical predictions; and 4) parsimony and generality of the theory that refers to the use of fewer and widely applicable variables to explain the phenomenon in a simplified and generalizable manner.

4.5 Philosophical positioning of the research

The thesis primarily investigates the ability of experienced product developers to design PSSs, using systematic and transferable methods of investigation such as protocol analysis. Therefore, the research carried out in this thesis can be considered as a form of the science of designing (based on Cross (2006)). Furthermore, this thesis also investigates the characteristics and extent of design knowledge and its application to PSS design problems, the development of

appropriate structures for the PSS design process, and the development and application of new PSS design support. These aspects also fall under the purview of the sciences of designing; based on Cross (2006). However, it is important to note here that this type of research is not the same as the physical sciences, as the sciences of designing the artificial are not considered to be the same as the natural sciences; rather, they are considered to be dichotomous forms of knowledge (Simon, 1969). Simon suggested that while the natural sciences mainly address the study of universal truths, the science of designing is concerned with the study of the development of the artificial, which is shaped by subjective goals, purposes, and human actions (Schaathun, 2022).

The dimension of the design research presented in this thesis that aims to descriptively observe and analyse the phenomenon of PSS designing on the cognitive level of the individual designers can be considered a form of phenomenology (based on Bryman (1984)). The other dimension of the research aims to prescriptively support PSS designing grounded on the empirical understanding of the phenomenon developed in the first dimension of this research and, subsequently, to evaluate such support. The epistemological nature of these two dimensions of research can together be considered as a form of pragmatism, as the employed research methods include both positivistic (quantitative and deductive) and interpretivist (qualitative and inductive) characteristics (based on Wahyuni (2012)). Furthermore, the knowledge developed in the prescriptive dimension of the thesis is an outcome of the analysis of the observable phenomenon of designing with subjective interpretivism and has practical applications, which according to Wahyuni (2012), also represent the characteristics of pragmatism. The pragmatic nature of the prescriptive dimension of design research is also acknowledged in the state of the art (Schønheyder & Nordby, 2018).

4.6 Disciplinary position of the research

The descriptive dimension of the research presented in this thesis investigates the cognitive nature of PSS designing by experienced practitioners from the product manufacturing industry, with a focus on enhancing the environmental sustainability of design solutions. This dimension of research is grounded in multiple disciplines, such as engineering design, cognitive, environmental, and social sciences. Further, since this dimension of research also integrates the views of these specific disciplines to create new knowledge, the research can be classified as inter-disciplinary (based on (Stock & Burton, 2011)), in line with the views of Cross (2006).

The prescriptive dimension of the research presented in this thesis aims to support the practitioners of the manufacturing industry to design environmentally benign PSS solutions, based on the developed understanding of their design cognition. Development of this kind of support and its subsequent evaluation involves the active collaboration of the recipients of such support, the

practitioners who are essentially non-scientific stakeholders. Since the overarching objective of this research is to support the practitioners within the product manufacturing industry to design PSS-based solutions with relatively lower environmental impacts than existing product-centric offerings, the research topic being addressed is a real-world challenge. Thus, the overall research is a form of transdisciplinary (Stock & Burton, 2011) or, more specifically, Transdisciplinary type 2 research (Sakao & Brambila-Macias, 2018).

Chapter 5 – Understanding product-service system designing

5.1 Overview

The empirical results presented in this chapter contribute towards the characterization of the cognitive processes underlying PSS designing, by comparing it with product designing. Thus, this chapter contributes to answering RQ1: *How do practitioners experienced in domains associated with product development conceptually design product-service systems on the cognitive level?* (Section 1.4). Initially, highlights of the findings of DS-1 carried out in Paper 1 are presented. In Paper 1, differences between the cognitive characteristics of designing specific to a typical product and a PSS design problem were inductively predicted (based on Proposition 1 and Hypotheses 1-4 of CAM-D, Section 3.3.3) and empirically investigated. Thus, the findings presented in this section, also contributes to the assessment of the falsifiability of Proposition 1 of CAM-D (Section 3.3.3). Then, additional findings of DS-1 carried out in Paper 2 are also presented in which domain-specific insights into the cognitive characteristics of PSS designing were explored.

5.2 Comparing the underlying cognitive processes of designing specific to product and product-service system design problems – Paper 1

5.2.1 Overview - Paper 1

Paper 1 aimed to theoretically predict, empirically measure, compare, and characterize designing specific to a typical product and a typical PSS design problem on the cognitive level. The findings of this study contribute to the answering of RQ1 of this thesis. At its start, this study implicitly utilized the hypotheses formulated in the conceptual model of CAM-D grounded in the augmented generic design hypothesis (Section 3.3.3) and the theoretically prescribed characteristic differences between product and PSS design problems and respective design objects (see Section 2.3.2 for a summary). Subsequently, differences between design cognition specific to the two distinct types of domain-specific designing were predicted.

To test these predictions, protocol analysis of two independent sets of design sessions was carried out in laboratory settings (see Section 4.2.2 and 4.3 for more details). The cohort for both sets were pairs of experienced product developers. One set of design sessions was treated to Condition A, where the task was to address a typical product design problem. The other set was treated to Condition B, where the task was to design a PSS, based on an existing

product, with increased resource-efficiency (reduced environmental impacts). These two conditions are representative of the phenomenon of interest of this thesis which aims to support designers in transition from designing typical products to PSSs based solutions with relatively lower environmental impacts.

The protocols were analysed with the following frames of analysis: 1) a coding scheme based on the FBS ontology to model the distribution of cognitive effort on design state issues and design state transformation processes (see Section 2.6.3) and 2) a coding scheme based on the hierarchical levels of systems abstraction (HSA) to model distribution of cognitive effort on levels of systems abstraction, problem decomposition and re-composition. This model includes the following three hierarchical levels of abstraction: 1) system level (the highest level of abstraction), 2) sub-system level (intermediate level of abstraction) and 3) element level (the lowest level of abstraction). Transitions from higher to lower levels are regarded as problem decomposition, while the inverse transitions are considered as problem re-composition (based on Section 2.6.4, see Paper 1 for more details).

5.2.2 Predicting the differences between design cognition specific to product and product-service system design problems

Based on the hypotheses of CAM-D (Section 3.3.2) grounded in the theoretical arguments concerning the domain-specific generic design hypothesis and preliminary empirical evidence corroborating it (Jiang et al., 2014; Simon, 1969; Visser, 2009), it is inductively hypothesized in Paper 1 that the differences in structuredness of domain-specific design problems in terms of tangibility (scope) and social embeddedness (scale of social actors associated with the object/activity being designed) of the design object to be developed may significantly influence design cognition. More specifically, it is inferred that designing specific to problems entailing the development of design objects with higher degrees of intangibility tends to be problem-oriented, while the ones with higher degrees of tangibility tend to be solution oriented. This inference was made based on a qualitative comparison of the findings of the studies of designing specific to a relatively more tangible domain of mechanical design (Lloyd & Scott, 1994), the more intangible domain of industrial design (Kruger & Cross, 2006) and a comparison of the two (Jiang et al., 2014).

A dissection of product and PSS design literature carried out in Paper 1 (see also Section 2.3.2) reveal that PSS design objects tend to have a higher degree of intangibility and qualitative aspects than typical product design objects. Additionally, the literature indicates that PSS designing requires comparatively more focus on functional modelling and simulation of expected system behaviour than typical product designing, partly based on Maussang et al. (2009), Meier et al. (2010) and Sakao et al. (2020). Based on the above-

presented theoretical arguments, the empirical evidence and partly on the hypotheses formulated in an exploratory study by Sakao, Gero, and Mizuyama (2020) concerning differences between product and PSS designing, the following predictions were (P) formulated:

P1 : Conceptual PSS designing expends more cognitive effort in the *problem space* than product designing.

P2 : Conceptual PSS designing expends more cognitive effort on the problem space-based design issue *function* than product designing.

P3 : Conceptual PSS designing expends more cognitive effort on the problem space-based design process *formulation of expected behaviour from function* than product designing.

It is often argued in the PSS design literature that PSSs tend to have comparatively higher degrees of heterogeneous system element interactivity, social construction, and thus complexity than typical products; therefore, the application of a comparatively higher systems perspective during PSS designing is deemed to be crucial (Brissaud et al., 2022; Ceschin & Gaziulusoy, 2016; Sakao et al., 2020). This argument, coupled with the claim that the level of systems perspective to be applied tends to rely on the scale or the degree of social embeddedness of the design problem (based on Simon (1969)), led to the formulation of the following prediction in Paper 1:

P4 : Conceptual PSS designing expends more cognitive effort on the *highest level of systems* abstraction than product designing.

Here, conceptual product and PSS designing, refers to the designing specific to the product and PSS design tasks addressed by the cohort. These design tasks are expected to be representative of the respective design domains, based on the defined constraints of conditions A and B respectively.

5.2.3 Empirical findings - Paper 1

In Paper 1, the research approach described in Section 4.3 is followed to investigate the differences between conceptual product and PSS design cognition. Statistical analysis of the distribution of cognitive effort on the different FBS ontology-based design issues revealed that PSS designing expended significantly more cognitive effort on the design issues Function and Design description, while product designing expended more effort on Structure and Requirements. The average distribution of cognitive effort on the different FBS ontology-based design issues for both product and PSS design sessions is illustrated in Figure 10.

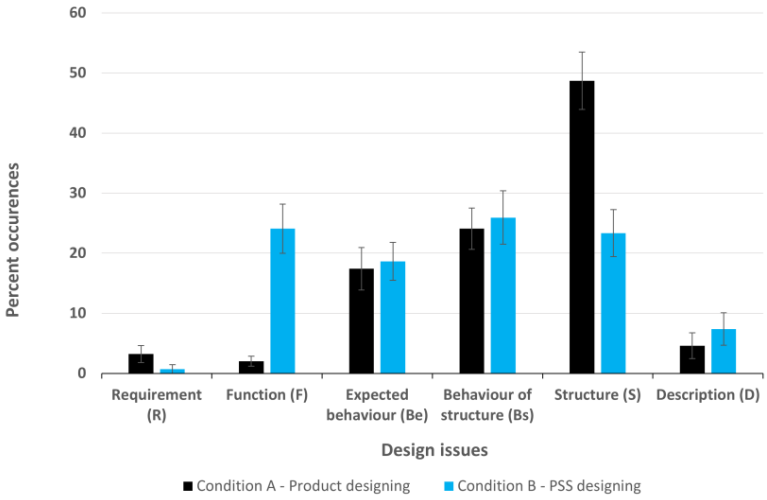


Figure 10. Mean values of occurrences of FBS design issues in percentage between product designing (Condition A, N=18 pairs) and PSS designing (Condition B, N=10 pairs). Note: The error bars indicate the standard deviations of the mean values. Adopted from Paper 1 (Neramballi et al., n.d.-b).

The distribution of cognitive effort on the different FBS ontology-based design processes for both product and PSS design sessions is illustrated in Figure 11.

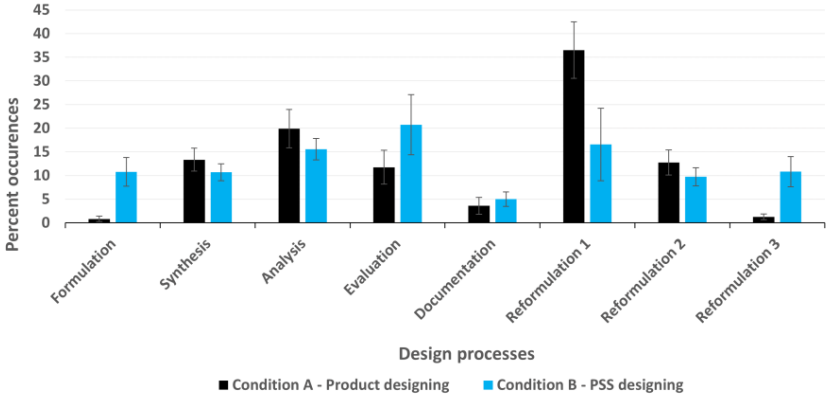


Figure 11. Mean values of occurrences of FBS design processes in percentage between product designing (Condition A, N=18 pairs) and PSS designing (Condition B, N=10 pairs). Note: The error bars indicate the standard deviations of the mean values. Adopted from Paper 1 (Neramballi et al., n.d.-b).

The findings of the statistical analysis of the distribution of cognitive effort on the different FBS ontology-based syntactic design processes suggest that PSS designing expended statistically more cognitive effort on Formulation, Evaluation and Reformulation 3. At the same time, product designing expended more cognitive effort on Reformulation 1, Analysis, Synthesis and Reformulation 2. The results further suggest that the values of both P-S design issue and design process indices for PSS designing were higher than product designing with statistically significant differences. This suggests that during PSS designing, designers applied a more problem-focused approach. These findings suggest that the three predictions (P1, P2 and P3) are corroborated and thus, the linked hypotheses of Proposition 1 (H1, H2 and H3) of CAM-D are also corroborated (Section 3.3.2).

The Markov transitions of both intra- and inter-hierarchical systems' level of abstraction for product and PSS designing are illustrated in Figure 12.

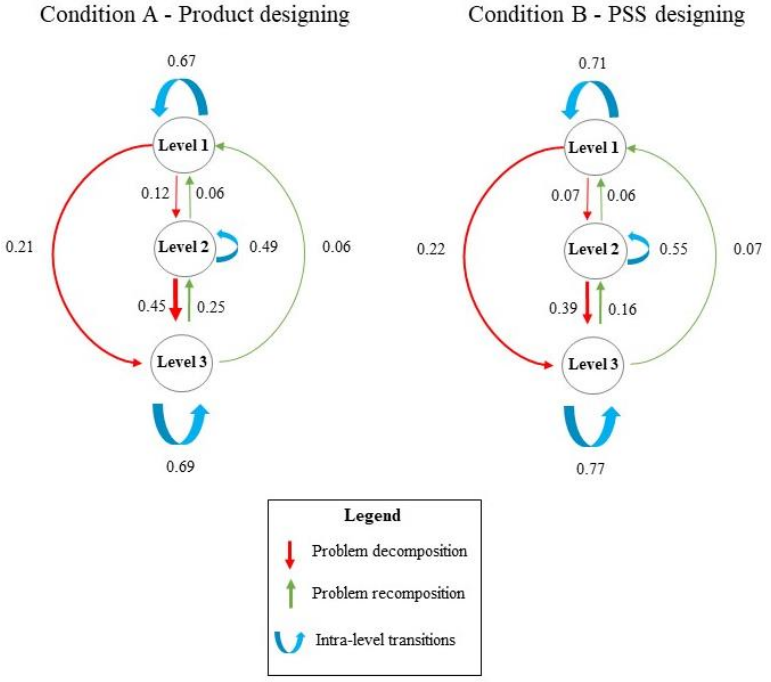


Figure 12. Mean values of Markov transitions within (intra-level) and between (inter-level) different hierarchical levels of systems abstraction during product designing (Condition A, N=18 pairs) and PSS designing (Condition B, N=10 pairs). Note: The thickness of the arrows representing inter- (problem decomposition and re-composition) and intra-level transitions are relative to the respective mean values of Markov transitions. Adopted from Paper 1 (Neramballi et al., n.d.-b).

The findings of the statistical analysis of the distribution of cognitive effort on the different levels of systems abstraction revealed no statistically significant differences in the focus expended on the highest and lowest levels of systems abstraction by product and PSS designing. Thus, the fourth prediction and the linked fourth hypothesis of Proposition 1 of CAM-D were rejected. Incidentally, the results revealed statistically significant differences in terms of the cognitive effort expended on the sub-systems level of abstraction and on the problem decomposition and re-composition to the sub-systems level, with product designing expending higher focus than PSS designing.

5.3 Design approaches to address product-service system design problem – Paper 2

5.3.1 Overview - Paper 2

Paper 2 conducts an in-depth protocol study of how practitioners experienced in typical product development-related domains address a typical PSS design problem (see Section 4.2.2 and 4.3 for more details). The findings of this study also contribute to answering RQ1 of this thesis, by providing an in-depth view into domain-specific design strategies applied during PSS designing. These results were obtained after analysing the same PSS design sessions performed by the same sub-cohort, as the one described in Paper 1, a PSS design domain-specific framework, termed PSS architecture, in addition to the two frameworks used in Paper 1. The model to represent PSS architecture is described in the following section.

5.3.2 A model to represent product-service system architecture

From an engineering perspective, three of the most basic categories under which the highly varied dimensions of a PSS can be categorized are the tangible product elements, the intangible service elements and the interactions between such elements from a systems perspective (Meier et al., 2010; Tischner et al., 2002; Tukker, 2004). Based on this categorization, a model of the PSS design object was proposed in Paper 2. The key design aspects (characteristically distinct design aspects, the values of which the designers can manipulate) of the PSS are: 1) product aspects (prescribed the code P), 2) service aspects (prescribed the code S), 3) interactions between product and service aspects (prescribed the code I), and 4) the aspects concerning the overall PSS (prescribed the code PSS). This framework is visualized in Figure 13.

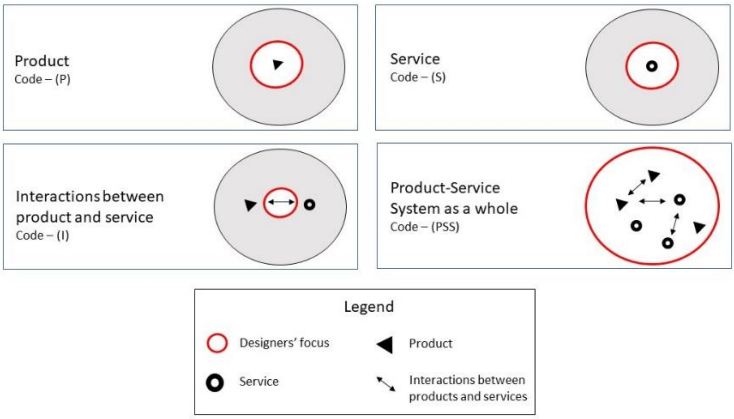


Figure 13. Model to represent product-service system architecture (based on Neramballi et al. (2022)).

5.3.3 Empirical findings - Paper 2

The results of the measurement of the distribution of cognitive effort on the different dimensions of the PSS architecture are presented in Table 3. These results suggest that designers expend most of their focus on the product aspects while expending significantly less on the service aspects, interactions between the two types of aspects, and aspects concerning the overall PSS. This finding potentially points towards product-centric strategies applied by the designers while addressing the PSS design problem.

Table 3. Mean values and respective standard deviations of the distribution of cognitive effort on aspects of products (P), services (S), interactions (I) between P and S, and product-service system (PSS) from 10 PSS design sessions, expressed as percentages, along with their coefficients of variation. Retrieved from Paper 2 (Neramballi et al., 2022).

Aspect	Mean	Standard deviation	Coefficient of variation
P	64.00	6.41	0.10
S	8.70	4.03	0.46
I	6.70	1.57	0.23
PSS	20.50	4.55	0.22

Note: P - Product, S - Service, I - Interactions between P & S

PSS - Overall system

5.4 Discussion - Characterizing product-service system designing

Based on the empirical findings of DS-1 comparing product and PSS designing (Paper 1) it can be inferred that due to the differences in the characteristics of the two domain specific design problems, cognitive processes underlying the

respective domain specific design processes (and thus, the respective design knowledge schemas being constructed or automated) vary significantly. These offer preliminary support for the first three hypotheses of Proposition 1 of CAM-D. These characteristic differences in terms of differing patterns of distribution of cognitive effort on the design cognition variables of the design process are summarized in Table 4.

Table 4. Comparison of cognitive characteristics of product designing (Condition A, N=18 pairs) and PSS designing (Condition B, N=10 pairs) in terms of patterns of information processing within the design micro-frame.

Design micro-frame variable	**Product design – focuses on**	**PSS design – focuses on**
Problem-solution design space	Solution space	Problem space
States and transformation processes of design information	Structure - Analysis - Synthesis - Reformulation 1 - Reformulation 2	Function - Formulation - Evaluation - Reformulation 3

First, differences in the patterns of distribution of cognitive effort on the variable problem-solution design space, between product designing and PSS designing are discussed. PSS designing expended relatively more cognitive effort on the problem space than product designing. Next, the differences in the patterns of distribution of cognitive effort on the variables design state issues and transformation processes, between product designing and PSS designing are discussed. PSS designing expended relatively higher effort on the design state issue Function and the design state transformation processes Formulation, Evaluation and Reformulation 3 compared to product designing. It is inductively inferred that these observations could potentially be linked to the relatively higher degree of intangible nature of the PSS design object compared to the product design problem (based on the augmented generic-design hypothesis and related discussions by Visser (2009)).

The cohorts of the empirical study comparing product and PSS designing (Paper 1) included experienced product developers with relatively limited or no experience in PSS designing. The empirical findings presented in this comparative study (Paper 1) revealed that the designers applied similar levels of cognitive effort on the highest (and lowest) level of systems abstraction during both product designing and PSS designing (Section 5.2.3). From this finding, it can be suggested that the focus on the highest level of systems abstraction may not necessarily depend on the characteristics of the design problem. Thus, it is abductively inferred that the focus on the highest level of systems abstraction may

have been influenced by the pre-existing knowledge of the designer (in line with the alternative viewpoint of (Goel & Pirolli, 1992; Visser, 2006, 2009)).

Furthermore, the distribution of cognitive effort on problem decomposition and re-composition to the sub-systems level and also on the sub-systems level of abstraction, was reported to be statistically lower in PSS designing than in product designing. This observed strategy of more dominant problem decomposition to the sub-systems level exhibited by the cohort performing product designing is consistent with the breadth-first approach exhibited by domain experts, as opposed to the depth-first approach (decomposing more frequently to the element level) typically exhibited by domain-novices (Cross, 2006; Ho, 2001; Song et al., 2016). Since both cohorts have extensive background experience in domains associated with product development and relatively limited experience in PSS designing, the observed differences in search or decomposition strategies could be abductively linked to the level of expertise within the respective domain-specific problems.

The findings presented in Section 5.3.3 (Paper 2) provided a closer view of the distribution of designers' cognitive effort on the different aspects of the PSS domain-specific design approaches, revealing that PSS designing expended majority of the effort on product aspects, while underemphasizing service aspects, interactions between the sub-systems or elements and the overall PSS. This over-emphasis on the product aspects within the element level and the under-emphasis on the other aspects of the PSS architecture could again be abductively linked to the lack of PSS design domain-specific knowledge of the cohort and the influence of the available product design knowledge.

Based on the above described characteristics of PSS designing in relation to product designing by the studied cohort of experienced product developers, it can be inferred that PSS designing may indeed be a deviation in terms of the routine practice of designing for the cohort. Daalhuizen (2014) suggests that prescriptive design support can potentially function as useful resources for effectively addressing such non-routine or unfamiliar design problems. More specifically, it is suggested that such unfamiliar design problems may introduce epistemic uncertainty (Ball et al., 2010; Daalhuizen, 2014). Such subjective uncertainty can restrict the ability of the designer to retrieve appropriate knowledge to fully scope and interpret the problem, and also to take appropriate actions required to effectively and efficiently reach the design goals (Daalhuizen, 2014). Daalhuizen (2014) further asserts that dedicated design support can function as mental tools to effectively scope such problems, and to identify, prioritize and structure possible courses of action. Therefore, it is inferred that experienced product developers may need such PSS domain-specific design support to effectively design PSSs.

Chapter 6 – Supporting product-service-system designing

6.1 Overview

The findings and discussions presented in this chapter answer RQ2: *How could conceptual product-service system designing be supported based on an understanding of its cognitive characteristics?* (Section 1.4). Highlights of three prescriptive design support catered to the cognitive characteristics of PSS designing that were developed and documented in appended Papers 3, 4 and 5 are presented. The support developed in Paper 3 is a design process support (based on Gericke et al. (2016)) that is expected to provide PSS domain-specific procedural guidance for the designers. The support developed in Papers 3 and 4 are LCA integrated design support methods (based on Gericke et al. (2016)) that are expected to provide contextual informational support for the designers.

6.2 Design process support for product-service-system designing – Paper 3

6.2.1 Overview of product-service-system design schema - Paper 3

Paper 3 aimed to develop a PSS domain specific design process support. The findings of DS-1 suggested that the cognitive characteristics of PSS designing is significantly different from that of product designing (Section 5.4). Furthermore, it was abductively inferred from the findings of DS-1 that while addressing the PSS design problem, the experienced product developers were mainly focusing on product aspects, underemphasizing the aspects concerning service, interactions between products and services, and the overall PSS. Based on these findings, it is inferred that designers without PSS design expertise may need PSS domain-specific design support to effectively structure and apply the literature prescribed strategies of PSS design (see Section 2.3.2).

It is suggested that the cognitive processes of novices can be significantly supported by introducing procedural design frameworks (partly based on Curry (2014) and Daalhuizen (2014)). The PSS design schema was developed as a part of PS (Paper 3), to support the cognitive processes of designers without PSS domain-expertise. It is a meso-level prescriptive procedural design process model of the conceptual PSS design process developed by integrating the steps prescribed by PBSA (Section 2.9) with the essence of prescriptive PSS design literature. Paper 3 initially carried out a meta-analysis and synthesis of reviews (Section 4.2.1) of prescriptive PSS design literature to distil the essence of the best practices of PSS designing. The distilled essence of prescriptive PSS designing includes the following key facets: 1) functionality-oriented designing, 2) identification of all relevant actors along the PSS value chain, 3) value

proposition, 4) development and integration of system elements (including products and services) and 5) examination of the balance of integration (for more details, see Paper 3 or Section 2.3.2). Subsequently, this essence of PSS designing was poured into the mould of PBSA for conceptual engineering designing (Pahl & Beitz, 2007) in order to obtain a consolidated PSS design schema. This model includes a series of ten iterative steps and is depicted in Figure 14.

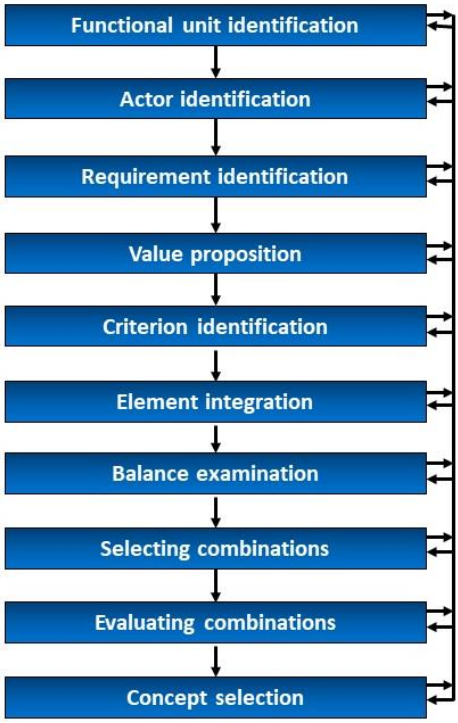


Figure 14. Visual representation of the PSS design schema. Adapted from Paper 3 (Sakao and Neramballi, 2020).

6.3 Design support methods for product-service-system designing

6.3.1 Overview – Papers 4 and 5

Based on the improved understanding of the cognitive characteristics of PSS designing (Section 5.4), it can be inferred that designers need more support to effectively address the PSS design problem space, in order to design environmentally benign design outcomes efficiently and effectively. While the PSS design schema provides procedural support on a higher level to structure and guide

the conceptual PSS design process, more granular informational design support methods for the specific steps within the design process prescribed by the schema are necessary to reduce the uncertainty surrounding the novel design problem. More specifically, literature suggests that during PSS designing, the designers need to be supported for 1) the process of requirement identification in terms of measuring environmental impacts, prioritizing, and defining environmental sustainability criteria for PSS design and 2) the process of element development, integration and examination of the balance and evaluation of the solutions in order to develop environmentally sustainable alternatives (see Sections 2.3.2 and 2.11). Below, highlights of two LCA-based micro-level prescriptive design support methods, (Section 2.8) developed as a part of PS (Papers 4 and 5), that can offer support for the above prescribed processes are presented. These two design support methods are lifecycle function deployment (LFD) and procedure for systemic lifecycle engineering, documented in Papers 4 and 5, respectively. On a higher level, these design support methods are expected to provide informational support for the development of environmentally sustainable design outcomes.

6.3.2 Overview of lifecycle function deployment – Paper 4

In Paper 4, principles of ADR were implicitly applied in a case study of a product manufacturing company (see Sub-section 4.2.3 and Section 4.3 for more details). The company was actively seeking to transition from designing typical product offerings towards designing PSS-based environmentally sustainable solutions. Also, it was uncovered that the company needed practical support to measure the environmental impacts of their existing product offerings.

Furthermore, through a series of workshops with the cohort of the ADR, it was uncovered that the practitioners in the company needed support to utilize such environmental information to redesign their product offerings efficiently and effectively to PSS-based solutions with relatively lesser environmental impacts. Subsequently a generic and transparent design support termed LFD was developed by catering to the company's stated requirements and the knowledge gap uncovered in the prescriptive PSS design literature (Section 2.11). LFD is an LCA integrated QFD-based PSS design method that is expected to support designers in translating the LCA results of existing products into actionable design insights that can guide the development of PSS-based solutions with relatively lesser environmental impacts. The overall flow of the different steps of LFD is depicted in Figure 15.

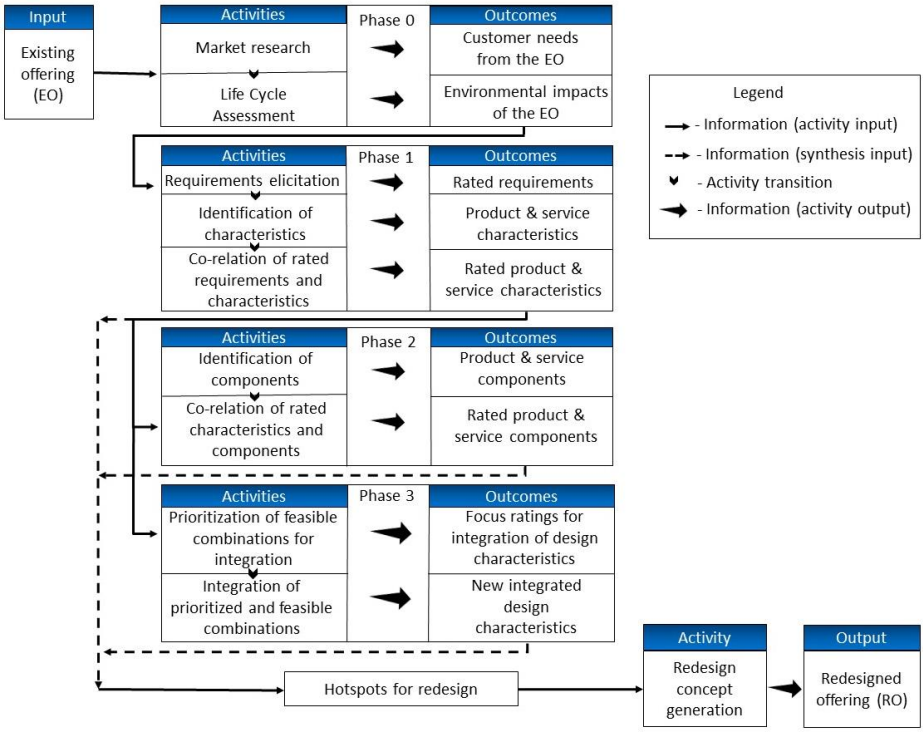


Figure 15. Visual representation of the flow of different steps of lifecycle function deployment (LFD). Adapted from Paper 4 (Neramballi et al., 2020).

LFD has a series of steps that are categorized into four phases. Phase 0 entails gathering information about customer needs through typical market research approaches and the environmental impacts of the offering being designed or redesigned through LCA. This information is subsequently channelled into Phase 1, which includes 1.1) a novel systematic procedure to translate the outcomes of LCA and market research into quantitatively normalized and prioritized design requirements (linked to the PSS design schema step *requirement identification*); 1.2) formulation of product and service design characteristics (linked to the PSS design schema step *element development*); and 1.3) a QFD for PSS-based procedure to translate the prioritized design requirements into prioritized product-service characteristics that designers need to focus their design efforts on. The outcomes of Phase 1 flow into Phase 2, which includes the following step: 2.1) translation of prioritized product-service characteristics into prioritized product-service components for design or re-design (linked to the PSS design schema step *element development*).

Phase 3 utilizes the outcomes of Phase 1 as well, as it helps in the consideration of interactions and the prioritization of the optimization of exchangeable product-service design characteristics. Such an optimization of the exchangeable characteristics might have a greater influence on the prioritized design requirements, than only introducing changes to their discrete parts (linked to the PSS design schema step *examination of balance of element combinations*). The overall outcomes of LFD are a list of hotspots of prioritized design parameters for designing PSS solutions with comparatively lesser environmental impacts.

6.3.3 Overview of the procedure for systemic lifecycle engineering – Paper 5

In Paper 5, principles of ADR were also implicitly applied in a different case study of a different product manufacturing company (see Sub-section 4.2.3 and Section 4.3 for more details). The company was seeking to optimize product-service interactions and performance from a systems and lifecycle perspective. One way of improving the lifecycle performance of product systems is through the identification of interdependencies between product-service sub-systems or elements and integrating them in the conceptual design stage, so that they can contribute to the systemic reduction of environmental impacts throughout the lifecycle (Section 2.3.2).

Similar to LFD, this procedure is expected to address multiple steps prescribed by the PSS design schema such as requirement identification, integration, and examination of balance of elements and evaluation of combinations of integrated elements. More specifically, this procedure aims to identify systemic design opportunities that can be used to improve environmental lifecycle performance by utilizing a knowledge base of product-service interdependencies documented from several other industrial cases of PSS design. This procedure entails a series of iterative steps.

The first step of the procedure entails performing an LCA of the current offering. Subsequently, in the second step, the LCA results and the product (P) – service (S) interdependency knowledge base is used to derive critical design parameters (DPs) of the overall PSS offering (linked to the PSS design schema step *requirement identification*). Subsequently in Step 3, with the aid of the P-S interdependency knowledge base, designers are supported to identify latent interdependencies between product and service DPs that are relevant to the identified PSS DPs from the perspective of the LCA results. In Step 4, the knowledge base is used to integrate the identified interdependent P-S DPs. This is done by supporting designers to analogically compare similar instances of P-S interdependencies documented in other cases of PSS design, and thus, to generate systemic design improvements for the current offering (linked to the PSS design schema step *element integration*). Finally, in Step 5, the conceptual design

improvements are evaluated with an LCA and compared with the LCA results of the current offering (linked to the PSS design schema step *evaluation of element combinations*). Based on the evaluation, if needed, the procedure is repeated iteratively. The overall flow of the procedure is visualized in Figure 16.

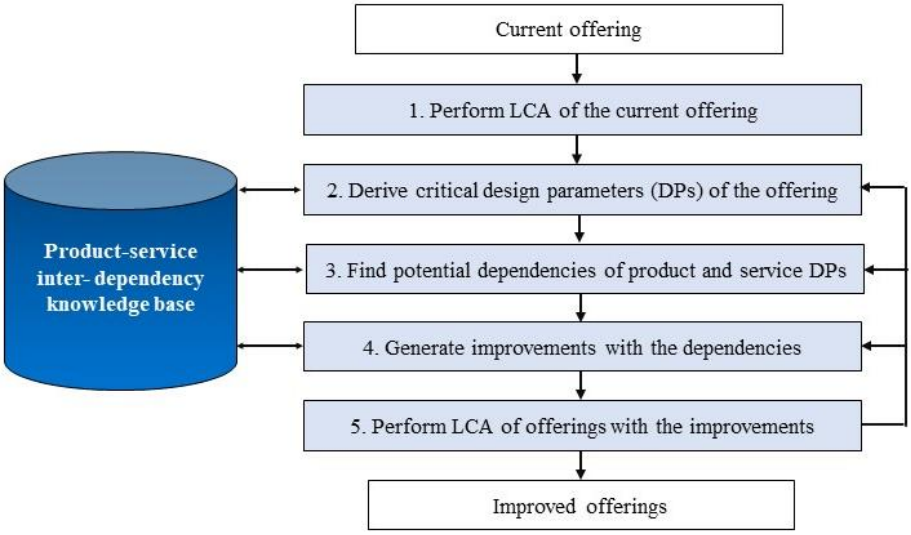


Figure 16. Visual representation of the procedure for systemic lifecycle engineering. Adapted from Paper 5 (Sakao *et al.*, 2022).

Grounded on the principles of DbA (Section 2.8), the aim of the knowledge base is to first analogically support designers to filter PSS-level DPs that have a significant influence on the LCA results, then to identify latent interdependencies between P-S DPs that are linked to the filtered PSS DPs and to integrate them through the introduction of suitable design changes. Integration refers to the effective utilization of the dependencies by interweaving the assignment of values of the P-S DPs, that is, the value of product DP is adjusted based on the value of the service DP, vice-versa or both. The knowledge base is an outcome of an extensive review, meta-analysis, and synthesis (Section 4.2.1) of multi-sectoral PSS design cases documented in the literature, which have reported multiple instances of P-S interdependencies.

A set of generic PSS, product, and service DP classes were abstracted from the recurring DP instances to ensure the classes' broad transferability. A total of 4 PSS, 5 product and 7 service DP classes were abstracted from 43 instances of P-S integration identified across the multiple cases (for more details, see Paper 5). The different product, service and respective linked PSS DP classes are presented in the P-S interdependency matrix (Figure 17).

Service DP class \ Product DP class	Serviceability	Affordability	Information transferability	Upgradeability	Temporal functionality
Labour intensity	**PbS** *Resource use*		**PbS** *Outcome / Affordability*		
Capability	**SuP** *Availability / Resource use*		**SuP** *Availability*	**SuP** *Availability*	**PuS** *Availability*
Responsiveness	**SuP** *Availability*				
Information transferability	**PuS** *Availability / Outcome / Resource use*		**PbS** *Availability / Outcome*	**PuS** *Availability / Outcome*	**PuS** *Outcome*
Affordability	**SuP** *Affordability*	**PbS** *Affordability*			
Frequency					**PbS** *Availability / Resource use*
Reliability					**PuS** *Outcome*

Figure 17. Product (P) – service (S) interdependency matrix. Adopted from Paper 5 (Sakao *et al.*, 2022). © Reused under CC-BY License.

In Figure 17, the intersecting cells in the matrix represent the linked PSS DP class and also the type of inter-dependency. The types of inter-dependencies between the abstracted product and service DP classes are broadly categorized under three archetypes: PuS – product DP is unilaterally dependent on service DP, SuP – service DP is unilaterally dependent on product DP and PbS – product and service DPs are bi-laterally dependent on each other (Figure 17). The null-grey cells suggest no inter-dependencies were identified between the respective P-S DP classes on the instance level. This DbA procedure and the knowledge base were operationalised with an excel based tool that is expected to support designers to efficiently and effectively integrate the important product-service DP classes to generate systemic lifecycle improvements.

Chapter 7 – Effects of design support for product-service-system designing

7.1 Overview

The empirical findings and discussions presented in this chapter answer RQ3: *How could domain-specific prescriptive support influence cognitive characteristics of conceptual product-service system designing and the environmental impacts of the resulting design outcomes?* (Section 1.4). Based on the improved understanding of the cognitive characteristics of PSS designing (Chapter 5), three prescriptive PSS domain-specific design supports were developed (Chapter 6). Initially, highlights of the empirical findings documented in Paper 6 concerning the effects of the PSS design schema on the cognitive characteristics of PSS designing are presented. Subsequently, highlights of the case applications of LFD documented in Paper 4 and of the procedure for systemic lifecycle engineering documented in Paper 5 and the empirical findings from the assessment of their effects are presented.

7.2 Effects of product-service-system design schema – Paper 6

7.2.1 Overview - Paper 6

Paper 6 aimed to explore the effects of introducing the PSS design schema (developed in Paper 3) on the cognitive characteristics of conceptual PSS designing. In order to answer RQ3, an exploratory protocol study of conceptual PSS designing with (experiment group) and without (control group) the support of the PSS design schema, was carried out in DS-2 and documented in the working Paper 6 (see Section 4.3 for more details). In this study, the following four frames of analysis were used: the framework based on FBS ontology (Section 2.6.3), the HSA framework (Section 2.6.4), a framework to represent the different lifecycle stages of solutions being designed (Section 2.7.2) and the framework to represent PSS architecture (Section 5.3.2). The preliminary results of the effects of the PSS design schema are presented as follows.

7.2.2 Empirical findings - Paper 6

Note that the results presented in this section are preliminary and statistical data analysis with a larger cohort size will be carried out in a future work. The results concerning the distribution of cognitive effort on FBS design issues by the control and experiment groups (N=3 pairs each) are visualized in Figure 18.

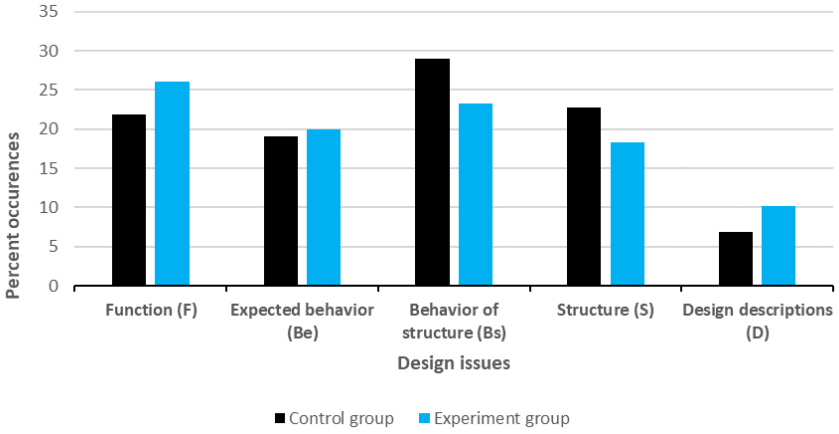


Figure 18. Mean values of occurrences of FBS design issues in percentage between the control group and experiment group (N = 3 pairs each). Adopted from Paper 6 (Neramballi et al., n.d.-a).

The preliminary results suggest that there were numerical differences in the distribution of cognitive effort on the FBS design issues between the control and experiment groups. While the control group expended the highest cognitive effort on the design issue Behaviour of Structure (M=28.97, SD=3.69), the experiment group expended the highest cognitive effort on Function (M=26, SD=0.57). Since the percent occurrences of Requirement was minimal for both groups, it was not visualized in Figure 18. The P-S issue index for the control group was numerically lower (M=0.80) than that of the experiment group (M=1.17).

The results of the distribution of cognitive effort on the different FBS design processes by the control and experiment groups (N=3 pairs each) are visualized in the following Figure 19. These preliminary findings of the percent occurrences of the FBS design processes suggest that both the control (M=23.30, SD=11.79) and experiment (M=24.83, SD=4.18) groups expended the highest cognitive effort on Evaluation. The differences in the distribution of cognitive effort on the design processes were marginal, with the experiment group expending slightly more cognitive effort on Formulation, Reformulation 1, and Evaluation, while the effort on Synthesis, Analysis and Reformulation 2 seems to have reduced. The P-S process index for the experiment group (M=0.50) was marginally higher than the control group (M=0.44).

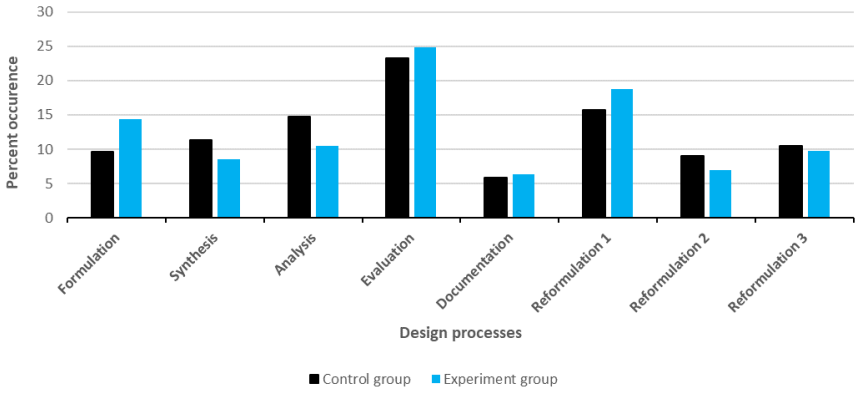


Figure 19. Mean values of occurrences of FBS design processes in percentage between the control and experiment groups (N = 3 pairs each).

The results of the distribution of cognitive effort on different levels of systems abstraction by the control and experiment groups (N=3 pairs each) are presented in Figure 20.

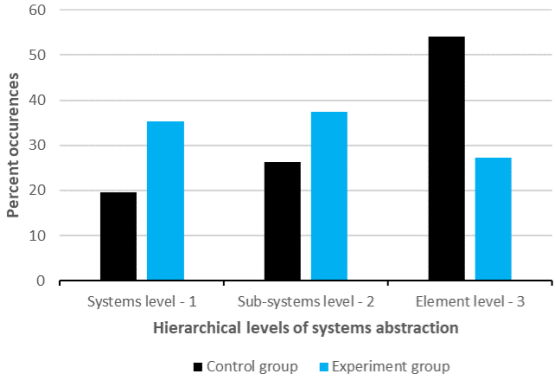


Figure 20. Mean values of occurrences of HSA between the control and experiment groups (N = 3 pairs each). Adopted from Paper 6 (Neramballi et al., n.d.-a).

The results presented in Figure 20 suggest that in the experiment group there is a significant increase in the systems and sub-systems levels of abstraction and a reduction in the focus on the element level. The empirical findings concerning the distribution of cognitive effort in the different lifecycle stages by the control and experiment groups (N=3 pairs each) are illustrated in Figure 21.

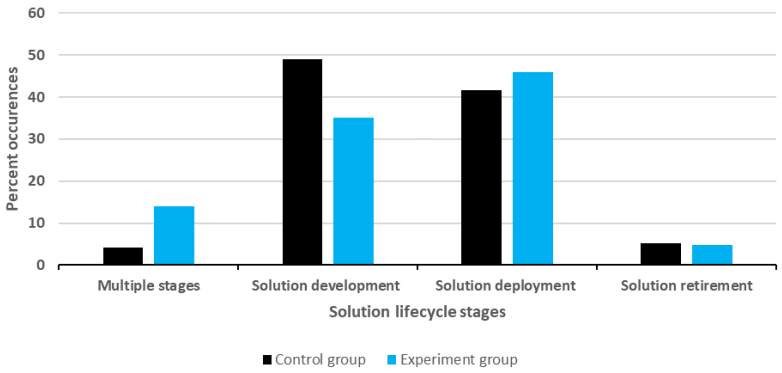


Figure 21. Mean values of occurrences of solution lifecycle stages in percentage between the control and experiment groups (N = 3 pairs each). Adopted from Paper 6 (Neramballi et al., n.d.-a).

The results indicate that the experiment group expended the highest effort on the lifecycle stage Solution Deployment (M=46; SD=4.56), while the control group expended the highest effort on the lifecycle stage Solution Development (M=48.95, SD=6.24). Furthermore, the cognitive effort on Multiple Lifecycle Stages has increased in the experiment group (M=14; SD=4.79) compared to the control group (M=4; SD=2.02). Also, in Paper 6, Markov models of the transition probabilities within and across the different solution lifecycle stages were developed for both the control and experiment groups (N=3 pairs each) and are presented in Figure 22.

Comparing the two Markov models, the results suggest that in the experiment group, probabilities of transitions within the lifecycle stages Solution Retirement (RET) and Multiple Stages (MUL) have relatively increased, while that of Solution Development (DEV) and Solution Deployment (DEP) have relatively decreased in comparison to the control group. While the transition probabilities from MUL to DEP, DEV to DEP, DEV to MUL, DEP to MUL and RET to MUL have relatively increased in the experiment group, probabilities of transitions from MUL to DEV, RET to DEV and RET to DEP have comparatively decreased.

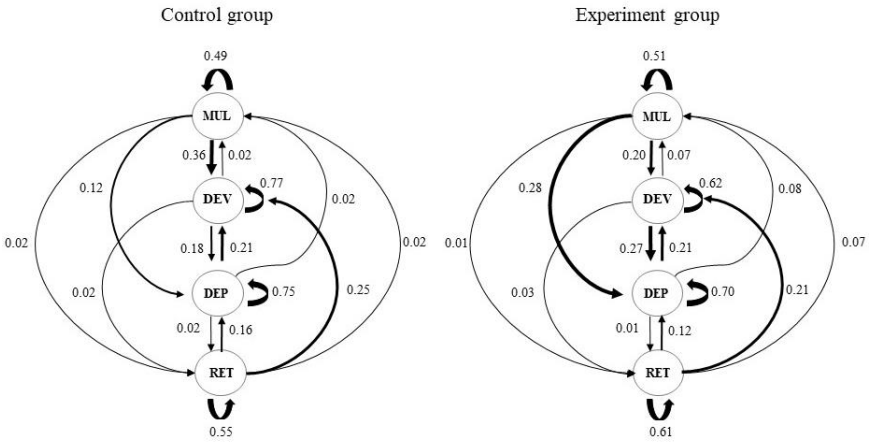


Figure 22. Markov transitions of cognitive effort within and between different solution lifecycle stages for the control and experiment groups (N=3 pairs each). Note: the thickness of the lines is proportional to the respective transition probability. Adopted from Paper 6 (Neramballi et al., n.d.-a).

The results of the comparison of the distribution of cognitive effort on the different aspects of the PSS architecture between the control and experiment groups (N=3 pairs each) are visualized in Figure 23.

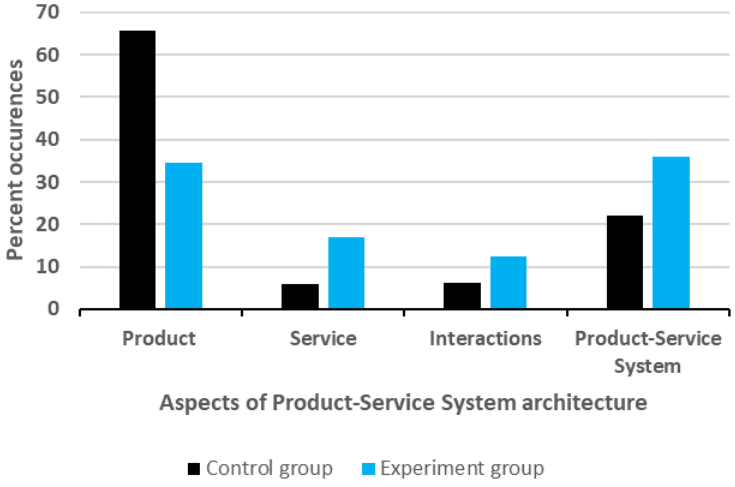


Figure 23. Mean values of percentage occurrence of aspects concerning product, service, and product-service system (N=3 pairs each). Adopted from Paper 6 (Neramballi et al., n.d.-a).

The results suggest that there is a starker increase in the cognitive effort on service aspects (M=16.89, SD=5.53) in the experiment group than in the control group (M=6.01, SD=2.73). In contrast, there is a significant drop in the effort on the product aspects in the experiment group (M=34.6, SD=2.92) compared to significantly higher effort observed in the control group (M=65.65, SD= 6.92). Also, the cognitive effort on the interactions between product and service aspects has doubled in the experiment group compared to the control group. The focus on PSS aspects has also numerically increased in the experiment group.

7.2.3 Discussion - Paper 6

The results indicate that the introduction of the PSS design schema may have contributed to a higher focus on the problem space and problem space-based design states and transformation processes (Figures 18 and 19, respectively). Furthermore, the results suggest that the experiment group may have applied a comparatively higher focus on the systems and sub-systems levels of abstraction (Figure 20). The findings also suggest that the experiment group have applied a wider lifecycle perspective due to the following reasons: 1) the experiment group reported a higher focus on multiple lifecycle stages and the solution deployment phase, while the control group mainly focused on the solution development (Figure 21); 2) the probability of transitions from the multiple stages and solution development lifecycle stages to the solution deployment stages are higher in the experiment group (Figure 22). Finally, there is an increased focus on the aspects concerning the service, interactions between product and service aspects, and the overall PSS in the experiment group, as opposed to the product-centric focus exhibited by the control group (Figure 23).

The results presented in this section address a criticism of the state of the art pointed out in several reviews that, prescriptive support for PSS designing is not sufficiently validated (Brissaud et al., 2022; Tukker, 2015; Vasantha et al., 2012). One example of an investigation of external support on conceptual PSS designing using protocol analysis is that of Bertoni (2013), which investigated the effects of the introduction of a CAD-based informational support on conceptual PSS designing carried out by student designers, but not of a prescriptive design process support as in this current work. While the study by Bertoni investigated the effects of different kinds of informational support on the focus spent on different types of problem-solving activities and PSS-related aspects of student designers, Paper 6 investigated the effects of a PSS domain-specific design process support on the design cognition of experienced practitioners.

Although there are several prescriptive procedural models of PSS design in the literature (Manzini & Vezzoli, 2003; Maxwell et al., 2006; Morelli, 2003), they present case-specific fragmented approaches of PSS designing. In order to consolidate the highly varied perspectives across different prescriptions,

Clayton, Backhouse and Dani (2012) synthesized the common design phases within the various approaches to PSS development. But that synthesized model is not on the meso- level, but rather it is on the macro-level (Section 2.4) and thus may not be suitable to guide the conceptual design activities. In contrast, the PSS design schema is a meso-level procedural design process support for conceptual PSS designing.

7.3 Effects of lifecycle function deployment - Paper 4

7.3.1 Overview of the case application - Paper 4

The case company who participated in the ADR documented in Paper 4, is actively seeking to transition from designing and offering consumable incontinence products to relatively more resource-efficient and environmentally benign systemic incontinence solutions (Section 6.3.2). The primary target users of the incontinence products are elderly residents of nursing homes. During the case application of LFD, outcomes of LCA and market research of their existing offering were used as input information by using the retrospective technique of document analysis (Section 4.3.3). Following the prescribed procedure of LFD, a list of hotspots of products and service DPs (in terms of characteristics and components) was narrowed down from a potentially large design space for the designers to focus their design efforts on. Based on this outcome, concepts for PSS-based solutions were derived and LCA was carried out.

7.3.2 Empirical findings and discussion - Paper 4

The concepts of PSS solutions generated using LFD were assessed with an LCA. More specifically, an LCA comparison of the existing product offering, and concepts of PSS offerings was carried out; the results are presented in Figure 24.

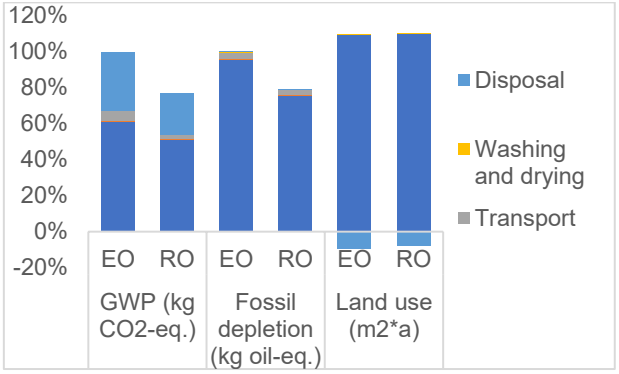


Figure 24. LCA comparison of existing offering (EO) and PSS-based redesigned offering (RO). Adapted from Paper 4 (Neramballi et al., 2020).

This comparison reveals that the environmental impacts of the redesigned PSS-based solutions obtained with the support of LFD are relatively less, compared to the existing product-centric offerings in terms of global warming potential (GWP) and fossil resource depletion. More specifically, the systemic design changes may have contributed to the reduction in the GWP and fossil resource depletion. These outcomes can be linked to the validation of support for PSS design schema steps of *requirement identification, development,* and *examination of the balance of PSS elements* and *the evaluation* of the environmental impacts *of the solutions*.

There are a few previous works that have demonstrated the exploitability of LCA (Amaya et al., 2014; Lelah et al., 2011) and LCA-integrated QFD (Fargnoli et al., 2018) to support PSS design with an environmental sustainability focus. However, the process of translating LCA results into actionable insights for guiding the conceptual design of PSSs has often been case-specific and non-generic. The novel procedure presented in LFD, to translate results of LCA, and market research coupled with the QFD for PSS based procedure, have contributed to a generic and transparent way to obtain actionable insights for conceptual PSS designing for environmental sustainability. Furthermore, the novel LCA-integrated procedure to investigate and leverage the exchangeability of product-service functionalities has offered a generic and systematic way to operationalize the equifinality property of PSSs for environmental sustainability (see Section 2.3.2 for more details).

7.4 Effects of systemic lifecycle engineering - Paper 5

7.4.1 Overview of the case application - Paper 5

The procedure for systemic lifecycle engineering was developed in Paper 5 using ADR methodology to address the practice-based requirements of another product manufacturing case company (Section 6.3.3). The case company who participated in the ADR in Paper 5 is seeking to reduce the environmental impacts and improve the lifecycle performance of its various offerings of vertical access equipment (VAE) and related services. This company was located in Spain and the existing offering which was the focus of the redesign of the study in Paper 5, was a vertical access equipment or service elevator for wind towers, also located in Spain.

Following the prescribed steps of the procedure (Section 6.3.3), the outcomes of an LCA of the existing offering performed in a previously linked research project were analysed using the retrospective technique of document analysis. The LCA results revealed that the major source of environmental impacts in terms of emissions (CO_2-eq) was linked to the use phase or more specifically, the maintenance activities of the VAE which contributed to around

82% of the total emissions. The assumed reason behind this LCA result is that during the maintenance of the VAE, the power-generating turbines within the wind tower must be shut down (due to safety regulations). Therefore, the stoppage in wind-based electricity production is compensated for by the local average energy mix of Spain, the country in where the wind towers are located.

Using this outcome of LCA as an input to the second step of the procedure prescribed by the design support, the critical PSS DP 'availability', which is directly linked to the LCA result, was shortlisted in a virtual focus group workshop with the study cohort (Section 4.3.3). This led to the filtering of all potential integrated P-S DP classes that together could affect the PSS DP class availability (Figure 17). Then, using the knowledge base, the potential interdependencies on the instance level of the shortlisted P-S DP classes documented in various cases of PSS design were analysed and analogically used to check the potential transferability of such interdependencies or integration possibilities to the current design case on the instance level. This led to the further filtering of P-S DP classes (3 combinations) and the respective instances (7 combinations out of 105 possible combinations) specific to the current case that can potentially influence availability of the PSS and thus, contribute to reduction of emissions via integration.

By analogically referring to the relevant cases documented in the knowledge base linked to the shortlisted P-S DP classes and respective instances, conceptual design changes to integrate those specific DP instances were made. These results were analysed using an LCA (see Paper 5 for more details).

7.4.2 Empirical findings and discussion - Paper 5

The LCA of the conceptual design changes developed using the design support method revealed that the redesigned PSS-based offering could potentially have 51% lesser emissions (CO_2-eq) than the existing offering. Based on these results, it can be qualitatively inferred that the method was successful in offering informational support linking LCA results (linked to the PSS design schema step *requirement identification*) to the various DPs of the offering. Maussang et al. (2009) proposed a design support method for PSS designing to support designers to consider the interactions between products and services from a systems perspective. However, it does not provide lifecycle based informational support to reduce the environmental impacts of the PSS from a systemic perspective.

There are a few existing LCA integrated design support methods that offer lifecycle based support for the design of product systems (Bonou et al., 2015; Zwolinski & Tichkiewitch, 2019) and for the design of PSS (Fargnoli et al., 2018). However, such methods do not offer systemic lifecycle engineering support to consider cross-functional product-service interactions, to reduce environmental impacts. The novel procedure for systemic lifecycle engineering

proposed in Paper 6 offered informational support for the designers 1) to focus on examining the P-S inter-dependencies, 2) to integrate combinations of P-S sub-systems or elements, and 3) to evaluate the potential environmental impacts of the combinations from a systemic and lifecycle perspective. These contributions offer detailed informational support to the PSS design schema steps *integration, examination of balance* and *evaluation* of combinations of PSS elements with respect to reduction in environmental impacts.

Chapter 8 – Refinement and extension of the conceptual framework

8.1 Influence of pre-existing design knowledge

8.1.1 Discussion - findings of Papers 1 and 2

The empirical findings of DS-1 (Paper 1) offered preliminary support for the predictions and linked hypotheses (H1-H3) concerning the differences in the distribution of cognitive effort on the design process variables problem-solution spaces, design state issues, and processes linked to the differences in the structuredness of product and PSS design problems (Chapter 5). However, the prediction concerning the differences between conceptual product and PSS design cognition in terms of the distribution of cognitive effort on the highest level of systems abstraction was rejected, along with the respectively linked H4 of Proposition 1. Based on this finding, it was abductively inferred that (in Section 5.4) systems abstraction might be linked to the previous knowledge and experiences of the designer (in line with the views of Goel & Pirolli (1992) and Visser (2009)).

Designers with varying types and levels of expertise have been shown to apply different design abstraction strategies and problem decomposition and re-composition strategies (Section 2.6.4). More specifically, it is reported that the design strategies in terms of levels of abstraction or search strategies such as problem-solution decomposition and re-composition, applied by the designers are linked to their pre-existing design knowledge developed during previous domain-specific design experiences (Akin, 2001; Goel & Pirolli, 1992; Visser, 2009). The level of expertise is also expected to significantly influence such search strategies, as differences between experts and novices in terms of breadth-first or depth-first decomposition are reported in literature (Cross, 2006; Ho, 2001; Song et al., 2016).

Specifically, expert designers are more likely to engage in breadth-first decomposition and novices are more likely to engage in depth-first decomposition (see Section 2.6.4). Incidentally, in DS-1 (Paper 1), it is observed that product designing applied a statistically higher focus on the sub-systems level of abstraction and was more likely to decompose the problem to the sub-systems level (breadth-first decomposition), than the cohort that addressed the PSS design problem. Since both cohorts have extensive product domain-based experience and comparatively less PSS domain-based experience, it is abductively inferred that these differences could be attributed to a lack of PSS domain-specific knowledge (see Section 5.4).

The existing literature suggests that although designers may be able to transfer strategies across different domains, it does not necessarily mean that they can effectively switch practice between different domains as the superior performance of experts tends to be domain-specific (Cross, 2006, p.75). Similar observations concerning domain-specific expertise are also made in other types of problem solving activities (Sweller, 1988; Sweller et al., 1998; Van Merriënboer & Paas, 1990). Using research on design fixation as a point of departure, Crilly and Cardoso (2017) raise several pertinent questions concerning the effects of domain knowledge on fixation, mechanism and effects of fixation (or the effects of unconscious use of prior knowledge) with relation to information processing during designing, co-relations between cognitive economy (balance between efficiency and effectiveness) and the use of prior knowledge.

While addressing new problems, designers might be fixated on their previous experiences such as prior interpretations of problems, previously applied design strategies and developed solutions (Crilly, 2015). Such prior experiences or knowledge may result in the development of an unconscious bias that may force the designer to carry out a restricted exploration of the design space (Crilly & Cardoso, 2017). The findings of DS-1 (Paper 2) revealed that the cohort addressing the PSS design problem afforded the majority of their focus on the product aspects, while affording significantly less focus on aspects related to services, interactions between products and services and the overall PSS. It is abductively inferred that the cohort might have been fixated on product aspects due to their extensive product domain-based experiences, in an effort to reduce cognitive load (see Chapter 3). As a consequence, their ability to effectively explore the PSS design space may have been restricted (Section 5.4).

8.1.2 Proposition 2

Based on these theoretical arguments and empirical observations, the CAM-D model (see Chapter 3) is refined and extended with Proposition 2, also with two conditions. These conditions postulate the interplays of states of information processing within the bth design micro-frame (F_b, where $b > 0$) that is created in the working memory space during the design process specific to domain *y*, and the type and extent of domain-based design experience. The designer is assumed to have pre-existing design experience specific to the domain *x*, which is characteristically different from the characteristics of the design problem currently being addressed (specific to domain *y*). These conditions are presented as follows:

Condition 1:

F_b is created when exogenous design data specific to domain *y* is pulled into the working memory space and interpreted as design information (*Ey*). If the degree of automation (*n=p)* of the knowledge schemas specific to domain *x*

is greater than the degree of automation ($m=z$) of the knowledge schemas specific to domain *y*, significant cognitive load (based on Section 2.5) in the working memory space will be expended on constructing Ky_m. If $Kx_{n=p}$ is transferable to the current design problem specific to the domain *y*, then it will be pulled into the working memory space along with $Ky_{m=z}$ to reduce the high cognitive load (based on Section 3.2.1) required to address the design problem. Kx_n will have a greater influence than Ky_m on the information processing within the design micro-frames as long as the former has a higher degree of automation (n > m) (see Section 3.2.1). Thus, the information processing within F_b will be a function of the knowledge schemas specific to domains *x* and *y*, and the design information (Ey) specific to domain *y*.

Condition 2: If the degree of automation of Ky_m is higher than Kx_n, experiential design knowledge schemas with sufficient automation, suitable configurations, and content (specific to domain *y*), to address the design problem will be available to the designer. Thus, the input knowledge i_{b-1} being pulled into F_b is assumed to only include $Ky_{m=z}$. Therefore, the information processing within F_b is assumed to be a function of the knowledge schemas specific to domain *y* and the design information (Ey) specific to domain *y*.

These two conditions of Proposition 2 can be represented as follows:

$$F_b = \begin{cases} f(Kx_{n=p}, Ky_{m=z}, Ey) & Condition\ 1: if\ n > m \\ f(Ky_{m=z}, Ey) & Condition\ 2: if\ m > n \end{cases} \quad (3)$$

The following hypotheses linked to Condition 1 of Proposition 2 are abductively formulated by relating the incidental empirical findings of DS-1 (Papers 1 and 2 in Section 5.4) with existing theories and the findings of other existing literature (see Section 8.1.1). Subsequently, the influence of the level of domain-specific design experience, on the distribution of cognitive effort on the variables of problem-solving search strategies (Section 2.6.4) during designing specific to the same domain, is abductively hypothesized as follows:

H5: The level of domain-specific design experience of the designer significantly influences the distribution of cognitive effort on the sub-systems level of abstraction during designing specific to the same domain of design.

H6: The level of domain-specific design experience of the designer significantly influences the distribution of cognitive effort on the search strategies during designing specific to the same domain of design.

Furthermore, the exploratory empirical findings of Paper 2 are related with other empirical findings and theoretical arguments concerning design fixation (see Section 8.1.1). Subsequently, the influence of the level of domain-based design experience on the distribution of cognitive effort on the variables of domain-specific strategies (e.g., PSS design architecture (Section 5.3.2))

during designing specific to a characteristically distinct but transferable domain, is abductively hypothesized.

H7: The level of domain-specific design experience of the designer significantly influences the distribution of cognitive effort on domain-specific design strategies, during designing specific to a characteristically different but transferable domain of design.

8.2 Role of design prescriptions based on CAM-D

8.2.1 Discussion - findings of Paper 6

Based on previous works in the domain of instructional science, Paas, Renkl and Sweller (2004) suggested that the cognitive load of domain-novices could be greatly mitigated by introducing appropriate domain-specific prescriptive support, affording them more capacity to effectively carry out the information processing pertaining to the task. Here, it is suggested that the cognitive load of designers without domain expertise, may also be alleviated by supporting their design activities with domain-specific prescriptive design support (partly based on Curry (2014)).

The organization of the design process or approach being considered by the designer while addressing domain-specific design problems, is expected to be shaped mainly by the designer's own knowledge, unless it is imposed by the external design problem or prescriptions (based on Visser (2009)). External prompts, cues and tools are shown to influence the cognitive characteristics of designing (Gero et al., 2013; Visser, 2009; Yu & Gero, 2016). More specifically, the empirical study by Gero, Jiang and Williams (2013) shows that the provision of a structured external design support can influence the distribution of cognitive effort on problem-solution spaces, design states and transformation processes. The empirical findings of DS-2 (Paper 6) suggest that the introduction of the PSS design schema may have influenced the distribution of cognitive effort on variables of the design process such as problem-solution spaces, design state issues and transformation processes (Section 7.2).

External design process support prescribes a formally specified sequence of activities to aid designers in developing a design or a class of designs specific to a domain, which will often be the customization of a methodology to a domain specific problem (based on Gerrike et al. (2015)). Building on the arguments of Roozenburg and Eekels (1995), Daalhuizen and Cash (2021) suggest that such procedural support can include knowledge concerning specific theories or typologies integrated into certain ways of working in terms of structure for the design process to unfold. Structure refers to a minimal set of logically and chronologically ordered actions that need to be undertaken by a designer to

perform a domain-specific design activity (based on Daalhuizen & Cash (2021)).

Thus, if there is a lack of domain-specific design knowledge schemas or, more in general, of domain expertise, the information processing within design micro-frames can be supported by exogenously providing domain-specific design process support (which can be linked to meso-level procedural design models (see Section 2.9)). The effectiveness of such process support can be ensured by catering the prescriptive internal domain-specific logic of the support to the intended goals of the support (based on Daalhuizen & Cash (2021)) and of the design problem being addressed. The empirical findings of DS-2 also suggest that the introduction of domain-specific design process support PSS design schema, may have influenced the distribution of cognitive effort on variables such as hierarchical levels of systems abstraction and on variables of domain-specific design approaches such as lifecycle stages and aspects of PSS architecture (Section 7.2).

8.2.2 Proposition 3

Based on the discussion of the empirical findings of the exploratory study carried out in DS-2 (Paper 6) (see Section 7.2) and the theoretical arguments concerning design process support (Section 8.2.1), the following proposition is inductively postulated:

Proposition 3*:*

> The introduction of a design process support, the internal logic of which is catered to the goals of the domain-specific design problem currently being addressed by designers without domain-specific design experience, can significantly influence the information processing within design micro-frames.

The following hypotheses concerning the influence of design process support on the distribution of cognitive effort on problem-solution design spaces, design state issues and transformation processes (Section 2.6.3), hierarchical levels of systems abstraction (Section 2.6.4), and domain-specific design approaches (Section 2.3.2 and 2.7) , are inductively formulated:

H8: The use of a domain-specific design process support by a designer without domain-specific design experience, significantly influences the distribution of cognitive effort on problem-solution spaces during designing specific to the same domain.

H9: The use of a domain-specific design process support by a designer without domain-specific design experience, significantly influences the distribution of cognitive effort on design state-issues during designing specific to the same domain.

H10: The use of a domain-specific design process support by a designer without domain-specific design experience, significantly influences the distribution of cognitive effort on design state-transformation processes during designing specific to the same domain.

H11: The use of a domain-specific design process support by a designer without domain-specific design experience, significantly influences the distribution of cognitive effort on levels of systems abstraction during designing specific to the same domain.

H12: The use of a domain-specific design process support by a designer without domain-specific design experience, significantly influences the distribution of cognitive effort on domain-specific design approaches during designing specific to the same domain.

Chapter 9 – Concluding discussion

9.1 Assessment of research quality and limitations

9.1.1 Research quality and limitations of the empirical work

In this section, the criteria for research quality assessment outlined in Section 4.4 are used to discuss the quality of the empirical work of this thesis. The protocol studies in DS-1 and DS-2 aimed to 1) cognitively characterize conceptual PSS designing with respect to product designing by experienced product developers, and 2) study the effects of the PSS design schema on the cognitive characteristics of the design process involving a cohort with the same characteristics. Thus, the characteristics of the cohort were treated as a control variable in both DS-1 and DS-2. To do so, a criterion-based sampling (Cash et al., 2022) approach was utilized to select the cohort. Although on an abstract level, the characteristics of the cohort (in terms of type and level of experience) were controlled, the participants may have had differing roles, responsibilities, experiences, and idiosyncrasies (based on Dinar et al. (2015)). This issue can be addressed in future works by including a large enough sample size to enable randomization (ibid).

Such differences may have posed as confounding variables, potentially challenging the internal validity or credibility of the findings. This variability is expected to be especially high in the case of experienced practitioners in relation to other types of cohorts such as student designers. Experienced practitioners tend to serve multiple organizational functions within their own work settings and have different educational backgrounds, while the levels of knowledge and experiences of students tend to be uniform based on the stage and type of their education. However, since insights into design expertise can only be accessed through the study of experienced practitioners (Cross, 2006), they were chosen as the focus of the studies.

Another source of threat to the internal validity of the studies in DS-1 is the differences in the details of the design tasks given for the product and PSS design sessions, which may also pose as confounding variables (based on Dinar et al. (2015)). Here, a trade-off between experimental control and transferability is made, as the respective design tasks were chosen in a bid to represent typical product and PSS design challenges that may be encountered by practitioners. More specifically, the PSS design task has a higher level of detail compared to the product design challenge. The PSS design task was designed to represent the reality of the design challenge faced by practitioners within manufacturing companies transitioning from designing products to environmentally benign PSSs,

who tend to build upon existing product-centric offerings while designing PSSs (based on Isaksson et al. (2009)).

Moreover, this trade-off is also made by qualitatively considering the consistency in the similarity of the cognitive characteristics of product designing measured from different previous protocol studies entailing different product design tasks (see the comparison of different protocol studies of product designing in Neramballi et al. (2018)). To further strengthen the credibility of the empirical findings, the predictions need to be tested in future works with a factorial experimental design. Such studies should introduce different design challenges specific to the two domains, having different levels of detail. However, in DS-2 (Paper 6), this threat to internal validity was limited as both the control and experiment groups were given the same design task and the independent variable was the PSS design schema.

As discussed previously, to facilitate transferability or external validity, the design tasks for the respective domain-specific design problems are chosen to be representative of the respective challenges faced in the real world. However, since the design sessions were carried out in laboratory settings as opposed to industrial ones, the validity of the findings in industrial settings in DS-1 and DS-2 may be restricted (Ball & Christensen, 2018; Cash et al., 2013). However, this limitation is considered as a necessary trade-off to be able to facilitate statistical analysis and thus, the dependability or reliability of the empirical findings, which is crucial for theory building and testing (Dinar et al., 2015; Hay et al., 2020).

Furthermore, to ensure the dependability (or objectivity) of the protocol analysis-based empirical research in both DS-1 and DS-2, the Delphi protocol (Gero & Mc Neill, 1998) was applied, which entails the use of multiple independent coders and an expert arbitrator to carry out the analysis of the protocol data. The inter-coder reliability was measured against the codes given by the expert arbitrator. Furthermore, several of the used frames of analysis are widely utilized coding schemes, and the entire process of data collection and analysis has been transparently described to enable reproducibility (Dinar et al., 2015; Gero & Milovanovic, 2020). The DS-1 and DS-2 of this research do not consider the influence of co-design, knowledge sharing, sketching, uncertainty, and other designer-specific variables. Such issues have been explored in other previous works (Cash & Kreye, 2017; Gero & Milovanovic, 2021; Hay et al., 2017).

The ADR carried out in PS and DS-2, mainly use a mix of qualitative techniques such as focus group workshops, document analysis and the quantitative technique of LCA. This research mainly aimed to develop and validate design methods to support practitioners of two manufacturing companies seeking to transition from designing typical products to PSSs with relatively reduced environmental impacts. The research also aimed to cater to the development of the design-specific needs of the practitioners to successfully realize such a

transition. ADR was chosen to specifically ensure that the support being developed, applied and tested would cater specifically to the end users of the support. The credibility or internal validity of these findings is expected to be high, as AR tend to be high in internal validity and tend to provide in-depth insights (Bhattacherjee, 2012)

More specifically, the use of ADR enabled the active involvement of the practitioners during the method development process. The empirical data collected during the focus group workshops were consistently reviewed by all the research participants to ensure credibility and dependability. However, it is important to note here that due to the use of ADR, the degree of active involvement of researchers within the empirical field may be higher when compared to typical case studies or observational studies. Such active involvement of the researcher might have affected the dependability of the research (Bhattacherjee, 2012), which is inherent in prescriptive design research. Thus, it is considered a necessary trade-off.

Although typical AR tends to have limited transferability (Bhattacherjee, 2012), the use of the systematic and transparent procedure prescribed by ADR is expected to facilitate the transferability of the prescriptions developed during the research process to other similar cases, while simultaneously ensuring the internal credibility of the case company (based on Sein et al. (2011)). The inherent generic limitations of LCA (Finnveden, 2000) and the challenges specific to LCA of PSSs (Kjaer et al., 2016) are expected to prevail. However, since such issues are beyond the scope of this thesis, they are not discussed further.

9.1.2 Quality assessment of the conceptual model

Using the theory quality assessment criteria (based on Bhattacherjee (2012), Chakrabarti & Blessing (2015) and Eckert & Stacey (2014)) presented in Section 4.4, the quality of the developed conceptual model of CAM-D and the role of design support are assessed as follows. In order to ensure logical consistency, the constructs, propositions, variables and hypotheses outlined in the developed conceptual model were built upon previous empirical observations, widely accepted theories and models from both cognitive science and design literature (which is deemed as a necessary criterion for design theory building (Chakrabarti & Blessing, 2015)). Further, a theory should demonstrate explanatory ability, predictive ability and falsifiability (Bhattacherjee, 2012; Cash, 2018; Chakrabarti & Blessing, 2015).

All the propositions of the proposed theoretical model were utilized for either predicting or explaining the observed phenomenon of the cognitive characteristics of PSS designing with respect to product designing, to guide the development of cognitively catered PSS design support and to predict or explain the effects of the developed support within the boundary conditions of the set

assumptions. Further, the falsifiability of all the propositions was operationalized with either the inductive or abductive formulation of empirically testable hypotheses. The predictive ability of the hypotheses of Proposition 1 of CAM-D, concerning the influence of characteristics of design problem on the cognitive characteristics, was assessed with the testing of respective predictions in the context of this research (catering to the recommendations of developing falsifiable theory (Bhattacherjee, 2012; Eckert & Stacey, 2014)).

Based on the incidental findings of DS-1 (Papers 1 and 2) and the preliminary findings of DS-2 (Paper 6), the conceptual model CAM-D was refined and extended in Chapter 8 (in line with the suggestion for theory building by Cash (2018)). To refine and extend the conceptual model, additional propositions and testable hypotheses concerning the influence of pre-existing knowledge schemas on PSS designing based on the previous design experiences of the cohort were abductively formulated. Furthermore, the effects of the PSS design schema on the cognitive characteristics of the design process, were inductively formulated based on the preliminary findings of DS-2. These hypotheses when tested in future works can assess the predictive ability of Propositions 2 and 3. Rigorous empirical validation and assessment of the propositions, hypotheses, and respective predictions, with both larger cohort sizes, enhanced controls, and differing contexts, are necessary to further strengthen the internal and external validity of the model.

Since, the hypotheses linked to the three propositions were defined in a generic (non-domain specific) manner, the generality of the presented model is expected to and also shown to be high as they were partly (H1-H4) used to model the cognitive characteristics of product and PSS designing. Also, the theoretical model and the correspondingly developed design process support were shown to be applicable to a large pool of sample of practitioners working within the different manufacturing sectors, which highlighted the breadth of the generality of the model, a criterion deemed necessary by Eckert and Stacey (2014). It is important to note that other issues germane to the design activity such as the sketching, co-design and knowledge sharing are out of the scope of this model.

9.2 Implications to the state of the art and practice

9.2.1 Contributions and implications to design science

The main contributions and implications of this thesis to design science are discussed as follows:

- ✓ This thesis presents empirical insights into design expertise from a pool of 68 experienced practitioners from the product manufacturing industry.

Insights into designing by such a cohort were deemed crucial to advance the understanding of design expertise in general (Cross, 2006) and, more specifically, in the domain of PSS designing (Vasantha et al., 2015). Further, this thesis has adopted a theory-driven empirical research approach. Such an approach was adopted in response to the growing consensus in the design science community concerning the need to transition from typical descriptive and exploratory studies towards the development of theory-driven research (Cash, 2020; Dinar et al., 2015; Hay et al., 2020). Therefore, a theoretically ingrained conceptual framework was developed and used to guide the empirical research of this thesis (Chapter 3).

- ✓ This thesis presents a theoretically-based conceptual framework of the cognitive architecture and mechanisms underlying the variant nature of design cognition, termed CAM-D.

CAM-D is grounded in the well-established theory of cognitive architecture and cognitive load theory (Sweller, 1988; Sweller et al., 1998), the augmented generic design hypothesis (Visser, 2009), and other relevant design theories (Section 2.6). It has been observed in the reviews of the state-of-the art that there is a lack of, and a need for, a comprehensive and generic theoretical model of design cognition (Cash, 2018; Hay et al., 2017). Cash & Kreye (2017) have proposed a theoretical formalism of the design activity termed the uncertainty-driven action (UDA) model, which links the constructs of design action and cognition. However, several other issues of relevance remained unaddressed. With the development of CAM-D, this thesis has attempted to address several of the following issues outlined by Hay et al. (2020):

On a higher level, Hay and colleagues note that there are major opportunities to build upon the rich exploratory design research available in the state of the art and on existing cognitive science theories to develop theoretical models of the design activity by 1) identifying, mapping, formulating, and testing predictions between the relationships between the major variables underlying design cognition and 2) refining and extending scientific formalisms of design cognition. Among other aspects, the potential of theorizing about inter-linked topics, such as knowledge and memory, cognitive processes, design processes, and design objects and their attributes, were highlighted with questions such as (based on Hay et al. (2020)):

- Which cognitive science theories and models are applicable to design cognition research, and how can they be transferred?
- How can the cognitive processes underlying design activities and methods be mapped?
- How can cognitive science theories be leveraged to understand the interactions of design objects and their attributes with the designers?

- How can models and theories of memory be linked with design cognition?
- How are different types of memory and knowledge utilized during design tasks?

With the conceptualization of CAM-D, this thesis has attempted to address several of the above-highlighted issues in order to improve the understanding of the cognitive characteristics of conceptual PSS designing for environmental sustainability. More specifically, generic predictive propositions and hypotheses were developed as a part of CAM-D, which are expected to be applicable to a wide range of design domains.

✓ The influence of the characteristics of domain-specific design problems on design cognition was theoretically predicted and empirically assessed. This contribution can potentially be used to predict the differences in design cognition specific to characteristically distinct domains of design.

More specifically, with Proposition 1 and the associated hypotheses (Section 3.3) of CAM-D, the thesis has attempted to map and predict the inter-relations between the attributes of domain-specific design objects, information processing within the working memory, and the knowledge or expertise development within the long-term memory underlying the process of designing. The predictive ability of the first proposition of the model has been partially assessed in the context of conceptual product and PSS designing of DS-1 of this thesis.

The protocol analysis-based research covered in DS-1 (Papers 1 and 2) provides rich insights into differences in design cognition specific to the two characteristically distinct domains of product and PSS designing by practitioners experienced in product development. Further, it provides in-depth insights into the domain-specific design strategies applied by the cohort while addressing the PSS design problem. Thus, on a general level, this research provides insights into how practitioners experienced in one domain approach a design challenge specific to a characteristically distinct but transferable domain. Such empirically tested theoretical insights into the variant nature of design cognition were previously deemed a relevant topic of interest for design research (Visser, 2009).

✓ This thesis also contributes towards the development of a framework that could potentially predict and explain the interplays of memory, pre-existing design knowledge, cognitive load and the phenomenon of design fixation.

Based on the incidental empirical findings of DS-1 (Papers 1 and 2) and the well-established theory of design fixation, abductive inferences were drawn to formulate Proposition 2 and associated hypotheses (5-7) of CAM-D (Section

8.1), thus extending the model. Proposition 2 and the associated hypotheses conceptualize and predict the influence of both the lack of pre-existing design knowledge specific to the domain of the design problem being addressed (PSS designing in the context of this current research) and fixation-type effects due to the retrieval of pre-existing design knowledge specific to a characteristically distinct domain of design (product designing in the context of this research), in order to mitigate cognitive load. Thus, an empirically grounded and predictive conceptual framework is developed that could contribute towards developing an improved understanding of the mechanisms underlying the interplays of memory, domain-specific knowledge, cognitive load, and design fixation. These issues have been highlighted as areas of interest for future research by Crilly & Cardoso (2017) and Hay et al. (2020). Liikkanen (2010) has previously utilized similar cognitive constructs, such as memory and knowledge and associated processes, to present a cognitive model of creative idea generation during designing. However, the model does not explore the phenomena of domain-specific variances and domain-specific expertise in design cognition, in detail.

- ✓ This thesis also contributes towards the development of a conceptual framework that can potentially predict and explain the effects of a domain-specific design process support on design cognition.

Based on the exploratory empirical findings of DS-2 (Paper 6) and relevant theoretical arguments, Proposition 3 and associated hypotheses (H8-H12) concerning the effects of domain-specific design process support on design cognition were formulated. Although one of the main meta-goals of design cognition research is to support design practitioners in improving their design practice, there is a growing divide between the two areas (Zielhuis et al., 2022). Zielhuis et al. (2022) suggest a need to make the results of design cognition research more accessible to the design practice community. Moreover, design researchers seldom utilize insights from design cognition research to guide the development of design support (see Section 1.3). Thus, on a philosophical level, this thesis attempts to bridge the divide between phenomenological approach of descriptive design cognition research and the pragmatic approach of the prescriptive design research (Section 4.5). Therefore, Proposition 3, along with the rest of the conceptualization of CAM-D, is expected to contribute towards supporting design researchers to more effectively utilize such empirical insights to provide cognitively catered support for design practice.

9.2.2 Implications to design research and practice for environmental sustainability

The growing importance of a societal transition towards an environmentally sustainable and circular economy has incentivised the manufacturing industry to constantly adapt. Thus, the design research community needs to be cognizant of

the need to support the manufacturing industry with methods fit for the evolving industrial practice (Gericke et al., 2020). The potential of PSSs to support the industrial transition towards an environmentally sustainable, resource-efficient, and circular economy has been widely heralded by academia (Roy, 2000; Tukker, 2015). In order to effectively support the manufacturing industry in designing such environmentally sustainable solutions, it is vital to develop empirical insights into how such design processes are carried out on the cognitive level of the individual designers (based on Brissaud et al. (2022), Ramani et al. (2010) and Vallet et al. (2013)). More specifically, it is necessary to understand what cognitive models designers do and should bring while designing for environmental sustainability (based on (Klotz et al., 2019)). Such empirical insights are crucial to effectively support typical product manufacturing companies to design and offer PSS-based solutions with relatively less environmental impacts.

- ✓ Based on the empirical findings of DS-1 (Paper 1) of this thesis, it was inferred that the challenge of designing environmentally sustainable PSSs may be significantly different on the micro-scaled cognitive level than designing specifically to a typical product design challenge.

This empirical finding challenges the assumption that product and PSS designing are similar activities in terms of problem-solving (Vasantha et al., 2015). Furthermore, based on the contributions of Papers 1 and 2, this thesis developed and presented an improved understanding of the cognitive characteristics of PSS designing performed by experienced product developers of manufacturing companies. These findings pointed towards the need for PSS domain-specific design support for experienced product developers to effectively design PSSs with relatively lower environmental impacts.

- ✓ Based on the insights obtained from the improved understanding of the cognitive characteristics of PSS designing, three design prescriptions that provide support for the different aspects of PSS designing for environmental sustainability were presented in the PS of the thesis (Papers 3, 4, and 5).

First, a PSS domain-specific design process support, termed PSS design schema, was developed (Paper 3). Its effects on design cognition were exploratorily investigated in a protocol study in DS-2 (Paper 6).

- ✓ Preliminary findings indicate that the introduction of the PSS design schema, as a domain-specific design process support for the designers, might help them to apply a higher systems perspective and a wider lifecycle perspective (more focus on Multiple Stages and Solution Deployment) to focus more on service aspects, interactions on the product-service aspects, and the overall PSS (Paper 6).

These observed changes in design strategies are in line with the literature-prescribed strategies that are suggested to be required to develop more environmentally sustainable PSS-based solutions (based on Aurich et al. (2006), Brissaud et al. (2022) and Ceschin & Gaziulusoy (2016)). For example, it is hypothesized that cumulative environmental impacts of specific types of products may significantly increase in the use or Solution Deployment phase (Section 2.2, Figure 3). PSSs with higher service degrees are hypothesized and also shown in some cases to have superior environmental performance (Section 2.3).

The first design support method, LFD, was developed to provide detailed guidance for designers within the typical product manufacturing industry to conceptually redesign their existing product-centric offerings to PSS-based solutions with comparatively lower environmental impacts. This design support addressed a vital gap in the prescriptive design literature concerning the lack of systematic and transparent support for designers to effectively utilize LCA results in the context of design for environmental sustainability in general (Ramani et al., 2010) and also, more specifically, in the context of PSS design for environmental sustainability. The procedure for systemic lifecycle engineering (Paper 5) was developed to guide designers to identify, narrow down, and integrate critical product and service DPs that can together address the overall system functions or PSS DPs, which can significantly improve the systemic lifecycle environmental performance.

- ✓ The assessment of the validity of the application of the two design support methods (Papers 4 and 5) provided two accounts of the potential of conceptually designing PSS-based solutions with relatively lower environmental impacts compared to the existing product-centric offerings of the companies.

This validation of the applications of the design support adds to the limited yet needed empirical evidence for the claim concerning the potential of PSS-based solutions to have relatively lower environmental impacts than typical product-based ones (Boehm & Thomas, 2013; Brissaud et al., 2022; Tukker, 2015). Furthermore, the existing offerings redesigned in the two cases are substantially different: one is a relatively simple, consumable incontinence product, while the VAE is a relatively complex, long-lasting sub-system within a larger renewable energy-producing wind tower system. These two case applications highlight the potential to develop environmentally benign PSS-based solutions across a wide range of industry sectors. Moreover, the empirical findings from the validation of the design support methods preliminarily suggest that designers without PSS design expertise can indeed be guided to develop environmentally benign PSS solutions by providing appropriate informational support for the different steps of the conceptual PSS design process.

- ✓ The conceptual model of CAM-D developed in the thesis can potentially be utilized to characterise the cognitive processes specific to the designing of structurally more complex systems, such as STS and policies for environmental sustainability, with respect to PSS, product, and other more relatively structured domains of designing.

The potential for realizing environmental sustainability is hypothesized to increase with the consideration of larger systems with higher degrees of social embeddedness as design objects (e.g., spatio-social systems, STS, policies) (Ceschin & Gaziulusoy, 2016). CAM-D can potentially be used to improve the understanding of the cognitive characteristics of the processes of designing such systems. As demonstrated in this thesis, such insights can be potentially used to develop effective support for designing such systems.

Chapter 10 – Conclusion and ways forward

10.1 Conclusion

Grounded on the theories of cognitive architecture and cognitive load (Sweller et al., 1998), the situated notion of designing (Gero & Kannengiesser, 2004), and the augmented generic design hypothesis (Visser, 2009), the conceptual model of CAM-D was developed (Chapter 3). Proposition 1 and associated hypotheses concerning the influence of domain-specific design problems on design cognition were inductively formulated. Subsequently, these hypotheses were used to predict the design cognition differences between conceptual PSS designing by experienced product developers with respect to conceptual product designing (Chapter 5). In order to test these predictions, a protocol study comparing product and PSS designing was carried out as a part of DS-1 (Paper 1).

The empirical findings suggest statistically significant differences between conceptual designing specific to the two domain-specific design problems. PSS designing was reported to be more problem-focused than product designing. The in-depth protocol analysis of designing specific to the PSS design problem, also carried out as a part of DS-1 (Paper 2), revealed that the cohort focused mainly on product aspects while affording substantially lower focus on the service aspects, interactions between product and service aspects, and aspects concerning the overall PSS. These empirical insights from DS-1 together answer RQ1.

Exploiting this improved understanding, one PSS domain-specific design process support and two information-based design support methods were developed as a part of PS. The PSS design process support, termed the PSS design schema, was developed based on PBSA and five PSS essences. The two design support methods, termed LFD and procedure for systemic lifecycle engineering, were developed by implicitly following the principles prescribed by the ADR method in cooperation with two separate manufacturing companies, respectively. These prescriptions developed in the PS together answer RQ2.

The effects of the PSS design schema on PSS designing were exploratorily investigated with a protocol study as a part of DS-2. The results point towards differences in the distribution of cognitive effort on FBS design issues, design processes, levels of systems abstraction, different solution lifecycle stages, and different aspects of PSS architecture. Due to the relatively small cohort size, statistical analysis of these findings was not done, but the findings were used to inductively formulate hypotheses concerning the effects of domain-specific design process support. The two design support methods were applied in the two respective case companies by implicitly following the principles of ADR. PSS design changes to their existing product-centric offerings were introduced using the support in both cases. The effects of these two design

support methods were assessed in terms of LCAs of the developed design outcomes. In both assessments, a substantial reduction in the environmental impacts of the design changes was observed. These empirical findings of DS-2 answer RQ3.

The conceptual model of CAM-D was refined and extended to develop two additional propositions and associated hypotheses based on the empirical findings of this research (Chapter 8). The overall contributions of CAM-D to design science are briefly summarized as follows: Three propositions were postulated, and their falsifiability was operationalized with empirically testable hypotheses. Proposition 1 and associated generic hypotheses concerning the influence of the exogenous domain-specific design problem on design cognition were deductively assessed in DS-1 (Paper 1). Proposition 2 and associated generic hypotheses concerning the influence of the pre-existing design knowledge of the designer on design cognition were abductively postulated, partly based on the findings of DS-1 (Paper 2). Finally, Proposition 3 and the associated generic hypotheses concerning the influence of the externally provided domain design process support on the cognitive characteristics of the conceptual design process were inductively postulated partly based on the preliminary empirical findings of DS-2 (Paper 6).

This thesis also contributes towards the advancement of the knowledge on design for sustainability as the empirical findings, the proposed model of CAM-D, and the role of design support together contribute towards an improved understanding of how traditional product developers address PSS for environmental sustainability-related design challenges. Further, guidelines towards how to effectively support the design of such systems are presented and assessed. It is increasingly being recognized that to effectively address sustainability challenges, it is necessary to design relatively more complex and socially embedded systems. Thus, the findings of this thesis can potentially contribute towards improving the understanding of the cognitive processes underlying the designing of such systems. Grounded on such an improved understanding, effective support for designing such systems can also be developed.

10.2 Future research directions

A series of potentially interesting future research avenues emerge from the conclusions of this thesis. Although some preliminary evidence supporting the predictive ability of the proposed conceptual model of the CAM-D is presented, further research in other contexts is needed to extend, refine and validate the theoretical model (based on Cash (2018)). A natural next step would be to extend, refine, and rigorously test the predictive powers of the proposed model. In terms of extension, more granular constructs of the design micro-frame and design knowledge schemas can be developed, and associations between them can

be mapped. Furthermore, protocol studies with larger cohort sizes are needed to strengthen the generalizability of the developed cognitive model of PSS designing. Protocol studies of different types of cohorts, such as practitioners experienced in PSS designing and novice student designers, can enrich the empirical understanding of the cognitive characteristics of PSS designing.

Further protocol studies with larger cohort sizes and statistical analysis are needed to carry out hypothesis testing of the effects of the PSS design schema on the cognitive characteristics of PSS designing. Multiple case study applications of LFD and the P-S interdependency knowledge base involving different cohorts are needed to increase the generalizability of the validation findings. It would also be interesting to study the effects of the two presented design support methods on the different types of cognitive loads (Sweller et al., 1998) and the distribution of cognitive effort on the different variables of the design process by the designers.

Cognitive science research has used a wide range of other, more advanced methods, such as behavioral experiments, psychometrics, and more invasive neurophysiological imaging methods (Hay et al., 2020). The contributions of this thesis can also be used as a basis to carry out neuroimaging studies to investigate and compare the neuro-physiological characteristics specific to PSS designing with respect to other domains of design. The potential of utilizing such neuroimaging techniques to bring rich empirical insights into design cognition has been highlighted by reviews of the state-of-the art (Gero & Milovanovic, 2020; Hay et al., 2020).

Furthermore, in order to improve the sustainability potential of design approaches, it is necessary to consider larger and more socially complex systems as objects of design (Ceschin & Gaziulusoy, 2016). An interesting future work would be to apply the conceptual model of CAM-D in studying and comparing the cognitive characteristics underlying the designing of such larger and more socially complex design objects, such as STSs or policies for environmental sustainability and the circular economy. Based on the above-described future research avenues, the following RQs for future research are formulated, and the design science community is invited to explore them:

1. What are the differences and similarities in the neuro-physiological activations of product and PSS designing?

2. What are the cognitive characteristics of STS designing for a circular economy?

3. What are the cognitive characteristics of policy designing for a circular economy?

4. How can STS and policy designing for a circular economy be effectively supported based on an understanding of the cognitive characteristics of the respective design processes?

"*Revolution is not in the social or the economic field but at a much greater depth; It is in the very structure of the human consciousness.*"

– Jiddu Krishnamurthi

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What does this book offer?

- A generic conceptual model of cognitive architecture and mechanisms of designing.
- Empirical insights from a pool of 68 experienced product developers from the manufacturing industry.
- Empirical insights into cognitive characteristics of product-service system designing for environmental sustainability.
- Empirical insights into potential influence of past product development knowledge on product-service system designing for environmental sustainability.
- Three product-service system domain specific design support and empirical insights into their effects on product-service system designing for environmental sustainability.

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